

DETERMINANTS OF GEOGRAPHIC LABOR MOBILITY,
ELECTRIC VEHICLE CHARGING STATION DEPLOYMENT, AND
MORTGAGE REFINANCING

A dissertation presented

by

Arvind Sharma

to

The Department of Economics

In partial fulfillment of the requirements for the degree of Doctor of

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In the field of

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ABSTRACT OF DISSERTATION

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Abstract

Chapter 1: Impact of Occupational Licensing on Interstate Migration of Workers

Labor mobility offers workers the opportunity to improve their personal and financial situations, and plays a large role in affecting unemployment and underemployment across the country. There are several factors that limit a worker's mobility, including family ties and potential inertia as well as external factors arising from regulation. One external factor that has gained prominence in recent decades is the use of (state-specific) occupational licensing requirements. Combining data from the Current Population Survey with information from the Annual Social and Economic Supplement for all occupations and states over 5 years, I study the impact of occupational licensing on a worker's interstate mobility. Using k-means algorithms to account for potential reporting issues and running a set of fixed effects models, I find little evidence that occupational licensing limits interstate mobility. Thus, the concern that occupational licensing contributes to mismatches in the labor market seems unsubstantiated.

Chapter 2: Pricing & Competition in Electric Vehicle Charging Infrastructure (with Nehan Naim)

Electric vehicle charging infrastructure is a major driver (or barrier in case of lack thereof) for electric vehicle adoption. Unlike conventional vehicles, electric vehicle (EV) owners can use various commercial as well as residential options for charging their vehicles. This expanded choice set has an important bearing on the decision of commercial EV charging station offerings, where these stations in the US differentiate across their offerings of charging technologies, including speed (AC, DC) and compatibility (DC Connector types), as well as pricing. United States has more than 25000 public EV charging stations, mostly operating as part of networks, like ChargePoint, Electrify America, Tesla, etc., or as independent or un-networked stations. Recently, some networks have been partnering with each other to allow 'roaming' to customers between these network stations. This paper empirically investigates the effect of 'roaming' service and product/technology

differentiation on business pricing by combining individual station data with market characteristics. We find stations part of a roaming network are more likely to charge a positive price, and stations use product/technology differentiation to soften competition.

Chapter 3: Impact of FinTech firms on the Housing Mortgage Market

I study the effect of exclusively online high-technology home mortgage originators (“FinTech” firms) on three different types of housing loans originating in the US from 2011 to 2017, namely home purchases, home improvements and home refinancing. Unlike traditional banks, FinTech firms are much less regulated and usually do not have physical stores, and like many other mortgage originators, they have an originate-to-distribute business model – which allows them to be more flexible in residential refinancing markets. However, while FinTech firms could be causing more refinancing, they could also be simply filling in the refinancing need better through enhanced prediction of potential customers and aggressive expansion in geographic areas wanting to refinance, say through online advertising channels – essentially creating an endogeneity issue. Using a year-county fixed effects model with an instrumental variables approach (state-level Bartik instrument for the number of approved applications by FinTech firms in a county), I find a significantly higher number of approved home refinancing and home purchases applications in counties with a higher FinTech presence, even after controlling for aggregated loan, borrower and county level controls. This paper builds upon prior work on the role of technology in mortgage lending and contributes by better addressing the endogeneity concern, though my sample is currently limited to the largest 20 lenders in the US as of 2016 only.

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Table of Contents

Abstract	3
Acknowledgments	5
1 Impact of Occupational Licensing on Interstate Migration of Workers	10
1.1 Introduction	10
1.1.1 Motivation	10
1.1.2 Some Pros and Cons of Occupational Licensing	12
1.1.3 Certification versus License	13
1.1.4 Paper Overview and Organization	14
1.2 Literature Review	15
1.2.1 Some Occupational Licensing Studies on Geographic Mobility	17
1.3 Data	21
1.3.1 Sources	21
1.3.2 Overview	22
1.3.3 Construction of Occupational Licensing Variable	25
1.3.4 Data Cleaning and Transformations	32
1.3.5 Summary Statistics	42
1.4 Specification	47
1.5 Results	50

1.5.1	Robustness Check: k-means algorithm identifies licensed individuals . . .	55
1.5.2	Robustness Check: Alternative Specifications	57
1.5.3	Making sense of the results	59
1.6	Conclusion	60
1.6.1	Method	60
1.6.2	Results	61
1.6.3	Future Work/Potential Extentions	62
1.6.4	Contribution	63
1.7	Bibliography	65
1.8	Appendix	69
1.8.1	Summary Statistics of Non-Transformed Data	69
1.8.2	Migration to Another State (Transformed Data)	77
1.8.3	k-means with 3 clusters	81
2	Pricing and Competition in the Electric Vehicle Charging Infrastructure	83
2.1	Introduction and Motivation	83
2.2	Pricing and Competition Literature	86
2.3	Data	89
2.3.1	Summary Statistics: Key Variables	93
2.4	Model 1: Service and Product/Technology Differentiation	94
2.4.1	Specification 1	95
2.5	Model 2: Extended Spatial Competition	100
2.5.1	Specification 2	100
2.6	Conclusion and Future Work	104
2.7	Bibliography	107
2.8	Appendix	110
2.8.1	Figures	110

3	Impact of FinTech Firms on Housing Mortgage Market	115
3.1	Introduction	115
3.1.1	Classifying FinTech Firms	117
3.2	Literature Review	119
3.3	Data	120
3.3.1	Sources	120
3.4	Specification	122
3.4.1	Empirical Methodology: IV with Time and County FE	125
3.4.2	Summary Statistics	126
3.5	Results	129
3.6	Conclusion	131
3.6.1	Further Work	132
3.7	Bibliography	134
3.8	Appendix	136
3.8.1	Figures	136
3.8.2	Tables	137

Chapter 1

Impact of Occupational Licensing on Interstate Migration of Workers

1.1 Introduction

1.1.1 Motivation

There are two strong motivations to study occupational licensing (OL).

First, about one-quarter of the U.S. workforce needed a license to work in their occupation in 2019 as per National Conference of State Legislature (2019). Not only is this large in magnitude but occupational licensing already impacts more people than both union memberships and federal minimum wage coverage in US, which affected 10.3% and 1.8% of the employed Americans in 2019 (Shierholz 2020; Release 2020). In fact, as of July 2015, there were at least 60 occupations that were licensed in every state and over 1,100 occupations that were licensed in at least one state, as per Treasury Office of Economic Policy, Economic Advisers, and Labor (2015).

Furthermore, occupational licensing as a labor market institution has been steadily gaining ground over the years. Back in 1950s only 5% of the employed people were required to hold a

license (Morris M. Kleiner 2010). This five-fold increase in share of licensed workers is not primarily driven by a growing share of workers self-selecting into heavily licensed occupations i.e. changing composition of the workforce. Instead, about two-thirds of the growth of licensed workers comes from increase in the number of occupations that require a license (Treasury Office of Economic Policy, Economic Advisers, and Labor 2015). Additionally, de-licensing of an occupation rarely occurs, and recent attempts in nine states to collectively de-license groups of occupations have shown more potential but have been almost uniformly unsuccessful (Thornton and Timmons 2015).

The second motivation to study occupational licensing is that it can potentially explain why both geographic mobility and people switching between occupations have been decreasing simultaneously. Since 1980s, economists have documented a secular decline in migration measured at interstate, intercounty and intracounty level, and a decline in labor market transitions - i.e., people are not changing employers (jobs), occupations, and industries as much as they did in the past (Molloy, Smith, and Wozniac 2014). Furthermore, these two stylized US labor market facts cannot be concurrently accounted for by trends popularly studied in the labor literature such as concerns of dual-career households¹, changes in the distribution of job opportunities across space², and polarization in the labor market³ (Molloy, Smith, and Wozniak 2011).

However, if individuals in licensed occupations (who usually make more wages than unlicensed individuals) have a higher opportunity cost of moving out of their licensed jurisdiction, or face additional moving/relicensing costs then occupational licensing is one potential reconciliatory

1. “Rising share of dual-earner households” which are likely to experience a joint co-location problem with a spouse and are thus likely to stay in a city and not change jobs/industry/occupations only rose from 2% in the 1980s to 3% in the 2000s, which is too small of a change to meaningfully affect aggregate migration patterns.

2. “Range of occupations and industries has become more uniform across metropolitan areas and has reduced the incentive of people to migrate.” This only explains the decline in geographic mobility trend but it need not explain decline in labor churning trend.

3. “Expanded use of computers, greater ease of automation and off-shoring increases demand for high-skill jobs (e.g. professional, managerial, and design jobs) but reduces demand for the middle-skill jobs (defined as office administration, sale and production jobs)”. Such structural shifts in the occupation distributions can potentially have restrict migration if historically low skilled workers in food service, personal care services or cleaning services are more likely to move to a different labor market to take middle-skill jobs, and there was an elimination of large shares of middle-skill jobs that could limit the “migration worthy” jobs for less educated workers. Unfortunately, again there is no not much empirical support for this “hollowing-out effect” hypothesis.

factor explaining both why geographical fluidity is decreasing (largely across state boundaries) and why there is less churning (between occupations).

The current large magnitude and high growth of occupational licensing laws in the recent decades has started drawing more policy attention. Occupational licensing has come in the lime-light alongside other pressing labor market issues like immigration, changes in demographic structure of workforce, technological advances and globalization of competition. In fact, Institute for Justice concluded in a 2018 report that “occupational licensing is widely recognized as one of the most important labor market issues in the United States, and the topic warrants some more research”.

1.1.2 Some Pros and Cons of Occupational Licensing

Proponents of occupation licensing usually highlight its benefits such as better product and quality control, safeguarding public health and safety through focusing on higher, protecting consumers by guaranteeing minimum educational requirements and industry oversight, supporting career development or pathways for licensed workers, and enhanced professionalism for licensed workers (National Conference of State Legislatures 2019). In particular, they emphasize that occupational licensing can solve the classic lemons problem in service markets and thus avoid market failure due to adverse selection - thereby making labor markets more competitive when forces like reputation or litigation fail to achieve desired outcomes.

However, opponents of occupational licensing often highlight reducing geographic mobility of workers, increasing rent seeking behavior, reducing wages for unlicensed workers relative to their licensed counterparts, reducing employment in licensed occupations, reducing market competition and innovation, increasing the price of goods and services, and disproportionately burdening immigrants, military veterans and their families, people with a criminal history⁴ and dislocated or unemployed workers (National Conference of State Legislatures 2019). In particular, they have

4. 1 in every 3 American adults has a criminal record, as per Bureau of Justice Statistics, 2016.

argued that occupational licensing goes against free market principles as it restricts movement of labor across the country, depriving workers of opportunities not only in their current state of residence but also other “greener economic pastures” (Sean E. Mulholland 2016).

Since occupational licensing may hurt the economy of gains from trade in labor services that are based on comparative advantages or variation in market opportunities, some today are raising the call for occupational licensing system reforms (Sean E. Mulholland 2016 and Sanders 2020). However, it is interesting to note that de-licensing of an occupation rarely happens, and attempts in nine states to collectively de-license groups of occupations in 2015 has been almost uniformly unsuccessful (Thornton and Timmons 2015).

1.1.3 Certification versus License

The three terms “license”, “certification”, and “registration” are often used interchangeably causing confusion, but have more formal and very specific meanings.

Registration usually requires individuals only to list contact information with a specific agency, usually a designated government body, and as a general rule does not require the person to pass an exam or meet hold pre-determined standards (Sanders 2020). Thus, they are the least restrictive regulation out of the three basic forms. Certification on the other hand give the holder a protected occupational title, and individuals without the certificate may practice the occupation as long as they do not use the protected title. However, a license restricts individuals from indulging in “scope of practice” without permission from a government agency, and can be used to refuse people the legal opportunity to earn a living in a chosen field (Sanders 2020).

More formally, according to Current Population Survey (CPS), certifications and licenses are both non-degree, time-limited credentials that need to be renewed periodically, and demonstrate a level of skill or knowledge needed to perform a specific type of job (Labor Statistics 2017).

The difference between the two is based on who issued the credential. More specifically, “certification is a credential awarded by a non-governmental certification body based on an individual

demonstrating through an examination process that they have acquired the designated knowledge, skills, and abilities to perform a specific job” (Labor Statistics 2017). In particular, certificate does not convey a legal authority to work in an occupation. For example, Project Management professional certifications by Northeastern University, or Python/R data science certifications on LinkedIn. A certification can be highly valuable too, like Chartered Financial Analyst (CFA) designation.

In contrast, “license is a credential awarded by a governmental licensing agency based on pre-determined criteria, which could include some combination of degree attainment, certifications, educational certificates, assessments (including state-administered exams), apprenticeship programs, or work experience” (Labor Statistics 2017). More importantly, license conveys a legal authority to work in an occupation. For example, a K-12 teaching license, a nursing license or even a cosmetology license.

Current Population Survey started collecting data on these two non-degree credentials from 2015, given its paucity and their distinct nature from traditional educational attainment data, like a high school degree or a bachelor’s degree, to help policymakers/ researchers study how these two types of credentials affect labor market success.

1.1.4 Paper Overview and Organization

In this paper, I use three recently added non-degree credential questions in Current Population Survey to estimate whether a person is licensed by the state government to work in their primary occupation or not from 2016 to 2020. I have more than 400 unique occupations spread across 50 US states and Washington, D.C.. Within an occupation-state at a given period of time, all individuals should either be licensed or unlicensed. However, I find that the individual license estimate is noisy, and thus use k-means unsupervised machine learning algorithm to correct for the misreporting in survey, and estimate an individual’s licensing status in both current and last year. The algorithm suggests that within a state-occupation-time cluster where less than 22% individuals report they are

licensed, everyone should be unlicensed. Similarly, in occupation-state-time clusters where more than 69% individuals report they are licensed, everyone should be licensed.

I find individuals in k-means suggested licensed occupations do have higher wages, just like theory and past empirical work suggests, and this serves as a sanity check. Then, I run a state-occupation-time fixed effects model with detailed individuals demographic controls and find that occupation licensing does not have a negative impact on worker geographic mobility between states, and occupational licensing also does not impact geographic mobility within the state. This preferred specification is quite robust to choice of controls and alternative specifications.

The rest of the paper is organized as follows:

Section (1.2) discusses literature, in particular the four outcome variables that have been typically examined in occupational licensing literature and specific geographic mobility results.

Section (2.3) describes the two data sources (1.3.1), provides an overview of data transformations required to be able to run the preferred specification (1.3.2), the three certification questions and construction of the licensing variable from them (1.3.3), correcting misreporting in the self-reported licensing variable using k-means unsupervised machine learning algorithm (1.3.4), changing the interpretation of migration outcome variable to estimate my preferred specification (1.3.4), followed by the summary statistics (1.3.5).

Section (1.4) describes the identifying equation and the fixed effects model preferred specification. Section (1.5) presents and discusses the results, while Section (1.6) concludes.

1.2 Literature Review

Historical Occupational Licensng Study Categories

Occupational licensing studies in the past have mostly looked at individual occupations and studied them in depth in one the following four distinct classes of outcome variables, namely -

(1) Impact on Quality, Health and Safety, (2) Impact on Wages and Employment, (3) Impact on Consumer Prices, and (4) Impact on Geographic Mobility (Treasury Office of Economic Policy, Economic Advisers, and Labor 2015).

First, occupational licensing studies on quality, health and safety are usually motivated by how occupational licensing can potentially reduce consumer uncertainty about the labor service quality, and thereby increase consumer's demand for the licensed service (Arrow 1971). For example, if it takes consumers too much time, effort or they simply cannot distinguish between high and low quality electrical work, then competition may crowd out the high quality electricians from the market and only the "lemons" or low quality electricians will serve the entire market. Thus, while occupational licensing can theoretically fix the lemon problem, most empirical evidence does not find that stricter licensing requirements improve these outcomes. For example, teaching licenses usually have no effect on student scores (Thomas J. Kane 2008), tighter dentistry licenses requirements have no effect on dental deterioration or amount of repair required (Kudrle 2000), and floristry license requirement has no effect on floral arrangements ratings by florist judges (Carpenter 2012).

Second, occupational licensing studies on wages and employment usually motivated by regulatory capture theory, where practitioners capture monopoly rents, restrict labor supply and drive up prices of labor. Thus, a wage gap gets created between those who manage to enter licensed occupations and those who fail to enter the licensed occupations, usually estimated to be about 5 to 10 percent (Treasury Office of Economic Policy, Economic Advisers, and Labor 2015). Also, the wage premia differs by occupations with different licensed occupations experiencing different levels of wage growth, and the highest wage growth is documented in some of the occupations that also have the greatest proportions of licensed workers, such as health care support and practice, education, and legal occupations (Treasury Office of Economic Policy, Economic Advisers, and Labor 2015). Furthermore, employment grows at a slower rate in licensed occupations. For example, states with the least restrictive regulations of nurse practitioners (NPs) had more licensed NPs

and patients were more likely to receive primary care from NPs when physician primary care was lacking (Goodwin et al. 2013).

Third, occupational licensing studies on consumer prices theorize that prices should be higher as supply of new professionals is reduced and competition between providers is less vigorous. Many studies find that more restrictive licensing laws lead to higher prices for consumers. For example, regulating the number or functions of mid-level dental-care providers (hygienists and assistants) increases price of filling a cavity or dental visit (price index of 12 services) (Liang and Ogur 1987 and Brian Isom 2019), additional state mortgage broker bonding/net worth increases the probability that mortgage is high priced (Morris M. Kleiner 2007), and medium or high level of nursing license regulation increases the price of well-child medical exams (Marier et al. 2014).

Fourth, and finally, occupational licensing studies on geographic mobility are usually motivated by costs associated with re-allocation which may prevent migration. For example, additional monetary and non-monetary time costs like time to meet new qualifications in destination state, such as education, experience, training or testing, can discourage individuals from relocating. Alternatively, if the opportunity costs are higher for licensed individuals to move out of their home state compared to non-licensed individuals say because licensed individuals have higher wages, we can also see lower geographic mobility for people in licensed occupations as a result.

1.2.1 Some Occupational Licensing Studies on Geographic Mobility

First, there has been relatively little research on effects of occupational licensing on geographic mobility (Treasury Office of Economic Policy, Economic Advisers, and Labor 2015), and the little research is largely descriptive in nature because of the lack of available information on changes in licensing requirements that could be used in a traditional causal framework (Johnson and Kleiner 2020). Also, there was a shortage of data on non-degree credentials till about 2010, and Kleiner and Krueger (2010) conducted the first national study using specially designed Gallup survey to study occupational licensing (though their study focused on wage dispersion).

Second, in the past, emphasis has been on a few occupations like nurses and lawyers only, while many other occupations have been largely not studied. For example, lower interstate mobility of lawyers could be traced to investments in state specific law and procedure (estimated to have a small effect), and to licensing through state bar exam (Pashigan 1977).

More recently, as states started adopting reciprocity agreements (which lower re-licensure costs), occupational licensing's impact on lawyers and nurses has been studied through a difference-in-difference design. However, we get mixed results across occupations. For instance, mobility of nurses did not increase following staggered adoption of Nurse Licensure Compact⁵ (NLC) by states, even among residents of counties bordering NLC states that should be most impacted (DePasquale and Stange 2016). On the other hand, there is some evidence that the adoption of reciprocity agreements increases the interstate migration rate of lawyers (Janna E. Johnson 2017).

Furthermore, while universal licensure⁶ is gaining some popularity now, with as much as 11 states passing laws to mandate their occupational boards to allow individuals with out-of-state licenses to obtain a valid occupational license to practice in 2021, there are still significant limitations (Follett, Hentze, and Herman 2021). Almost always this process is not automatic, and not only do licensees must meet certain residency, testing, background check and other requirements, but as well as pay all applicable fees in order to practice in the new state.

Third, apart from certain occupations, geographic mobility occupational licensing studies examine certain groups of people like military spouses and immigrants. Military spouses often struggle to find and maintain employment due to frequent moves (an average of once every three years), and are particularly vulnerable to licensing requirements (Hentze 2021). Similarly immigrants, described as “grease on the wheels of the labor market” since they help labor market efficiency by making geographic choices based on wage differences across locations, are adversely affected too (Borjas 2001). Licensing reduces the mobility of immigrants, may exacerbate worker shortages

5. An interstate compact that allows registered nurses and licensed practical and vocational nurses to practice across state lines in participating states without acquiring a new license.

6. Defined as a state recognizing a person's occupational license granted in another state as valid.

and can slow the entire economy (Hugh Cassidy 2019). Even low-skilled immigrants workers are affected and disperse less as a result of occupational licensing regulations, as evidence from from Vietnamese immigrants who came to US after Vietnam War and entered the manicure industry (Maya N. Federman 2006).

Fourth, there have been very few attempts to generalize occupational licensing geographic mobility results to all occupations. In 1977, Pashigan found that occupational licensing has had a quantitatively large effect in reducing interstate mobility of 34 professionals occupations (Pashigan 1977). More recently, there are only three works that I am aware of. Out of these, Kliener and Johnson (2020) do the most rigorous study and single out causal effects of occupational licensing on interstate migration by disentangling the decision to move into three components - risk aversion, loss of local capital, and relicensing costs.

To get to the casual effects, Kliener and Johnson (2020) assume that moving more than 50 miles destroys all local clientele/business reputation effects, and people who stay in their home state are fundamentally higher risk averse individuals who value family ties and friendships much more (Johnson and Kleiner 2020). Thus, by throwing out these two sub-groups of people, they can look at actual costs of occupational licensing on interstate migration as occupational licensing in US is primarily at state level. Furthermore, Kliener and Johnson (2020) prefer to rely on American Community Survey over Current Population Survey (CPS) due to three reasons, namely - (1) CPS does not have sub-state place of residence (required to disentangle local client effect), (2) CPS does not report state of birth, but only country of birth (required to disentangle risk aversion), and (3) CPS has much smaller sample size (Johnson and Kleiner 2020)⁷.

Therefore, using American Community Survey data from 2005 to 2018 and looking at individuals who move 50 or more miles and reside outside their state of birth, their preferred specification shows that individuals in 16 state-specific universally licensed occupations⁸ move be-

7. One advantage of Current Population Survey over American Community Survey is that it allows one to differentiate between new entrants and continuing members in an occupation.

8. Example - bar for lawyers or the pharmacy jurisprudence exams, which are state-specific but nonetheless these occupations are licensed in all states.

tween states at a 7 percent lower rate compared to members of 6 quasi-national licensed occupations⁹(Johnson and Kleiner 2020).

They get similar results when they use Current Population Survey as a robustness check and compare interstate migration in the 16 state-specific universally licensed occupations (treatment group) against all other occupations (excluding 6 quasi-licensed occupations). This specification is close to my empirical work. However, to estimate who is licensed and who is not, they simply assume that only individuals in the 16 state-specific universally licensed occupations are licensed from 2005 to 2018. Thus, assuming licensing is not increasing over time is a very strong assumption, especially given that we know that new occupations tend to get licensed. I improve upon this licensing variation by using the three new certification questions added questions in Current Population Survey from 2016 to 2020 and k-means algorithm to better capture the growth in occupational licensing over time and states. Current Population Survey comes with certain disadvantages too though, namely that disentangling different components of migration cannot be determined.

Second, a White House report in 2015 that tried to replicate upcoming Kliener and Johnson (2020) approach found adverse (state) occupational licensing restricting interstate migration too. Using American Community Survey data from 2010 to 2013 and looking at people from age 25 to 65, they found that interstate migration rates for workers in the most licensed occupations are lower by almost 15% of the average migration rate compared to those in the least licensed occupations (Treasury Office of Economic Policy, Economic Advisers, and Labor 2015). Difference in within-state migration rates for workers was much smaller - only about 3% of the average migration rate. These impacts were also much larger for younger (under 35) licensed workers. Furthermore, the using in Survey of Income and Program Participation (SIPP) over the eight-month period starting in late 2012, licensed workers were about 20% less likely non-licensed to change states (Treasury Office of Economic Policy, Economic Advisers, and Labor 2015).

Third, Nunn (2016) used the three certification questions in Current Population Survey to

9. Occupations that have a national exams with a single passing standard, such as for nurses, physicians, and social workers who must also transfer their license when they move between states, although at a slightly lower cost.

look at how occupational licensing affects wages, employment and migration (within state and across state moves of individuals). He presents simple graphs displaying his results but does not report statistical significance or regression tables. After adjusting for observable individual¹⁰ characteristics (work experience, education, gender, and race), licensed workers appear somewhat more likely than their unlicensed counterparts to move within a state, but are much less likely to move outside their state (Nunn 2016). In particular, he does not use geographic/state or occupation fixed effects, which are likely driving his results¹¹. Furthermore, Nunn (2018) published using the new data in Current Population Survey again to look at occupational licensing, this time with a richer set of individual level and geographic controls, but only focuses largely on the wage margin, operationalised by wage distribution, premia, inequality, and omits the geographic mobility/migration outcome this time (Nunn 2018).

Thus, there is still some value in using the new Current Population Survey data to study occupational licensing and its impact on worker geographic mobility.

1.3 Data

1.3.1 Sources

I obtain data from two surveys - Current Population Survey (CPS) and the Annual Social and Economic Supplement (ASEC).

CPS is a representative sample of the civilian household population in the US surveying occupants of about 60,000 households every month, and follows a 4-8-4 rotation pattern where every household/address is interviewed once a month for 4 months straight and then interviewed again after a gap of 8 months (1 year later) in order to reduce the inconvenience to each household while allowing for some yearly or monthly comparisons (Bureau 2019). This in-person and telephonic

10. Aged 25 to 64

11. My own work shows the importance of controlling for these, and occupation fixed effect is even more important than geographic/state fixed effect term. Since he published as soon as the first round of certification data was released, he does not need time fixed effect

interview was launched in 1940s and continues till today, providing many official labor force statistics for the US as it collects both labor market health and demographic data, including on income and poverty issues.

While CPS collects labor force data on a monthly basis, additional topical data are often collected via supplements to the monthly survey. Annual Social and Economic Supplement is one premiere supplemental survey to Current Population Survey which is annual (as the name suggests) and the ASEC data is mostly collected in March. Thus, ASEC is also called the March Supplemental to the Basic Monthly March CPS, and is in fact a widely used dataset.

1.3.2 Overview

The data section has three important ideas - construction of occupational licensing variable from three certification variables and correcting this noisy measure with k-means clustering algorithm, merging of CPS and ASEC records, and finally transforming migration variable interpretation in our clean data to form an out-migration specification instead of an in-migration specification (a slightly more interesting hypothesis). In this subsection, I will provide a brief overview of why these three steps are required, and then talk about each transformation in more depth later in the separate subsections.

First, I merge individual records across the two datasets (for March of every year) because our outcome variable (migration status) exists only in ASEC, while our key variable of interest (self-reported occupational licensing status) can only be constructed from three certification questions added in CPS.

Also, ASEC sample size is larger than CPS by construction, and thus I use Current Population Survey as the master data¹². I merge March Basic Monthly CPS survey (master data) with March

12. CPS has been expanded over the years to get better estimates on certain sub-populations, like persons of Spanish (Hispanic) origin and low-income children who do not have health insurance coverage to increase reliability of estimates and help answer certain policy question. However, ASEC sample had to be expanded at a faster rate to supplement the CPS, in particular due to people dropping out of CPS in lieu of its inherent 4-8-4 rotation pattern.

Supplemental (Annual Social and Economic Supplement) using “MARBASEID”¹³ identifier constructed by IPUMS¹⁴ to generate our pooled cross section data (Flood et al. 2021). Only less than 5% people in CPS are not matched with people in ASEC, and 281,428 individuals are matched. I drop occupations that have less than 10 individuals, and am left with 281,404 individuals. Section 1.3.4 will discuss the Current Population Survey master data cleaning process, along with the merge results with ASEC in more depth.

Second, using the three certification questions in CPS, I create the key parameter - active government licensing, which informs us if an individual is licensed by the state government or not to practice their primary job in their occupation within the state. Construction of this variable is discussed in Section 1.3.3.

Furthermore, I correct for misreporting within occupation-state clusters, as everyone within an occupation-state cluster at a given period of time should ideally give a uniform answer, either yes or no, about being licensed by the state. Since I find that even within occupations that we know are licensed by all states, like registered nurses where only 82% people report they are licensed, and this misreporting occurs to varying degrees in all states, I conclude that the active government license variable generated from the three certification questions is a noisy measure of being licensed (by the state in a particular occupation). Thus, I reduce the misreporting error by using k-means clustering algorithm to determine which occupations-state clusters are licensed or not at any given period of time.

In particular, I collapse individual level license dummy variable (281,404 observations) in the merged CPS ASEC data by state, occupation and time to get proportion of people who are

13. “MARBASEID” was created specifically to correctly link the ASEC and the March Basic Monthly data and saving researchers duplication effort. The match is otherwise complex for two reasons Sarah M. Flood and José Pacas, “Using the Annual Social and Economic Supplement as Part of a Current Population Survey Panel,” *Journal of Economic and Social Measurement* 42, nos. 3-4 (May 2018): 225–248. First, the variables required to link the ASEC to CPS monthly files are not available for all years on the ASEC. As a result of omitted linking keys and the ASEC over-sample, duplicate and false matches are problematic. Second, Census Bureau transition to a computer-based interview resulted in more prominent data quality issues for linking across months even if they did not compromise the integrity of each individual month of data.

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licensed in each occupation-state-time clusters (58,117 observations)¹⁵. Then, I classify each of these 58,117 collapsed state-occupation clusters into 3 categories (licensed, unlicensed and not sure) or 2 categories (licensed, unlicensed).

When I created 3 categories, k-means algorithm suggested state-occupation-time clusters where less than 22% people have an active government license are unlicensed occupation-state-time clusters, and state-occupation-time clusters where more than 70% people have an active government license are licensed state-occupation-time clusters. The in-between state-occupation-time clusters are put in the ‘not sure’ about state-occupation license status category. Thus, I standardize or correct for the individual license status (active government license variable response) within all state-occupation-time clusters, or for all 281,404 individuals in the dataset.

Similarly, if I create only 2 categories, k-means algorithm suggests that state-time-occupation clusters where less than 42% people have an active government license should be non-licensed state-time-occupation clusters. Regardless of creating 2 or 3 cluster classifications, the k-means algorithm correction reduces the number of people who have an active government license slightly. More details will be discussed in Section 1.3.4.

Third, I prefer to test that if one is currently licensed by the state, they did not move to another location over the next one year (out-migration specification), rather than to test if one is currently licensed by the state, they moved from another location over the last one year (in-migration specification).

The default migration question in ASEC asks individuals if they moved from abroad, moved from another state, moved from another county within state, moved within a county, or did not move over the last one year, which without any further transformations only allows me to test the in-migration specification. Moreover, since ASEC provides not only individual’s current occupation and state of residence but also individual’s last year occupation and state of residence too, I can

15. Note that I have 5 years of data (2016 to 2020) with 51 states and 442 unique occupations, so the maximum number of occupation-state-time clusters that could have been formed is 112,710. However, the actual data only has 58,117 occupation-state-time clusters because there are occupations that have few people and thus may not span across all states, and at time not even across all years.

test the out-migration specification only if I can estimate the individual's last year licensing status too¹⁶. Here, k-means algorithm allows me to estimate individual's last year licensing status too, and thus get to my preferred specification.

In particular, I apply k-means algorithm decision rule to identify which individuals were licensed last year only when I have licensing data on these individual's last year occupation-state-time cluster in my data. Since the certification questions data in CPS was only released from 2016 onward, I cannot estimate last year licensing status for individuals in 2016, and lose one full year of data. Similarly, I lose a few more people because there was no one in their last year occupation-state-time cluster who had answered the three certification questions. Section 1.3.4 will discuss this, and I will finally be left with about 200,000 individuals when I create licensed/unlicensed classification, and about 157,000 individuals when I create licensed/unlicensed/do not know classification.

The rest of the section is organized as the following. I will first elaborate about the construction of our occupational licensing variable in Section 1.3.3, then about Current Population Survey data cleaning process and merge results with Annual Social and Economic Supplemental in Section 1.3.4, followed by k-means correction in Section 1.3.4, transforming the dataset to generate the preferred migration interpretation in Section 1.3.4, and finally present the summary statistics in Section 1.3.5.

1.3.3 Construction of Occupational Licensing Variable

CPS added 3 certification questions in its monthly surveys since January 2015 based on the work of the Interagency Working Group on Expanded Measures of Enrollment and Attainment (GEMEnA¹⁷) (Allard 2016).

Together, these 3 questions (profcert, statecert, jobcert) allow us to create a proxy for whether

16. The three certification questions in CPS are monthly

17. Members included senior staff from the U.S. Bureau of Labor Statistics (BLS), the Census Bureau, the Council of Economic Advisors, the U.S. Department of Education's Office of the Under Secretary, the National Center for Science and Engineering Statistics, and the OMB.

a person requires license to operate in a particular occupation within a state or not. The questions were -

Profcert

1. *“Do you have a currently active professional certification or a state or industry license? Do not include business licenses, such as a liquor license or vending license.”*

The first question (profcert) is asked of everyone above 15 in Current Population Survey, excluding those with an employment status of “Armed Forces” (as they are not part of civilian institutional population).

Second, profcert is used to identify people with certifications or licenses (working definitions for certifications and licenses are in Section 1.1, not business licenses issued to companies. Examples of occupational licenses issued to individuals giving them the right to practice include a real estate license, a medical assistant certification, a teacher’s license, or an IT certification (Thornton and Timmons 2015).

In particular, some credentials that are not considered a certification or license in the Current Population Survey include U.S. Bureau of Labor Statistics, “Frequently Asked Questions about Data on Certifications and Licenses,” *Labor Force Statistics from the Current Population Survey*, April 2017, - (1) Licenses to engage in personal, as opposed to professional, activities. For example, regular driver’s licenses are excluded, but commercial driver’s licenses (CDLs) are included. (2) Licenses issued to businesses, such as liquor licenses or vending licenses. (3) Certificates of attendance or participation in short-term training. (4) Educational certificates awarded by an educational institution—such as a community college, a 4-year college or university, or a trade school—based on completion of all requirements for a program of study. While this training may help in the performance of a specific job, educational certificates are not necessarily required or considered proof of qualification. These types of credentials are not time-limited. Examples of educational certificates include a digital arts certificate from an online university or a motorcycle mechanics

diploma from a community college.

Third, people may have more than one of these credentials.

Lastly, note that profcert does not distinguish between certifications or licenses.

Statecert

2. *“Were any of your certifications or licenses issued by the federal, state, or local government?”*

The second question (statecert) is asked of people who answered “yes” to the first question. According to the definition developed by The Interagency Working Group on Expanded Measures of Enrollment and Attainment (GEMEnA), licenses are issued by a governmental body and people who answered “yes” to the second question are classified as having a license. People with a license may also have a certification.

Four things to note about statecert.

First, the intent of statecert question is to determine the issuing authority of a credential and thus, to distinguish between certifications and licenses. Recall licenses are issued by a governmental body, unlike certificates. It is natural to ask this question after profcert because of the interest in distinguishing between certifications and licenses.

Second, while statecert does not help us distinguish between the level of government that issued the license, we know that most occupation licensure occurs at state level and not local or federal level. To substantiate this claim, I will quote Federal Trade Commission (2018) - “Occupational licensing, which is almost always state-based, inherently restricts entry into a profession and limits the number of workers available to provide certain services. It may also foreclose employment opportunities for otherwise qualified workers. This reduction in the labor supply can restrain competition, potentially resulting in higher prices, reduced quality, and less convenience for consumers...”(Goldman 2018).

Third, statecert is a conditional question. Only those who answered yes to profcert were asked this question. In other words, if a person answered no to profcert, they are not asked statecert question.

Fourth, statecert was modified to allow for the situation in which a person has more than one certification or license. Instead of “was it issued by ...” the question wording was modified to “were any of your certifications or licenses issued by ...” to allow for the possibility of multiple credentials. Such modification rules ensure questions are not misunderstood, and complex skip patterns using the responses to several questions answered in the past ensure that CPS interview stays short and clear.

Jobcert

3. *“Earlier you told me you had a currently active professional certification or license. (Is/was) your certification or license required for your (job/main job/job from which you are on layoff/job at which you last worked)?”*

The third question (jobcert) is asked later¹⁸ in the Current Population Survey interview of employed and unemployed people who were identified in the first question (profcert) as having a certification or license.

First, jobcert is also a conditional question like statecert but subject to more constraints - it has the smallest universe of out of the three certification question. In particular, unlike profcert and statecert, people with employment status of “Unemployed new workers” and “Not in the Labor Force” (NILF) (due to “Retirement”, “Inability to work”, or “Other reasons”) are not asked jobcert question. Thus, jobcert is only asked from everyone who answered “yes” to profcert question and falls in one of the three employment categories - “At work”, “Has job, not at work last week”,

18. After consideration, Bureau of Labor Statistics (BLS) decided that placing the first two questions after the educational attainment questions made the most sense. The primary reason behind this decision was the importance of collecting information from these two questions for all individuals, not just for those who were employed. Unlike the job-related questions, the educational attainment questions are asked of everyone. The third question (jobcert), which asked specifically about whether the credential was for the respondent’s job, did belong among other job-related questions.

or “Unemployed experienced worker”. Current Population Survey subject matter experts decided that it was important to ask jobcert of the unemployed about their past jobs. This is consistent with other labor force questions, such as industry and occupation, and allows for an examination of the unemployed who have previously worked.

Second, in jobcert question, “required” can mean either legally required (such as a license to practice medicine) or required by their employer (such as a computer maintenance certification). Some people may have credentials that they do not use for their current job, such as a computer technician who obtained a real estate license for personal interest or for a second job. Also, people sometimes have certifications or licenses that may be helpful for their jobs but are not required by law or by their employer. Thus, jobcert question allows us to create a strong working level, binding definition for our licensing variable.

Third, jobcert question wording differs by individuals’ labor force status/employment. Thus, the question is not as complex as it may initially seem. For example, people who are employed at only one job are asked about their job, while those with multiple jobs are asked about their main job. Likewise, unemployed individuals who are on layoff are asked about the job from which they are on layoff, and unemployed individuals who previously worked are asked about their last job.

Finally, we can construct an individual self-reported licensing status measure, active government license if a person answered yes to statecert and yes to jobcert. By construction, this is equivalent to if a person answered yes to profcert, statecert and jobcert. Table 1.1 above reports the results.

Since Current Population Survey is a high-quality national level survey implementing many best practices/wording choice when collecting data, I do not worry about serious misreporting. In fact, testing in about 25 cognitive interviews where respondents included individuals who had certi-

Table 1.1: Summary Statistics of Certification Questions

	Mean	Std. Dev.	Min	Max	Universe
Profcert: Has active prof-cert/license	0.251	0.43	0	1	281404
Statecert: Has govt issued prof-cert/license	0.907	0.29	0	1	70584
Jobcert: Cert required for job	0.839	0.37	0	1	70584
Licensed / Govt. issued prof-cert required for job	0.196	0.40	0	1	281404
Observations	281404				

Source: Current Population Survey (CPS), 2016-2020

Profcert Universe: 16-79 years old excluding people employed in Armed Forces.

Statecert Universe: Everyone who answered “yes” to Profcert. Those who answered “no” to Profcert are excluded.

Jobcert Universe: Everyone who answered “yes” to Profcert, excluding people “Not In the Labor Force” (NILF) and “Unemployed New Workers”. Again, those who answered “no” to Profcert are excluded.

fications or licenses as well as those who did not revealed no problems with the proposed placement or wording of the questions. Also, the cognitive testing did not find evidence that the new questions had an impact on responses to existing questions. Thus, Bureau of Labor Statistics decided to accept the proposal to permanently add the three questions about certifications and licenses to the monthly Current Population Survey. Furthermore, non-response on the questions is quite low, and regular imputation method used for other Current Population Survey questions was used on the three certification questions.

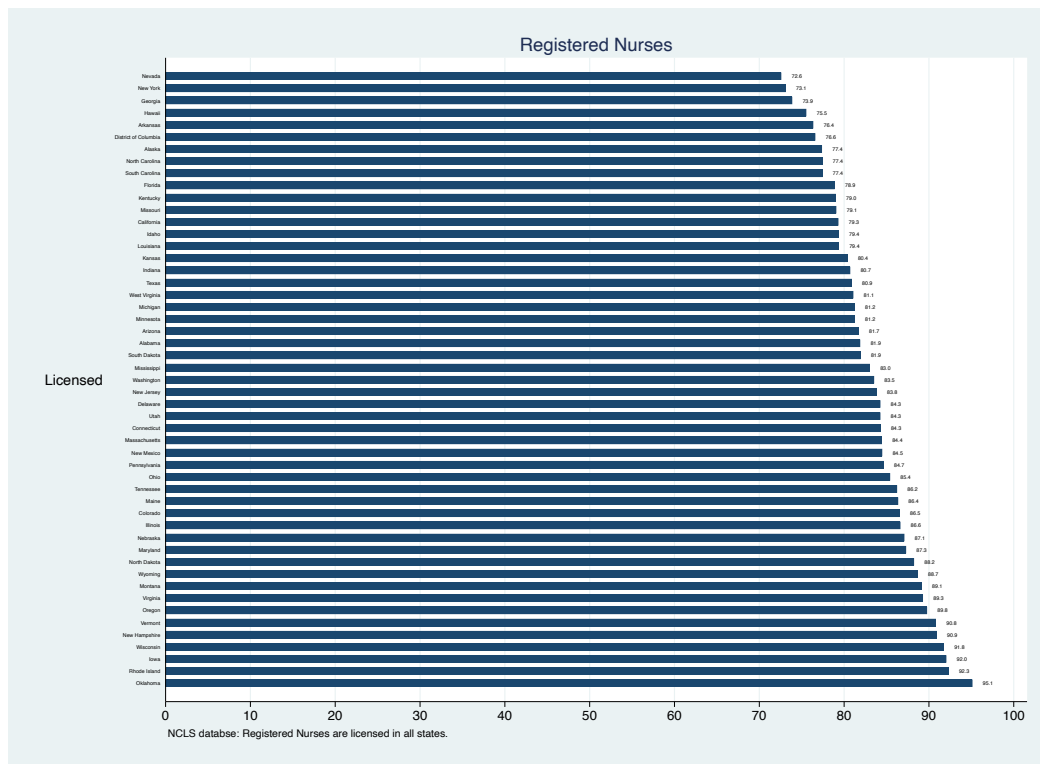
Nonetheless, examining the raw data/individual responses suggests there is in fact some misreporting. Note again that occupational licensing happens mostly at state level, so our key variable active government license can be interpreted as State Occupational Licensing. Thus, all individuals within a state-occupation-time should either answer “yes” or “no” unanimously, but instead we find that even in state-occupations that we know require a license for individuals to operate, not everyone answers correctly that they have a license. For example, even in occupations that that are licensed in all states like “Registered Nurses”, only about 82% registered nurses at the national level report they have an active government issued license (see Table 1.2 below), and the misreporting is spread across states (see Figure 1.1 below).

Table 1.2: Proportion of People Licensed in Occupations (ranked from top 20 to lowest 20 occupations)

Occupation	Proportion Licensed	Frequency
Chiropractors	.915	118
Optometrists	.899	79
Dentists	.879	282
Veterinarians	.876	170
Physicians and Surgeons	.857	1853
Pharmacists	.847	600
Occupational Therapists	.845	238
Registered Nurses	.825	6212
Lawyers, and judges, magistrates, and other judicial workers	.805	2411
Physical Therapists	.804	484
Aircraft Pilots and Flight Engineers	.802	248
Respiratory Therapists	.794	194
Physician Assistants	.791	196
Speech Language Pathologists	.786	294
Emergency Medical Technicians and Paramedics	.78	414
Psychologists	.774	389
Dental Hygienists	.754	357
Special Education Teachers	.741	729
Radiation Therapists	.735	34
Secondary School Teachers	.731	2093
...	...	...
...	...	...
...	...	...
Bank Tellers	.0071	567
Postal Service Clerks	.0049	203
Laundry and Dry-Cleaning Workers	.0033	303
Reinforcing Iron and Rebar Workers	0	10
Tire Builders	0	11
Communications Equipment Operators, All Other	0	14
Avionics Technicians	0	14
Paving, Surfacing, and Tamping Equipment Operators	0	20
Food & Tobacco Roasting, Baking, and Drying Machine Operators & Tenders	0	21
Textile Knitting and Weaving Machine Setters, Operators, and Tenders	0	21
Gaming Cage Workers	0	21
Rolling Machine Setters, Operators, and Tenders, metal and Plastic	0	21
Adhesive Bonding Machine Operators and Tenders	0	21
Helpers—Installation, Maintenance, and Repair Workers	0	26
Prepress Technicians and Workers	0	38
Textile, Apparel, and Furnishings workers, nec	0	40
Extruding & Drawing Machine Setters, Operators, and Tenders, Metal & Plastic	0	41
Credit Analysts	0	48
Pressers, Textile, Garment, and Related Materials	0	54
Extruding, Forming, Pressing, & Compacting Machine Setters, Operators, & Tenders	0	54
Ushers, Lobby Attendants, and Ticket Takers	0	76

Source: Annual Social & Economic Supplemental (ASEC), 2016-2020

Figure 1.1: Proportion of Registered Nurses reporting holding a License, by state. Registered Nurses are licensed in all states, yet individuals working in this occupation report they do not have a license.



Thus, there may be some under-reporting on average for occupations that are licensed and I will correct for this using the following methodology - if a high proportion of people within the state-occupation-time cluster hold an active government license, we can infer that the occupation is licensed by the state, and assign everyone in the state-occupation-time cluster with an active government license value of 1. Similarly, if a low proportion of people within the state-occupation-time hold an active government license, we can infer that everyone in the state-occupation-time does not hold a license. I discuss about this correction methodology in Section 1.3.4 below.

1.3.4 Data Cleaning and Transformations

First, I start with Current Population Survey survey as my master dataset, and apply three filters (year, employment status and age) so that we can start with a sample of civilian non-institutionalized working-age population which could be asked all three certification questions.

Firstly, I keep year 2016 onwards only (even though the three certification questions were introduced in January 2015) because jobcert question's responses are not reported for 2015¹⁹, and exclude Current Population Survey data for 2021 as 2021 Annual Social and Economic Supplement data is still not released by IPUMS.

Secondly, people with employment status of "Armed Forces" are dropped since they are asked none of the three certification questions (part of institutionalized population). Furthermore, people with employment status of "Not In Labor Force (NILF)" and "Unemployed New Workers" are dropped too as they are not asked jobcert question (but are asked the other two). It does not make sense to ask them about jobcert as they do not have a job yet (or much job history) - they are marginally attached with the labor market.

Thirdly, I restrict our analysis to a working age population, defined as ages 16 to 79. Current Population Survey asks these three questions about certifications and licenses of all people in the survey age 15 and older, but individuals between the ages of 80-84 are clubbed as 80 while anyone over 85 is clubbed as 85. (Results are robust to working age population definition of 16-65, and 25-55).

Thus, everyone in our sample is in the labor force and a potential candidate to answer all three questions asked the question (though everyone may not be asked statecert or jobcert due to the conditional nature of questions - respondents have to answer "yes" to profcert).

Second, I then create the primary (dummy) variable of interest named active government license which identifies if a person has an active government issued license to practice the profession, as discussed in Section 1.3.3 above in the Current Population Survey dataset. Again, our definition is if a person answered "yes" to statecert and "yes" to jobcert. By construction, these people also answered "yes" to profcert question.

Third, I merge Current Population Survey data (master) with ASEC data using the individual

19. In 2015, the first year of data collection, jobcert question was not asked in May and June 2015 of households in their first and fifth interviews due to a programming error, leading to lack of data for these 2 months

unique id created by IPUMS called “MARBASECID”. Only about 12,681 observations remain unmatched in the master dataset, and 281,428 individuals are matched. Thus, only about 4% of data did not match between Current Population Survey (March Basic) and Annual Social and Economic Survey (March Supplemental) for every year in the time period 2016-2020²⁰.

Most individual level controls come from Current Population Survey data such as age, gender, race, marital status, number of own children under 5 in household, but ASEC data contains year on year migration status (used to construct outcome variables), and total personal income data. Thus, I had to merge the two datasets.

Fourth, I drop occupation categories that have less than 10 people as they may be more prone to misreporting error due to the occupation group being formed out of less number of people. 24 observations falling in 4 occupations get dropped.

Fifth, I will now create a new measure of active government license that corrects for misreporting in license status within an state-occupation-time cluster so that all individuals within an occupation-state cluster at a period of time are either licensed or not (i.e. have uniform replies). Subsection 1.3.4 below talks about the process in depth, and I will end this section by talking about estimating the active government license status last year for individuals to change the migration variable interpretation in Section 1.3.4. Finally, I show summary statistics in Section 1.3.5.

Correcting Mismeasurement in Active Government License variable using k-means algorithm

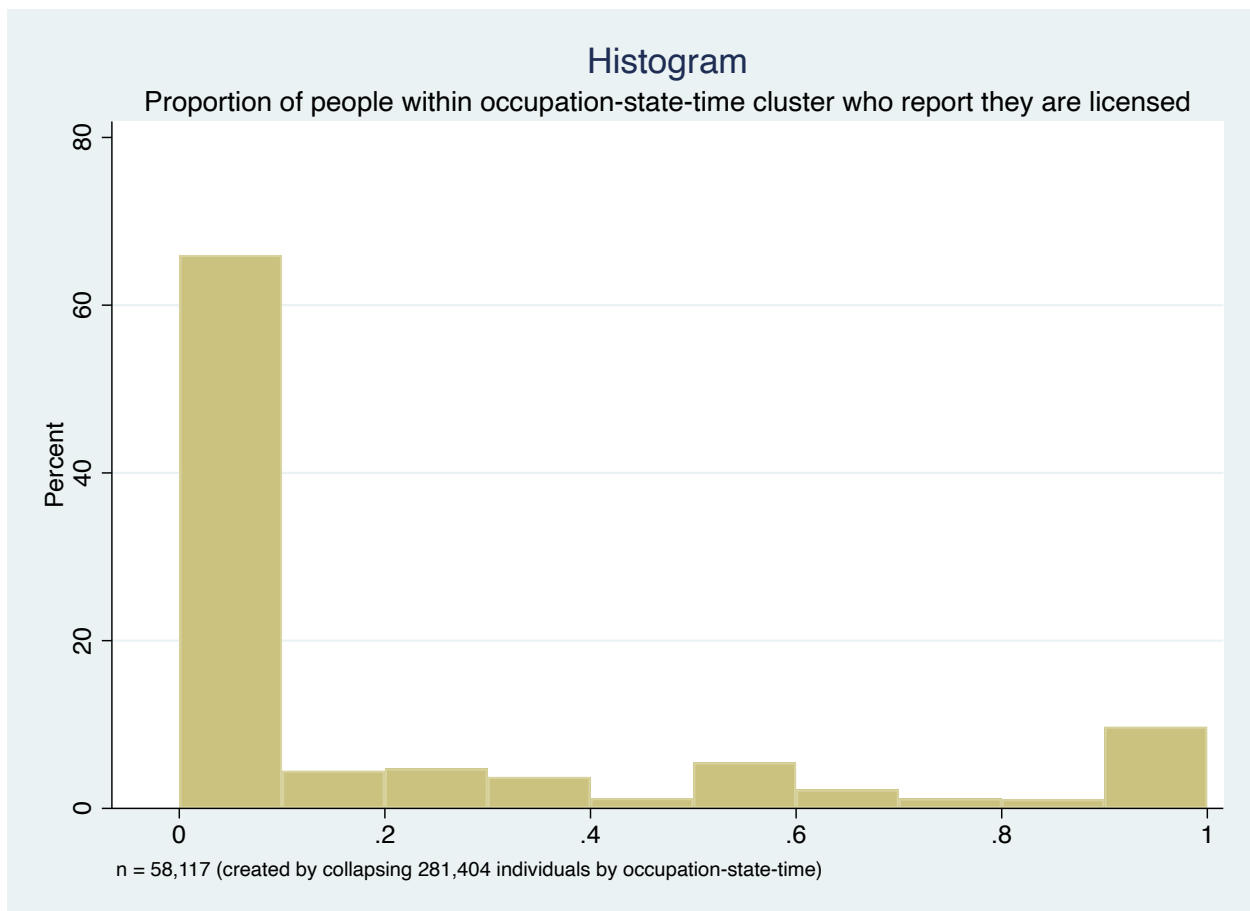
As mentioned in the Section 1.1, licensing has been increasing over time, though in our raw data time period from 2016 to 2020, licensing has been increasing very slowly. Moreover, de-licensing of an occupation rarely occurs (Thornton and Timmons 2015). Nonetheless, to be safe I control for the time dimension as some states could be switching occupation from unlicensed to licensed status over this time frame. Thus, I temporarily collapse the merged individual level data to state-occupation-time level (281,404 individuals collapsed into 58,117 state-occupation-

20. It may be possible to increase the match count even further by trying to marge March Supplemental data with Current Population Survey February Basic and Current Population Survey April Basic data.

time clusters) to examine the distribution of active government license dummy at state-occupation-time level (Figure 1.2).

The interpretation of the collapsed license dummy at state-occupation-time level is the following: proportion of people within an occupation-state-time that are licensed, and I use the distribution of these proportions using k-means algorithm to generate a decision making process so that I can unbiasedly conclude which occupations-state-time clusters are likely licensed or not, and thus rectify individual misreporting by superimposing the group wisdom on their individual “wrong” answers.

Figure 1.2: The maximum number of occupation-state-time clusters that we can have is 112,710 - since our data has have 5 years (March month only from 2016 to 2020), 51 unique states and 442 unique occupations. Since some occupations may have relatively few number of people not spread across all states (or even years), the actual number of occupation-state-time clusters in our sample is smaller.



The decision making process is based on the simple idea - if a low proportion of people within a occupation-state-time cluster report they are not licensed, it is likely that nobody is licensed within that occupation-state-time cluster. However, if a high proportion of people within a occupation-state-time cluster report they are licensed, it is very likely that everyone within that occupation-state-time cluster is licensed.

To chose the cutoff values for proportion of people within the occupation-state-time cluster beyond which everyone should be imputed a value of 1, and under which everyone should be imputed a value of 0, I use a simple unsupervised machine learning clustering algorithm called k-means.

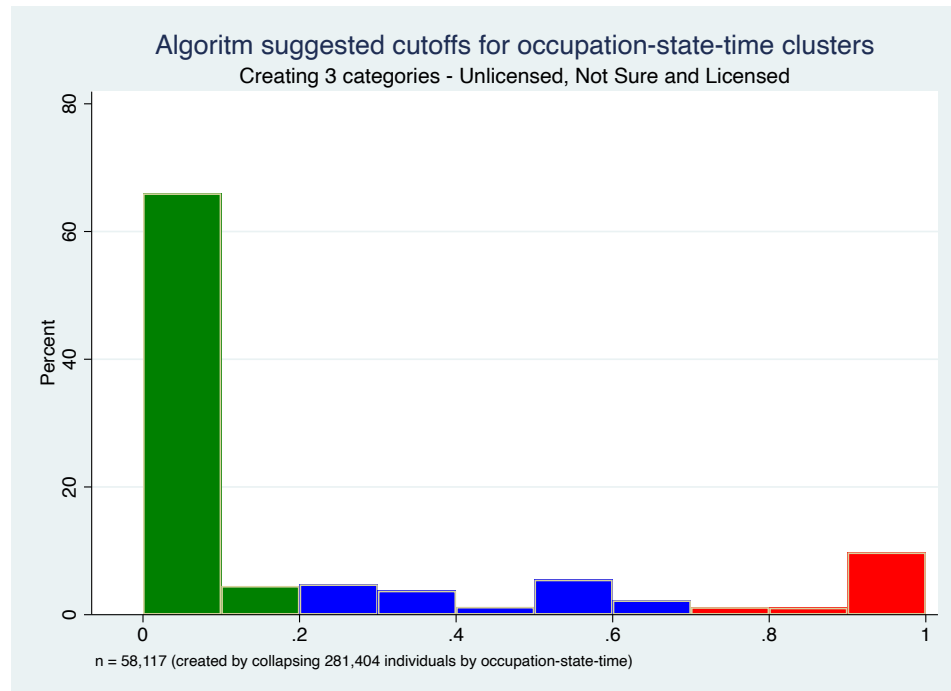
More formally, k-means clustering is an iterative procedure that divides the data into k groups or clusters. The procedure begins with k initial group centers, 2 in my case (for licensed and not licensed category), though my results are robust if I create 3 clusters (licensed, not licensed, or do not know category) too.

To implement the algorithm, we first assign all the observations to the group with the closest center. Second, the mean of the observations assigned to each of the groups is computed, and the first step is repeated again. These process continue until all observations remain in the same group from the previous iteration.

To avoid endless loops and attain algorithmic convergence, an observation is reassigned to a different group only if it is closer to the other group center. A tied distance between an observation and two or more group centers is resolved through assigning the observation to its current group if it is one of the closest, and to the lowest numbered group otherwise.

The two graphs, Figure 1.3 and Figure 1.4 below show how k-means generated cutoff points split the distribution of *proportion of people within-occupation-state-time clusters that are licensed* into 2 or 3 distinct categories. They show visually the decision rule of k-means clustering algorithm.

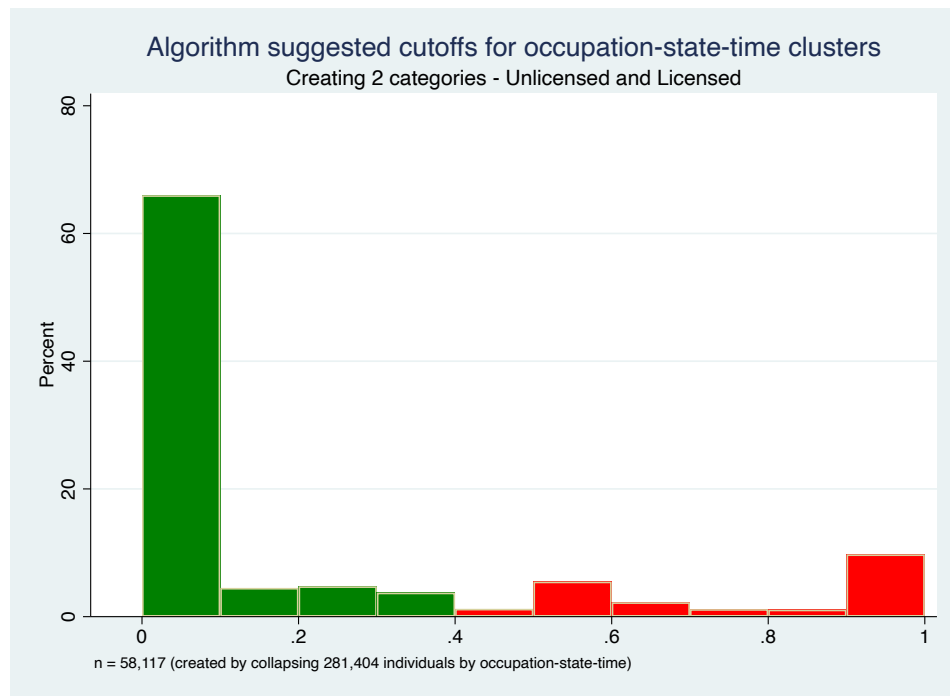
Figure 1.3: k-means algorithm with 3 clusters suggests that occupation-state-time clusters with average active government license response values below .219 are unlicensed, while occupation-state-time clusters with average active government license response values above .696 are licensed. The middle group is placed in a 'Not Sure' category, and we will drop these observations as we are not reasonably confident about their occupational license status due to noise.



Without this response correction, there is under-reporting of licensed status in licensed occupations, as discussed in the end of Section 1.3.3, and similarly over-reporting of unlicensed status in unlicensed states. In particular, looking at Figure 1.2, there might be systematic under-reporting of licensed status in licensed occupations over time, potentially due to three different reasons. First, there could be some recall bias towards not being licensed if people are not well informed about the licensing status of their own occupation, maybe because they took their exam long time ago and have poor memory. Second, systematic under-reporting could also be due to some people who are actually in training and thus not certified but report the occupation in survey anyway. Third, systematic under-reporting error could come from occupations that are broad enough to include individuals who must be certified and others who do not need certification (even though I am using 4 digit occupation coding scheme based which is quite detailed already).

Similarly, for occupations that we know are not licensed by states, there may be some sys-

Figure 1.4: k-means algorithm with 2 clusters suggests that occupation-state-time clusters with average active government license response values below .414 are unlicensed, while occupation-state-time clusters with average active government license response values above .416 are licensed.



tematic over-reporting: some people within these occupation-states-time clusters may misreport that they are licensed when in fact they are not.

There are two reasons for making this correction. First, this k-means response correction will reduce measurement error, though it will not completely eliminate the attenuation bias as k-means algorithm itself may make some classification mistakes. Calculating accuracy (correctly detecting true positives and false negatives), specificity/recall (correctly detecting true positive) and precision (correctly detecting false negative) exercises can be undertaken for a small subset of occupations in a given year using the National Occupational Licensing Database (which provides information on whether 48 occupations are licensed or not in a state). Since these occupations were chosen by analysts on a certain criteria - high labor growth occupations licensed in 30 or more states and requiring less than a 4 year college education, the sample seemed unrepresentative and thus I did not yet calculate the three tests which can potentially help approximate/bound accuracy of k-means algorithm. Beyond these three measures, I am not aware of a statistical test which informs us of

reduction in bias. Second, and the more important reason to use the k-mean response correction is to help us estimate an individual's last year occupation licensing status and thus test our preferred out-migration specification (see Section 1.3.4).

Changing Migration Variable Interpretation through Estimating Last Year Active Government License

Finally, after reducing the noise in active government license variable , I focus on the dependent migration variables.

Migrated variables in Annual Social and Economic Supplement (ASEC) indicates whether the respondent had changed residence in the past year. Non-movers, identified as individuals who were living in the same house as one year ago, were asked no more questions about migration over the past year. However, movers were asked where they lived last year in the US such as the city, county and state, or if they lived in a foreign country last year.

The category "Same house" includes both persons who did not move since the reference date (March 1 of the preceding year), and those who had moved but then returned to their earlier residence. On the other hand, movers are subdivided into the following 4 categories: those who had moved within the same county; those who had crossed county lines but stayed in the same state; those who those who had resided in a different state; and those who had migrated from abroad (Flood and Pacas 2018).

Appendix 1.8.1 provides more information on the original 280,404 individuals - Table 1.9 shows the summary statistics, Table 1.10 shows their current occupation frequency distribution and Table 1.11 shows their current state of residence frequency distribution.

Since our k-means corrected active government licensing variable tells us whether one is licensed or not by the state at the current moment, if I run a regression directly without any transformation, I would be answering is the following question - given that one is licensed by the state, did they move from another state/from abroad/from another county within state/from another house

within county, or not move at all.

However, as I mentioned in Section 1.3.2, I prefer to test the out-migration specification (given that a person is licensed by the state, do they move out of the state,...) over the in-migration specification (given a person is licensed by the state, did they move into the state...).

Since I can observe the individual's last year occupation and state of residence last year in Annual Social and Economic Supplement, I just need to estimate if these individuals were licensed last year or not to be able to test my preferred specification, and there are two ways to do this.

First, I could use the same decision rule (cutoffs proportions generated by k-means cluster algorithm) to classify all last year occupations-state-time as licensed or unlicensed too, and thus estimate the license status of individual's last year occupation in their last year state of residence. For example, if in 2018 a person was a nurse last year and lived in CA last year, they would be licensed if more than 41.6% nurses in 2017 in CA were licensed, otherwise they are not licensed. If we were generating three clusters, the person in 2018 was licensed last year if more than 69.6% nurses in CA last year were licensed, and unlicensed if fewer than 21.9% nurses were licensed. Thus, this is a simple rule to estimate last year occupation licensing status.

Second, I could use a 'slight variant of the first simple rule' and use the decision rule cutoffs generated by k-means cluster algorithm only for people whose last year occupation and last year state of residence was in the data. For example, if in 2018 I find person was a nurse last year and lived in CA last year, I will only attribute them to be licensed or unlicensed last year if there are nurses in 2017 in CA in my merged CPS ASEC data (as the k-means algorithm was trained on this data). However, if there are no nurses in 2017 in CA in my merged CPS ASEC data, I cannot ascertain the license status of this person and will drop these individuals. Thus, this second method would allow me to have more confidence in my estimate of a person being licensed last year, but I may lose some individuals if there are no people within their last year occupation-state-time cluster (which could happen with small sized occupations).

I will use this second approach as it is more appropriate, and gives me similar results as

the first simple approach. Thus, I transform the merged CPS ASEC data with corrected active government license variable for the 281,404 individuals one last time (this will be our master data going forward).

First, while every individual in our data has an occupation at time of survey, last year occupation is not known for some individuals. While there is no obvious pattern in the distribution of these ‘Not in Universe’ last year occupations across the years, occupations and other variables, almost 40% of these individuals are under 25, so presumably a lot of them were in school last year or not in the labor force last year. I drop these individuals whose last year occupations is not know (9,514 individuals).

Second, using the k-means generated cutoffs of 42% for 2 clusters and 22% and 70% for 3 clusters²¹, I create estimates for last year active government license status for all individuals in the master data if their last year occupation-state-time cluster exists in the master data. However, I cannot estimate last year license status for another 71,453 individuals, and thus drop them too.

A major chunk of these 71,453 individuals whose last year license status cannot be estimated happens by design - because we use ‘slight variant of the first simple rule’. Since my raw data starts from March 2016 onwards, I cannot estimate last year active government license for individuals in 2016 as there is no certification question data for March 2015 (jobcert question error in Current Population Survey). Thus, I loose 55,736 observations due to the design issue. Furthermore, I truly lose another 15,717 individuals over the other years because their last year occupation-state-time cluster does not actually exist in the master dataset, say because these are small sized occupations.

Thus, my last year active government license can only be estimated for 200,437 observations, down from 281,404 observations due to 9,514 missing last year occupation observations, and 71,453 individuals whose last year occupation-state-time cluster does not exist in my data (where all 55,736 individuals in 2016 are dropped).

I can however estimate last year active government license for people in March 2017 if there

21. An additional ‘not sure’ category.

are people in their last year occupation-state-time (March 2016) cluster²². Also, note that I label individual's current active government license as 'licensed (end of year)' and label individual's last year active government license as 'licensed (beginning of year)' in summary statistics tables.

1.3.5 Summary Statistics

Table 1.4 shows the descriptive statistics of what I call the 'transformed data' that I use for preferred out-migration specification, in contrast to the original/non-transformed data in Appendix 1.8.1 that can be used for in-migration estimation (which is likely to suffer from endogeneity).

Licensed (self-reported) binary variable takes on a value of 1 if the individual has an active government license. It was created from the three certification questions - profcert, statecert and jobcert, as shown in Table 1.3.

Table 1.3: Summary Statistics of Certification Questions (Transformed Data)

	Mean	Std. Dev.	Min	Max	Universe
Profcert: Has active prof-cert/license	0.254	0.44	0	1	200437
Statecert: Has govt issued prof-cert/license	0.913	0.28	0	1	50913
Jobcert: Cert required for job	0.845	0.36	0	1	50913
Licensed / Govt. issued prof-cert required for job	0.202	0.28	0	1	200437
Observations	200437				

Source: Current Population Survey (CPS), 2016-2020

Profcert Universe: 16-79 years old excluding people employed in Armed Forces.

Statecert Universe: Everyone who answered "yes" to Profcert. Those who answered "no" to Profcert are excluded.

Jobcert Universe: Everyone who answered "yes" to Profcert, excluding people "Not In the Labor Force" (NILF) and "Unemployed New Workers". Again, those who answered "no" to Profcert are excluded.

22. I do so by collapsing the master data by occupation-state-time cluster, and simply generating a new last year active government license variable that takes on a value of 1 if more than 42% people in the occupation-state-time cluster report they are licensed, and 0 otherwise (for the 2 cluster case). Similarly, I create a last year active government license variable with the 3 values (licensed, unlicensed and not sure). To merge these two variables back into the master dataset, I have to create a lagged time variable (last year) for all individuals in the master dataset, and rename the occupation-state-time clusters in the collapsed master data to last year occupation-state-time cluster. This allows me to do a m:1 match on the master data using the collapsed master data.

Licensed (self-reported) is added in the summary statistics to compare with our key parameter of interest - estimated active government license in last year, or equivalently active government license at beginning of year, as a sanity check. As already alluded, for the summary statistic table presentation, I will stick with the later interpretation/label for its relevance, and ease of contrast with non-transformed data. Also, note that licensed at beginning of year has fewer observations when it is constructed from k-means 3 clusters than 2 clusters as I drop the individuals in “Not Sure” cluster when calculating the summary statistics for the variable. The mean of k-means corrected active government license at the beginning of current year reduces slightly compared to the active government license variable, but the standard deviation is a bit higher. Estimated active government license at beginning of year will help us test for geographic mobility of licensed workers, which may be low (better wages and job prospectus/regulatory capture theory) or high (more educated/ambitious workers)

Second, from the licensed at beginning of year variable, we can construct a new variable measuring the ‘proportion of people licensed in an occupation-year’. This variable will be a proxy for licensing costs on average that people in an occupation-year will have to pay if they move, and thus help us test for whether occupational licensing restricts geographic migration or not. The reported statistic in Table 1.4 is constructed by taking the ratio of number of **licensed** individuals (k-means corrected with 2 clusters) in an occupation in a year divided by number of total individuals in an occupation in a year. Thus, it is no surprise that the mean of licensed at beginning of year (k-means corrected with 2 clusters) is the same as mean of proportion of people licensed in an occupation-year, and the standard deviation of the later is lower.

Third, while we had 5 migration dummies in the non-transformed data which had a “moved from” migration interpretation, in the transformed data with a “moved to” interpretation of migration variable we should not theoretically have a “Migrated: Moved to abroad/different nation” category (compare Table 1.4 with Table 1.9 in Appendix 1.8.1). This happens because only people

Table 1.4: Summary Statistics

	Mean	Std. Dev.	Min	Max	Count
Licensed (self-reported)	0.202	0.28	0	1	200437
Licensed at beginning of year (k-means corrected with 2 clusters)	0.188	0.39	0	1	200437
Licensed at beginning of year (k-means corrected with 3 clusters)	0.150	0.36	0	1	157862
Proportion of people licensed in an occupation-year	0.188	0.31	0	1	200437
Migrated: Moved to a different state	0.017	0.13	0	1	200437
Migrated: Moved to a different county within state	0.022	0.15	0	1	200437
Migrated: Moved to another house within county	0.065	0.25	0	1	200437
Migrated: No / Stayed in Same House (Non-Mover)	0.896	0.31	0	1	200437
Home: Owned (or being bought)	0.694	0.46	0	1	200437
Home: Rented	0.306	0.46	0	1	200437
Age	43.393	14.31	16	79	200437
Gender: Male	0.517	0.50	0	1	200437
Race: White	0.816	0.39	0	1	200437
Race: Black	0.095	0.29	0	1	200437
Race: Other	0.089	0.28	0	1	200437
Educ: Has some schooling	0.073	0.26	0	1	200437
Educ: High School completed or equivalent	0.260	0.44	0	1	200437
Educ: Some college, Associate or Bachelor Degree	0.524	0.50	0	1	200437
Educ: Has a Master Degree, Professional or Doctorate Degree	0.143	0.35	0	1	200437
Marst: Living with partner	0.552	0.50	0	1	200437
Marst: No partner: Single/never married	0.288	0.45	0	1	200437
Marst: Not living with partner	0.160	0.37	0	1	200437
# of own children in household	0.813	1.13	0	9	200437
Total Personal Income (in 1000s)	62.628	79.63	-16.5	2085.7	200437
State of Residence at beginning of year			2	52	200437
Occupation at beginning of year			1	443	200437
Time Period			2016	2019	200437
Observations	200437				

Source: Current Population Survey (CPS) and Annual Social & Economic Supplemental (ASEC), 2016-2020

within the US are surveyed - asked from where did move from since last year/where they were living last year, and thus we know if a person lived abroad last year. However, if a person moves abroad, they are automatically dropped from the Current Population Survey, and thus move to abroad variable will have null values only for this migration category and no individual control data. As expected, the 4 migration “move to” dummy variables add exactly to 1 in Table 1.4. Also, it is interesting to note that almost 90% individuals did not move houses, while almost 2% moved to another state.

Fourth, we have 8 categories of individual level controls such as home ownership status, age, gender, race, education, marital status, number of own children residing in the household (of any

age or marital status, irrespective of biological, adopted, or step-children), and total pre-tax personal income (from all sources for the last calendar year at the end of the year). These are observable at the end of year, which is not a major issue as most of these controls are largely time invariant (like race) or change at a relatively slow rate (like personal income). There is evidence though that home ownership and marital status may be important variables to influence migration decision for movers, so I will not include them in my preferred specification too as since they are measured at next year and could be affecting migration variable. My results are robust to including/excluding them though.

Fifth, occupations reports the person's primary occupation at the beginning of the year (one year ago). Respondents who held more than one job were to report the job at which they worked the largest number of hours. For persons who were employed at the time of the survey, occupation relates to the job worked during the preceding week; unemployed persons and those not currently in the labor force were to give their most recent occupation.

Occupation at beginning of year has 443 categories/occupations - one more than occupations at end of year, which is expected as the occupations at beginning of year are not restricted to occupations that have 10 or more people in an occupation at the end of the year (Section 1.3.4).

Furthermore, the unique number of categories of occupations at beginning of year/preceding year (shown in Table 1.4) does not differ much from unique categories of occupations at end of year/current year (shown in Table 1.9 in Appendix 1.8.1). Table 1.5 reports the frequency of occupations at the beginning of the year, and can be contrasted with Table 1.10 in Appendix 1.10 that shows frequency of occupations at end of year. Both these frequency tables also present the variation in occupational licensing across states in a randomly chosen year (2018). As expected, the number of observations within the unique occupation category at beginning of year/last year is lower than 10, unlike for occupations at end of year/current year which were restricted to have greater than 10 observations while setting up the k-means algorithm.

Sixth, states identifies the household's state of residence at the beginning of the year using

Table 1.5: Occupations Frequency Table (at beginning of year).

Proportion Licensed column shows proportion of people in the occupation that self-reported they require a license to practise their profession at the end of year.

Percent of licensed states in 2018 shows the share of total states that licensed the occupation in 2018 as per k-means algorithm, while total states in 2018 shows the count of all the states in which individuals practicing an occupation live at beginning of 2018.

As there are fewer people in an occupation, there may not be individuals in an occupation in a certain year, like ‘Rail-Track Laying and Maintenance Equipment Operators’.

Occupation	Freq- uency	Percent of total obs - 200,437	Proportion Licensed (self re- ported)	Percent of States Licensed (k-means corrected) in 2018	Total States in 2018
Managers, nec (including Postmasters)	7336	3.660	.145	0	51
First-Line Supervisors of Sales Workers	6452	3.219	.0743	0	51
Elementary and Middle School Teachers	5172	2.580	.712	94.1	51
Driver/Sales Workers and Truck Drivers	5123	2.556	.297	19.6	51
Registered Nurses	4913	2.451	.832	100	51
Retail Salespersons	4319	2.155	.029	0	51
Cashiers	4197	2.094	.0139	0	51
Secretaries and Administrative Assistants	3964	1.978	.0399	0	51
Chefs and Cooks	3537	1.765	.0535	0	51
Customer Service Representatives	3401	1.697	.0288	0	51
Janitors and Building Cleaners	3260	1.626	.0298	0	51
Waiters and Waitresses	2775	1.384	.0338	0	51
Laborers & Freight, Stock, & Material Movers, Hand	2744	1.369	.0285	0	51
Nursing, Psychiatric, and Home Health Aides	2721	1.358	.469	58.8	51
Construction Laborers	2634	1.314	.0654	1.96	51
Accountants and Auditors	2523	1.259	.221	7.84	51
Chief executives and legislators/public administra- tion	2405	1.200	.165	11.8	51
Computer Scientists and Systems Ana- lysts/Network systems Analysts/Web Developers	2328	1.161	.0608	0	49
...	...	...	...	...	
...	...	...	...	...	
...	...	...	...	...	
Paving, Surfacing, and Tamping Equipment Opera- tors	1	0.000	0	0	1
Rail-Track Laying & Maintenance Equipment Op- erators	1	0.000	0		.

Source: Annual Social & Economic Supplemental (ASEC), 2016-2020

the Federal Information Processing Standards (FIPS) coding scheme.

Similarly, state of residence at beginning of year has 51 unique categories/states (50 states plus D.C.) - it begins at 2 and ends at 52 because it has a placeholder category 1 for “Abroad”, and there are no individuals in our transformed data set that were living abroad at the beginning of year. This again is a sanity check since we do not have certification data on those who were living abroad last year as they were not part of Current Population Survey, and thus k-means algorithm cannot estimate their last year licensing status using our ‘slight variant of the first simple rule’ (see Section 1.3.4). Thus, there are no people who moved from abroad in our out-migration specification.

The unique number of state of residence at the beginning of the year (Table 1.4) does not differ from unique number state of residence at the end the year. Table 1.6 reports the frequency of state of residence at beginning of year for the 10 largest and smallest state categories, and can be compared to largest and smallest occupation categories at end of year (Table 1.11 in Appendix 1.11).

1.4 Specification

Recall that we have a pooled cross-sectional data for individuals from 2016 to 2019. There may be some people who are in the sample twice due to CPS 4-8-4 rotation pattern.

My first hypothesis is that if a person works in a state licensed occupation, they are unlikely to migrate to another state over the next one year because of higher opportunity costs than unlicensed workers, partially due to higher current wages²³ and better career prospectus in licensed job (compared to unlicensed jobs in terms of tenure, employment status and other job advantages)

23. Just like occupational licensing literature suggests, I too find that k-means algorithm suggested licensed workers have higher incomes, thereby mitigating concerns about correctly identifying licensed individuals.

Table 1.6: State of Residence (at beginning of year) Frequency Table

State	Frequency	Total	Percent
California	18237	200437	9.099
Texas	11923	200437	5.949
Florida	8710	200437	4.346
New York	8378	200437	4.180
Illinois	5780	200437	2.884
Pennsylvania	5780	200437	2.884
Ohio	5377	200437	2.683
Massachusetts	4816	200437	2.403
Michigan	4527	200437	2.259
North Carolina	4490	200437	2.240
...	...	...	...
...	...	...	...
...	...	...	...
Iowa	2545	200437	1.270
Kansas	2536	200437	1.265
Nevada	2419	200437	1.207
South Dakota	2398	200437	1.196
Alaska	2247	200437	1.121
Kentucky	2193	200437	1.094
Connecticut	2002	200437	0.999
Delaware	1948	200437	0.972
Rhode Island	1655	200437	0.826
Maine	1574	200437	0.785

Source: Annual Social & Economic Supplemental (ASEC),
2016-2020

(Nunn 2018). I call this ‘**cushy wage**’ hypothesis.

My second hypothesis is that if an occupation is more likely licensed across all states in the country, individuals will not migrate to other states over the next one year from time t due to facing additional re-licensing costs (like monetary, psychic or time). I call this ‘**re-licensing cost**’ hypothesis.

However, state occupational licensing should not affect individual migration within the state (to another county in same state, to another house in the same county, or not migrating at all/staying in the same house). Thus, the parameter coefficients for both these two hypothesis should be zero for within-state migration.

The model specification we have is Ordinary Least Squares (OLS) with a host of individual level controls, along with state, occupation, year, year-occupation and year-state fixed effects.

The estimating equations is the following, where i will index the individuals, t will index time, o indexes occupation and s indexes states -

$$\begin{aligned} Migration_{it} = & \beta Licensed_{ost} + \delta Proportion Licensed_{ot} + \\ & \gamma Controls_{it+1} + Occupation_o + State_s + Time_t \\ & + Occupation_o * Time_t + State_s * Time_t + \epsilon_{it} \end{aligned} \quad (1.1)$$

$Migration_{it}$ is a dummy variable taking a value of 1 if a person moves to another state (location more generally) over the next one year from beginning of time period t .

$Licensed_{ost}$ is a dummy variable taking a value of 1 if a person is licensed by state s to practice occupation o at beginning of time period t , and measures the ‘cushy wages’ hypothesis. It was created from three certification questions and its misreporting has been reduced by applying the 2-mean cluster algorithm to the noisy self-reported measure.

$Proportion Licensed_{ot}$ measures percentage of people within an occupation that are licensed at time t (proxy for probability of re-licensing required for an occupation), and measures ‘occup-

tional re-licensing costs' hypothesis.

$Controls_{it+1}$ include 5 categories of individual level control variables such as age, gender, race, education, and number of children under 5 years old in the household at end of time period t (or beginning of time period $t+1$). I have data on home ownership, marital status, and personal income at end of time period t too, which are reported in summary statistics Table 1.4, but I do not use them in my preferred out-migration specification as individuals tend to move due to first two factors and personal income may be a function of move, leading to over-controlling. Results stay robust to if I add these three controls though.

$State_s$ is state fixed effect term and controls for the (origination) state of residence at beginning of time period t . They are important because some states could have fundamentally different migration levels, say because of labor market conditions like unemployment or GDP, their political composition or even licensing propensity. Thus, I control for time invariant features of states.

$Occupation_o$ is the occupation fixed effect term and controls for individual's occupations at beginning of time period t . They are important because occupations can vary in terms of migration. For example, house builders are more likely to move across state borders for construction projects than say a massage therapist who may have a local clientele. We control for observable time invariant differences across occupations, and do not want a few occupations driving our results.

$Time_t$ is year fixed effect and controls for time period t . These are important because there may be waves of more or less migration going on over time, maybe due to political think tanks trying to lobby for policy changes before upcoming elections, structural changes in jobs availability, etcetra. These affect every occupation and every state equally.

1.5 Results

Table 1.7 shows the key results.

Column 1 tests the hypothesis that given that a person is licensed in a state-occupation cluster

at beginning of time period t , they would not move out to another state over the next one year as they would lose their ‘cushy’ licensed job benefits. Columns 2-4 serve as falsification tests, and the outcome variable changes to migration to another county within state, migration to another house within county, and to no migration (non-movers). Migration patterns within the state should not be adversely affected as individuals do not lose their state-license as long as they stay in the same occupations²⁴.

In column 1, we can see that if a person is currently working in a (state) licensed occupation, they are more likely to move to another state over the next one year by .11 percentage point. Since only about 1.7 percent people migrate to another states every year on average, individuals in state licensed occupations seems to be migrating out-of-state by about 6.5% more than non-licensed individuals on average, *ceteris paribus*. While this seems to be moderate in magnitude, this is not a statistically significant result even at 10% level of significance. I do not find any evidence that licensed individuals are moving less than across state borders due to losing their ‘cushy’ jobs.

Additionally, individuals in a licensed occupation in all states are less likely to move to another state by .41 percentage points compared to individuals in a unlicensed occupation in all states in a given year, *ceteris paribus*. Again, this is not statistically significant, and I do not find evidence for ‘occupational re-licensing costs’ reducing inter-state migration.

The coefficients on other parameters in column 1 are as expected though. One is less likely to migrate to another state as they get older (the positive quadratic age co-efficient is much smaller in magnitude than the negative linear age co-efficient), but once people are “old” enough they are more likely to migrate out of their licensed occupation-state. This suggests that maybe as people build local reputation or clientele, they are less likely to move out until they decide to retire. Also, people with more education are likely to move to another state than compared to people with only some schooling, and magnitudes are higher for people with more formal education, suggesting that people with more more ambition/skills are more likely to make inter-state moves. Furthermore, the

24. Robustness check where I limit to individuals whose last year and current year occupation was the same yields similar results.

Table 1.7: Affect of Occupational Licensing on Migration, 2016-2020

	(1) Moved to another state	(2) Moved to another county within state	(3) Moved to another house within county	(4) Did not move
Licensed (k-means corrected with 2 clusters)	0.0011 (0.0012)	0.0008 (0.0014)	-0.0013 (0.0023)	-0.0005 (0.0028)
Proportion Licensed (prob. of re-licensing costs)	-0.0041 (0.0068)	-0.0053 (0.0079)	0.0055 (0.0127)	0.0039 (0.0158)
Age	-0.0013*** (0.0001)	-0.0017*** (0.0002)	-0.0057*** (0.0003)	0.0086*** (0.0003)
Age Squared	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)	-0.0001*** (0.0000)
Gender: Male	-0.0010 (0.0007)	-0.0010 (0.0008)	-0.0019 (0.0014)	0.0039** (0.0017)
Race: White	-0.0017 (0.0013)	-0.0011 (0.0013)	-0.0031 (0.0022)	0.0058** (0.0027)
Race: Black	0.0014 (0.0016)	0.0009 (0.0017)	0.0113*** (0.0029)	-0.0136*** (0.0036)
Educ: High School completed or equivalent	0.0056*** (0.0011)	0.0046*** (0.0014)	0.0164*** (0.0024)	-0.0266*** (0.0029)
Educ: Some college, Associate or Bachelor Degree	0.0085*** (0.0011)	0.0076*** (0.0014)	0.0134*** (0.0024)	-0.0294*** (0.0029)
Educ: Masters, Professional or Doctorate Degree	0.0132*** (0.0015)	0.0071*** (0.0017)	0.0168*** (0.0029)	-0.0371*** (0.0036)
# of own children in household	-0.0037*** (0.0003)	-0.0027*** (0.0003)	-0.0054*** (0.0005)	0.0118*** (0.0006)
Regression	OLS	OLS	OLS	OLS
Controls	Yes (5)	Yes (5)	Yes (5)	Yes (5)
Time FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes	Yes
r ²	.014739	.0115908	.0316533	.0464488
N	200,429	200,429	200,429	200,429

Dependent variable (migration) is a dummy. Constant term is not reported.

* $p < .10$, ** $p < .05$, *** $p < .01$

more the number of own children in household, the less likely is one to leave the state, presumably because reallocation costs will be higher as family size increases.

In Appendix 1.8.2, Table 1.16 shows how the key parameters (licensed and proportion licensed) change as I add the 5 control variables one by one, and Table 1.17 shows how the key parameters change as I add fixed effects one by one. In the later table, I also add occupation-time interaction fixed effects term to controls for trends in occupations over the time period t^{25} and state-time interaction fixed effects term to controls for trends in state over time period t^{26} . I usually avoid adding state-occupation interaction terms in my baseline specification as we have corrected our active government license measure using k-means algorithm so that there is no variation in individual responses within occupation-state-time clusters, and after including the three plain vanilla fixed effects, adjusted R-square does not increase much²⁷ as I add the interaction fixed effects but the interacted fixed effects often results in one of our key parameters becoming infeasible to estimate.

Also, from Table 1.17 in Appendix 1.8.2, the first three columns show proportion licensed variable stays significant when we add time/year fixed effects alone, or state fixed effects alone, but the parameter loses significance as soon as we add occupation fixed effects. This suggests that there seems to be much more variation across occupations than states, and least variation across time. One potential reason there is positive and statistically significant coefficient on proportion licensed without occupation fixed effects, but not with occupation fixed effects, is that licensing is more appealing in occupations in which mobility is already low. Alternatively, states seem to be more consistent in their licensing laws, but there is quite a lot of variation in licensing by occupa-

25. Occupation-time fixed effects may be important because some occupations like truck driving may have been lucrative in 2016, but not so much in 2019 as they may have got saturated (excessive competition). These people may be migrating to other states in search of opportunities. If this co-incides with introduction of occupational laws, it will bias our active government licensing results downward.

26. State-time fixed effects may be important because states like New York or California may have been a great place to live in 2016, but a not so great in 2019, say due to cost of living or medical infrastructure being less available in these states. People may not have wanted to move out in 2016, but may want to move out in 2019. If some occupations in New York or California coincided with introduction of introduction of occupational licensing laws, it will bias our active government licensing variable downwards.

27. In column 7, R-square is 1.47% while adjusted R-square is 1.23%, and in column 9 R-square is 2.24% while adjusted R-square is 1.37%

tions that gets absorbed by the occupation fixed effects.

In column 2 (of Table 1.7), we can see that if a person is currently working in a (state) licensed occupation, they are more likely to move to another county within the state over the next one year by .08 percentage point. Since about 2.2 percent people migrate to another county within state every year on average, individuals in state licensed occupations seems to be migrating out-of-state by about 3.64% more than non-licensed individuals on average, *ceteris paribus*. This is neither large in magnitude nor statistically significant at 10% level of significance, and coefficients on controls variables follow the same pattern like in column 1 and are about similar in magnitudes too. I would expect the opportunity cost of licensed individuals is lower for within-state moves, but again find they are not moving significantly more between counties within the state. Additionally, individuals in a licensed occupation in all states are less likely to move to another state by .53 percentage points compared to individuals in a unlicensed occupation in all states in a given year, *ceteris paribus*. Again, this is not statistically significant, and I do not find evidence for ‘occupational re-licensing costs’ reducing intra-state county migration, as expected.

In column 3, we can see that if a person is currently working in a (state) licensed occupation, they are less likely to move to another house within county over the next one year by .13 percentage point. Since about 6.5 percent people migrate to another county within state every year on average, individuals in state licensed occupations seems to be migrating out-of-state by about 0.02% less than non-licensed individuals on average, *ceteris paribus*. Again, this is neither large in magnitude and nor statistically significant at 10% level of significance, and coefficients on controls variables follow the same pattern like in column 1 and are about similar in magnitudes too. Furthermore, since opportunity cost of licensed individuals is lower for within-state moves, I would expect them to move more within county in a state but do not find any statistical significant result. Additionally, individuals in a licensed occupation in all states are less likely to move to another state by .55 percentage points compared to individuals in a unlicensed occupation in all states in a given year, *ceteris paribus*. Again, this is not statistically significant, and I do not find evidence for

‘occupational re-licensing costs’ reducing intra-county house migration.

Finally, in column 4, we see that if a person is currently licensed, they are more likely to stay in the same house (within state) over the next one year by .05 percentage point i.e. not migrate. Since about 89.6 percent people stay in the same house on average, individuals in state licensed occupations seems to be staying in the same house by about 0.005% more than non-licensed individuals on average, *ceteris paribus*. Again, the magnitude of this estimate is very small, and the result is not statistically significant, suggesting some evidence for ‘cushy wage hypothesis’. Additionally, individuals in a licensed occupation in all states are more likely to not move to another place .39 percentage points compared to individuals in a unlicensed occupation in all states in a given year, *ceteris paribus*. Again, this is not statistically significant, and I do not find evidence for ‘occupational re-licensing costs’ reducing inter-state migration.

The signs on the certain control variables should flip, as the outcome variable is measuring if the person did not move now. This makes the story we have painted with the first three columns a bit more consistent, by construction. For example, as a person gets older or has more number of children in household, they are more likely to stay in the same house. Furthermore, if a person is more educated, they are less likely to stay in their own home. Also, it is interesting to note that individuals belonging to black race seem to be less likely to stay in the same household (as compared to all other races), and more likely to move to another house within the county in a year.

1.5.1 Robustness Check: k-means algorithm identifies licensed individuals

To test whether k-means algorithm correctly identifies licensed occupation-state time or not at a given moment of time, I can test for whether licensed individuals (as per k-means algorithm) have higher incomes or not, like occupational licensing literature argues and usually finds due to regulatory capture. Finding licensed individuals have higher wages would add confidence that my results are not contaminated by inability of the k-means algorithm to correctly identify licensed

individuals, and I will use the non-transformed/original data (281,404 observations in Table 1.9, Appendix 1.8.1) for this.

The estimating equations is the following, where i will index the individuals, t' will index time (end of year), o' indexes occupation (at end of year) and s' indexes state (at end of year) -

$$\begin{aligned} \text{Personal Income}_{it'} = & \beta \text{Licensed}_{o's't'} + \gamma \text{Controls}_{it'} + \text{Occupation}_{o'} + \text{State}_{s'} + \text{Time}_{t'} \\ & + \text{Occupation}_{o'} * \text{Time}_{t'} + \text{State}_{s'} * \text{Time}_{t'} + \epsilon_{it'} \end{aligned} \quad (1.2)$$

*Personal Income*_{it'} is the individual's total pre-tax personal income (or losses) from all sources for the last one calendar year at time t' . *Licensed*_{o's't'} measures if a person is licensed or not in occupation o' in state s' at time t' , constructed from the three certification questions and corrected for mis-reporting using k-means algorithm. *Controls*_{it'} are defined at the same time period (end of year) when income is measured, unlike in estimating Equation 1.1, and additionally include home ownership and marital status on top of age, gender, race, education attainment and number of own children in household. *Time*_{t'} controls for year fixed effects. *Occupation*_{o'} and *State*_{s'} are fixed effects controlling for individual's current occupation and state of residence at end of year t' , and *Occupation*_{o'} * *State*_{s'} and *State*_{s'} * *Time*_{t'} are interaction fixed effect terms.

Table 1.12 in Appendix Section 1.8.1 shows the regression results. Licensed individuals make about \$2,000 dollars more than unlicensed individuals as per k-means 2-cluster algorithm, and about \$4,000 dollars more than unlicensed individuals as per k-means 3-cluster algorithm. Coefficients on other parameters also make sense, like income increases by approximately \$1,000 dollars every year with age/experience/human capital, males makes about \$20,000 more females, white make about \$3,000 more while blacks make about \$1,000 less than other races, and individuals with advanced degrees (masters, profession or doctorate) make about \$40,000 dollars more than those with some high school degree, ceteris paribus.

Similarly, if I replace the dependent variable in equation 1.2 with employment status of workers, I find some preliminary evidence that licensed individuals are half to one percentage point less likely to be unemployed than their unlicensed counterparts (Table 1.13 in Appendix Section

1.8.1).

$$\begin{aligned}
 Unemployed\ Worker_{it'} = & \beta Licensed_{o's't'} + \gamma Controls_{it'} + Occupation_{o'} + State_{s'} + Time_{t'} \\
 & + Occupation_{o'} * Time_{t'} + State_{s'} * Time_{t'} + \epsilon_{it'}
 \end{aligned}
 \tag{1.3}$$

where $Unemployed\ Worker_{it'}$ takes on a value of 1 if the individual is unemployed, and $Controls_{it'}$ includes all variables in equation 1.3 and total personal income.

Furthermore, if I replace the dependent variable in equation 1.2 with log of total number of people within an occupation-state-time cluster and collapse my individuals level data by occupation-state-time cluster, I find some preliminary evidence that if an occupation-state-time cluster is licensed, there would be 5 to 14% fewer workers in them (Table 1.14 in Appendix Section 1.8.1).

$$\begin{aligned}
 LogN_{o's't'} = & \beta Licensed_{o's't'} + \gamma Controls_{o's't'} + Occupation_{o'} + State_{s'} + Time_{t'} \\
 & + Occupation_{o'} * Time_{t'} + State_{s'} * Time_{t'} + \epsilon_{it'}
 \end{aligned}
 \tag{1.4}$$

This suggests that regulatory capture effect that reduces the number of workers in licensed occupation dominates the adverse selection effect that increases the number of workers in licensed occupations.

1.5.2 Robustness Check: Alternative Specifications

The results are robust to alternative specifications. First, even when I use the corrected active government license variable with three clusters instead of two, I get similar results (see Table 1.20 for summary statistics and Table 1.21 for regression results in Appendix 1.8.3). Furthermore, the results are robust to using different filters on training data to determine the k-means algorithm decision rule, such as dropping occupations with less than 20 people or 50 people instead of occupations with 10 people only.

Second, adding housing, marital status and total personal income controls in the k-means 2 or

3 cluster approach does not not change the results either (over controlling). Table 1.18 in Appendix Section 1.8.1 shows the results for k-means 2 cluster.

Third, if I relax the definition of occupational licensing to the state license definition i.e. “yes” to statecert question only, and not “yes” to both statecert and jobcert question (which is what I used/stronger definition), the results do not change as the active government license variable stays almost the the same (results not attached). Similarly, if I relax the definition of occupational licensing from someone who answered “yes” to all the three certification questions (primary job is licensed) to someone who just answered “yes” to “profcert” question (holds an active license of certification) only, my results do not change.

Fourth, if I change the sample of individuals from 16 to 79 in my data to a working age sample of 25 to 54, it does not affect my results (results not attached).

Fifth, if I restrict myself to continuing workers only, defined as those whose last year occupation was the same as current year occupation (as opposed to new workers), I lose about 10% of observations but my results do not change (Table 1.19 in Section 1.8.1). This is an important result. as it allows us to rule out a particular reason to describe the why we get the results. Consider a toy model - assume that there are two types of workers - licensed and unlicensed who initially have similar wages and number of employed people. As licensing gains momentum (like it has in the US), labor supply of workers in licensed occupations shifts to the left (regulatory capture effects) and thus wages of licensed individuals will increase while the number of people employed in licensed jobs decreases (I find evidence for both these effects). Thus, while there are fewer people in licensed occupations, some of the individuals who lost their jobs due to regulatory capture could move to different states to pursue the same occupation, and this effect could potentially explain why I do not find evidence for re-licensing costs hypothesis. however, by focussing only on people who stay in the same occupation over the year, we can rule out this possible explanation to be driving my empirical results.

Sixth, weighting the regression by number of people in the occupation-state-time cluster also

does not change my results either (results not attached).

Finally, I change the preferred out-migration specification (equation 1.1) to an in-migration specification and test for the following hypothesis - given that an individual is currently self-reported licensed by the state in an occupation at end of year t , did the individual move into the state-occupation over the last one year? I find some evidence that licensed individuals are less likely to have migrated from another state or abroad (Table 1.15 in Section 1.8.1) suggesting there are licensing costs too, but the in-migration specification is likely to be endogenous given self-selection of individuals into unlicensed occupations while migrating to US state.

1.5.3 Making sense of the results

Table 1.8: Reason for Moving (asked from movers only)

Reason for moving	Frequency	Total	Percent	Proportion Licensed
Housing Related Reason	8347	20840	40.1	0.1675
Family Reason	6002	20840	28.8	0.1671
Job Related Reason	4542	20840	21.8	0.1808
Other Reasons	1949	20840	9.4	0.1380

Source: Annual Social & Economic Supplemental (ASEC), 2016-2020.
Non-movers are not asked why they moved.

The results can be partially explained by examining individual's reasons for moving since movers are asked why they moved. In our dataset of 200,437 individuals, about 89.6% people did not move, suggesting moving costs may be high, and thus only the remaining roughly 10.4% of $200,437 \approx 20,840$ people were asked this question.

Table 1.8 shows that about 40% people migrate for housing related reasons. This includes people who wanted to own home (not rent), wanted new or better housing, wanted better neighborhood, and moved for cheaper housing amongst other housing reasons. The high magnitude on the home ownership (as opposed to renting home) as well as statistical significance at 1% level of significance seems to suggest that people do not move if they own a home, and seem to be moving

to another state to own a home.

About 30% people move due to family related reasons, which include change in marital status, to establish their own household, and pursuing a relationship with unmarried partner amongst other family reason.

Only 20% people move for job related reasons, which includes a new job or job transfer, to look for work or lost job, for easier commute, and retirement amongst other job-related reason. Thus, it appears that migrating for work related reasons is not the highest priority for many individuals.

The remaining 9% people moved due to other reasons such as attending or leaving college (2.5%), health related reason (.9%), Change of Climate and Natural Disaster (.8%), and foreclosure/evictions.

Thus, it seems like people do not usually move for job related reasons, while other factors like owning a home or starting a family are more important reasons to move. Furthermore, from the geographic mobility literature, we know that moving is costly because of search costs, transitions/direct moving costs, and possible upheaval or psychic costs. In the presence of these costs, maybe occupational licensing is not the most important component.

1.6 Conclusion

1.6.1 Method

In summary, I use three recently added non-degree credential questions in Current Population Survey from 2016 onwards²⁸ to estimate whether a person is licensed by the state government²⁹ to

28. These three questions were - do you hold an active professional certificate or license (profcert), was the active professional certification or license issued by local, state or federal government (statecert), and whether this active professional certificate or license was required for your primary job (jobcert).

29. Since Federal Trade Communication Technical Report in 2018 affirms that “occupational licensing is almost always state-based”, if a person answers “yes” to the three certification questions we can infer they are likely licensed by the state government (Goldman 2018).

work in their primary occupation or not. The individual-level active government license dummy estimate based on responses to the three certification questions seems to suffer from misreporting³⁰, and I correct the misreporting by standardizing the responses of people within each occupation-state-time cluster using k-means unsupervised machine learning algorithm.

In particular, k-means clustering algorithm with 2 categories suggests that all individuals within occupation-state-time cluster where more than 42% people self-report holding a license should be licensed, while everyone else should be unlicensed. Alternatively, if one was to create 3 categories, the algorithm suggests that all individuals within occupation-state-time cluster where more than 69% people self-report to be licensed are likely licensed, while all individuals within occupation-state-time clusters where less than 21% people self-report to be licensed are likely unlicensed - and we cannot be sure of everyone else. Second, k-means algorithm allows me to estimate an individual's last year occupational licensing status, and thereby test my preferred out-migration specification instead of the default in-migration specification, based on data collected in CPS and ASEC.

1.6.2 Results

Like both theoretical and empirical literature suggests, I too find that occupational licensing creates a wage gap/inequality between licensed and unlicensed workers - thereby adding more credibility to my k-means algorithm decision rule that is correctly classifying occupation-states as licensed and unlicensed at any given moment of time.

I find that between state migration patterns of individuals are not affected if an individual currently works in a (state) licensed occupations. However, I do find that within state migration patterns of individuals are not affected if an individual currently works in a (state) licensed occupations, as one would expect. Thus, (state) occupational licensing does not seem to adversely affect geographic mobility for licensed professionals, and cannot help explain the stylized US labor

30. Within an occupation-state-time cluster, every individual should give the same uniform replies - either licensed or unlicensed. In particular, if some people report they are licensed while others report they are not suggests misreporting.

facts.

Second, I do not find much evidence for occupational relicensing costs too, just like I did not find support for cushy wages hypothesis. This could be simply because migration decision is very complex and there are major factors like home ownership or family reasons that trump job related reasons to migrate. We saw that only 20% movers moved for job related reasons, which is relatively lower compared to other factors like housing (40%) or family related reasons (30%).

My preferred out-migration specification results are quite robust to a wide range of specifications including choice of demographic controls or sub-sample analysis based on age, continuing/new workers, construction of number of clusters by k-means algorithm (2 or 3), choice of occupations-state to determine the k-means cutoff rule, weighting schemes, et cetra.

1.6.3 Future Work/Potential Extentions

While I find that licensed individuals are no different than their unlicensed counterparts on average in terms of their migration choices, and occupational licensing costs do not seem to be restricting inter-state geographic mobility of American workers, occupational licensing could be adversely impacting other margins of outcome like wage inequality, potential employment and employment status (for which I provide preliminary evidence). Additionally, occupational licensing could also be affecting quality/safety of products and/or product prices that could be future study topics.

Also, the study can be improved by controlling for interstate occupation reciprocity agreements that have gained popularity in a few professions like nursing that I do not currently account for.

Furthermore, I intend to improve Kliener and Johnson's (2020) causal estimates of occupational licensing on inter-state migration by using the k-means suggested licensed occupations in Current Population Survey (CPS) to create a mapping of occupations in American Community Survey (ACS) data so that their re-licensing costs can be more precisely estimated. Currently, their

licensing costs are within my 95% confidence interval, so my work does not necessarily contradict their study.

1.6.4 Contribution

My study looks at the impact of occupational licensing on geographic mobility of workers in pretty much the entire occupation set in US, unlike most other occupational licensing studies that focus on a single profession only and only try to see how licensing affects geographic mobility in a certain occupation. Simultaneous licensing of multiple occupations is closer to a general equilibrium effect whereas single occupation licensing effects is only a partial equilibrium effect. Furthermore, since almost one in four Americans are licensed today, this is an important question, especially if one believes in labor mobility is required for smooth functioning of the economy.

As far as I know, this is the first study that exploits the three certification questions in Current Population Survey to study the impact of occupational licensing on geographic mobility of workers. Unlike Kliner and Johnson (2010), who also look at the same question - causal impact of (state) occupational licensing on inter-state mobility, I do not assume away time and geographic variation in occupational licensing or focus on only a small subset of occupations. In particular, they did not exploit the certification questions in CPS and rely more on ACS data.

Unlike Nunn (2016) who did use the Current Population Survey certification questions to study migration behavior of licensed individuals in 2016, my results are not driven by unobservable differences between states and occupations as I account for these using fixed effects (not accounting for these explains his results), and I also use a stronger set of time varying controls. Furthermore, I even correct for misreporting in individual licensing status that can potentially drive geographic mobility results through the k-means unsupervised machine learning algorithm.

The results have significance for public policy makers trying to understand worker effect of labor certification process on worker decisions about geographic mobility. As occupational licensing gets more and more popular within political circles over time and space, understanding

its impact on workers can help politicians make more informed policy decisions. Also, since this study suggests that licensed workers seems to get positively affected on their wage and employment margin, policy makers have to weigh the freedom granted to licensed occupations to move between regions versus the negative externalities on non-licensed professionals in terms of wages and employment outcomes. Even if policymakers intend on simply increasing product quality or safety in product markets, they should consider evidence from this paper on worker outcomes, including in terms of mobility.

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1.8 Appendix

1.8.1 Summary Statistics of Non-Transformed Data

Table 1.9: Summary Statistics

	Mean	Std. Dev.	Min	Max	Count
Licensed (self-reported)	0.196	0.40	0	1	281404
Licensed at end of year (k-means corrected with 3 clusters)	0.121	0.33	0	1	224718
Licensed at end of year (k-means corrected with 2 clusters)	0.184	0.39	0	1	281404
Migrated: Moved from a different state	0.018	0.13	0	1	281404
Migrated: Moved from abroad / different nation	0.003	0.05	0	1	281404
Migrated: Moved from a different county within state	0.023	0.15	0	1	281404
Migrated: Moved within county in the same state	0.067	0.25	0	1	281404
Migrated: No / Stayed in Same House (Non-Mover)	0.889	0.31	0	1	281404
Home: Owned (or being bought)	0.688	0.46	0	1	281404
Home: Rented	0.312	0.46	0	1	281404
Age	43.076	14.41	16	79	281404
Gender: Male	0.521	0.50	0	1	281404
Gender: Female	0.479	0.50	0	1	281404
Race: White	0.815	0.39	0	1	281404
Race: Black	0.097	0.30	0	1	281404
Race: Other	0.088	0.28	0	1	281404
Educ: Has some schooling	0.079	0.27	0	1	281404
Educ: High School completed or equivalent	0.264	0.44	0	1	281404
Educ: Some college, Associate or Bachelor Degree	0.520	0.50	0	1	281404
Educ: Has a Master Degree, Professional or Doctorate Degree	0.138	0.34	0	1	281404
Marst: Living with partner	0.544	0.50	0	1	281404
Marst: No partner: Single/never married	0.296	0.46	0	1	281404
Marst: Not living with partner	0.161	0.37	0	1	281404
# of own children in household	0.807	1.13	0	9	281404
Total Personal Income (in 1000s)	59.149	76.50	-16.5	2085.7	281404
State of Residence at end of year			1	51	281404
Occupation at end of year (ASEC)			1	442	281404
Survey year			2016	2020	281404
Observations	281404				

Source: Current Population Survey (CPS) and Annual Social & Economic Supplemental (ASEC), 2016-2020
 Licensed binary variable takes on a value of 1 if the individual requires a govt. issued professional certificate for their primary job, and 0 otherwise.

Note that almost 89% individuals did not move, while slightly less than 2% moved from a different state. Even fewer (.3%) moved to US from abroad, and note that if somebody was to leave the US, I would not be able to observe where they migrated to (as they would not be included in Current Population Survey anymore).

Occupations reports the person's primary current occupation. If the individual was employed during the survey week, their occupation for the last week was recorded, while if he/she were not in the labor force or employed, their last occupation was recorded. Furthermore, if the individual held multiple jobs, the job where they worked the longest hours was recorded. **Table 1.10 below** reports the frequency of occupations for the 23 largest and smallest occupation categories.

States identifies the household's current state of residence using the Federal Information Processing Standards (FIPS) coding scheme. Table 1.11 below reports the frequency table of state of current residence.

Table 1.10: Occupations Frequency Table (at end of year).

Proportion Licensed column shows proportion of people in the occupation that self-reported they require a license to practise their profession.

Percent of licensed states in 2018 shows the share of total states that licensed the occupation in 2018, while number of states in 2018 shows the count of all the states in which individuals practising an occupation live.

Occupation	Freq- uency	Percent of total obs - 281,404	Proportion Li- censed (self re- ported)	Percent of States Licensed (k-mean2 corrected) in 2018	Total States in 2018
Managers, nec (including Postmasters)	8998	3.198	.142	0	51
First-Line Supervisors of Sales Workers	8163	2.901	.0757	0	51
Driver/Sales Workers and Truck Drivers	6541	2.324	.292	11.8	51
Elementary and Middle School Teachers	6530	2.321	.71	98	51
Registered Nurses	6212	2.208	.825	100	51
Cashiers	5772	2.051	.0125	0	51
Retail Salespersons	5607	1.993	.0294	0	51
Secretaries and Administrative Assistants	5071	1.802	.039	0	51
Chefs and Cooks	4710	1.674	.0512	0	51
Janitors and Building Cleaners	4361	1.550	.0268	0	51
Customer Service Representatives	4329	1.538	.0275	0	51
Laborers and Freight, Stock, & Material Movers, Hand	3616	1.285	.0271	0	50
Nursing, Psychiatric, and Home Health Aides	3574	1.270	.456	60.8	51
Waiters and Waitresses	3478	1.236	.0348	0	51
Construction Laborers	3368	1.197	.0591	0	51
Accountants and Auditors	3255	1.157	.22	7.84	51
Chief executives and legislators/public administra- tion	3025	1.075	.16	9.8	51
Computer Scientists and Systems Analysts	2949	1.048	.0627	1.96	51
Stock Clerks and Order Fillers	2709	0.963	.01	0	51
Software Developers, Applications & Systems Soft- ware	2703	0.961	.0281	2.04	49
Postsecondary Teachers	2655	0.943	.246	15.7	51
Maids and Housekeeping Cleaners	2643	0.939	.0114	0	51
First-Line Supervisors of Office & Admin Support Workers	2588	0.920	.0781	1.96	51
...	...	...	...	...	
...	...	...	...	...	
...	...	...	...	...	
Tool Grinders, Filers, and Sharpeners	10	0.004	.1	0	4
Average	641	.2278	.17850	18.52	27.59

Source: Annual Social & Economic Supplemental (ASEC), 2016-2020

Table 1.11: Current (end of year) State of Residence Frequency Table

State	Frequency	Total	Percent
California	24022	281404	8.536
Texas	15837	281404	5.628
Florida	11712	281404	4.162
New York	11182	281404	3.974
Illinois	8039	281404	2.857
Pennsylvania	7837	281404	2.785
Ohio	7367	281404	2.618
Massachusetts	6458	281404	2.295
North Carolina	6348	281404	2.256
Michigan	6259	281404	2.224
Georgia	6193	281404	2.201
Louisiana	6157	281404	2.188
Montana	5537	281404	1.968
New Jersey	5421	281404	1.926
Washington	5331	281404	1.894
West Virginia	5268	281404	1.872
Tennessee	5267	281404	1.872
District of Columbia	5257	281404	1.868
Alabama	5255	281404	1.867
Virginia	5217	281404	1.854
Indiana	4809	281404	1.709
Arkansas	4804	281404	1.707
Utah	4626	281404	1.644
Mississippi	4569	281404	1.624
Idaho	4516	281404	1.605
North Dakota	4452	281404	1.582
New Mexico	4382	281404	1.557
Oregon	4367	281404	1.552
Arizona	4352	281404	1.547
South Carolina	4274	281404	1.519
New Hampshire	4236	281404	1.505
Oklahoma	4205	281404	1.494
Wyoming	4197	281404	1.491
Wisconsin	4195	281404	1.491
Minnesota	4115	281404	1.462
Missouri	3987	281404	1.417
Hawaii	3964	281404	1.409
Vermont	3946	281404	1.402
Nebraska	3927	281404	1.396
Maryland	3866	281404	1.374
Colorado	3747	281404	1.332
Kansas	3692	281404	1.312
Iowa	3579	281404	1.272
Nevada	3542	281404	1.259
South Dakota	3454	281404	1.227
Alaska	3392	281404	1.205
Kentucky	3208	281404	1.140
Delaware	3056	281404	1.086
Connecticut	3049	281404	1.083
Rhode Island	2554	281404	0.908
Maine	2378	281404	0.845

Source: Annual Social & Economic Supplemental (ASEC), 2016-2020

Wages Regression: k-means identified licensed individuals have higher incomes

Table 1.12: Affect of Occupational Licensing on Personal Income, 2016-2020

	(1) Total Income (in 1000s of \$)	(2) Total Income (in 1000s of \$)	(3) Total Income (in 1000s of \$)
Licensed (k-means corrected with 2 clusters)	2.0531*** (0.5371)		
Licensed (k-means corrected with 3 clusters)		4.0999*** (0.9941)	
Licensed (self-reported)			5.5436*** (0.3925)
Age	1.4230*** (0.0612)	1.4286*** (0.0674)	1.4024*** (0.0612)
Age Squared	-0.0096*** (0.0007)	-0.0097*** (0.0007)	-0.0094*** (0.0007)
Gender: Male	18.9556*** (0.3081)	18.4024*** (0.3381)	18.8598*** (0.3081)
Race: White	3.1871*** (0.4814)	2.9239*** (0.5299)	3.0440*** (0.4814)
Race: Black	-1.0071 (0.6316)	-1.2228* (0.7029)	-1.0839* (0.6314)
Educ: High School completed or equivalent	5.5674*** (0.5305)	5.3032*** (0.5666)	5.4773*** (0.5304)
Educ: Some college, Associate or Bachelor Degree	14.5545*** (0.5300)	14.3116*** (0.5674)	14.3000*** (0.5301)
Educ: Has a Master Degree, Professional or Doctorate Degree	39.5273*** (0.6757)	39.3122*** (0.7477)	38.8812*** (0.6771)
Marst: Never married/single	-5.4201*** (0.3821)	-5.5949*** (0.4250)	-5.2928*** (0.3821)
Marst: Separated	-2.1338*** (0.3733)	-2.3715*** (0.4177)	-2.1442*** (0.3732)
# of own children in household	1.5621*** (0.1288)	1.6081*** (0.1443)	1.5571*** (0.1287)
Home: Owned (or being bought)	5.8405*** (0.2979)	5.7878*** (0.3291)	5.7523*** (0.2979)
Regression	OLS	OLS	OLS
Controls	Yes 7	Yes 7	Yes 7
Time FE	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes
Interacted Time-State FE	Yes	Yes	Yes
Interacted Time-OCC FE	Yes	Yes	Yes
r ²	.2526355	.2646967	.2531305
N	281,382	224,695	281,382

Dependent variable personal income (in 1000s of dollars). Constant term is not reported.

* $p < .10$, ** $p < .05$, *** $p < .01$

Employment Status Regression: k-means identified licensed individuals likely to be employed

Table 1.13: Affect of Occupational Licensing on Employment Status, 2016-2020

	(1) Unemployed Worker	(2) Unemployed Worker	(3) Unemployed Worker
Licensed (k-means corrected with 2 clusters)	-0.0054*** (0.0015)		
Licensed (k-means corrected with 3 clusters)		-0.0112*** (0.0029)	
Licensed (self-reported)			-0.0111*** (0.0011)
Age	-0.0009*** (0.0002)	-0.0009*** (0.0002)	-0.0008*** (0.0002)
Age Squared	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)
Gender: Male	0.0016* (0.0009)	0.0013 (0.0010)	0.0018** (0.0009)
Race: White	-0.0080*** (0.0014)	-0.0085*** (0.0016)	-0.0078*** (0.0014)
Race: Black	0.0175*** (0.0018)	0.0214*** (0.0021)	0.0177*** (0.0018)
Educ: High School completed or equivalent	-0.0107*** (0.0015)	-0.0109*** (0.0017)	-0.0105*** (0.0015)
Educ: Some college, Associate or Bachelor Degree	-0.0151*** (0.0015)	-0.0149*** (0.0017)	-0.0146*** (0.0015)
Educ: Has a Master Degree, Professional or Doctorate Degree	-0.0125*** (0.0020)	-0.0121*** (0.0022)	-0.0114*** (0.0020)
Marst: Never married/single	0.0194*** (0.0011)	0.0199*** (0.0013)	0.0191*** (0.0011)
Marst: Separated	0.0146*** (0.0011)	0.0144*** (0.0012)	0.0146*** (0.0011)
# of own children in household	0.0015*** (0.0004)	0.0018*** (0.0004)	0.0016*** (0.0004)
Home: Owned (or being bought)	-0.0089*** (0.0009)	-0.0095*** (0.0010)	-0.0087*** (0.0009)
Total Personal Income (in 1000s)	-0.0001*** (0.0000)	-0.0001*** (0.0000)	-0.0001*** (0.0000)
Regression	OLS	OLS	OLS
Controls	Yes 8	Yes 8	Yes 8
Time FE	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes
Interacted Time-State FE	Yes	Yes	Yes
Interacted Time-OCC FE	Yes	Yes	Yes
r ²	.0431333	.0470275	.0434213
N	281,382	224,695	281,382

Dependent variable is a dummy measuring if a person is unemployed (experienced worker), as opposed to employed/at work or has job (but not at work last week).

Constant term is not reported.

* $p < .10$, ** $p < .05$, *** $p < .01$

Log number of people in state licensed occupation is lower than in unlicensed occupations, suggesting regulatory capture effect dominates adverse selection effect

Table 1.14: Affect of Occupational Licensing on Number of People within an Occupation-State-Time cluster, 2016-2020

	(1) Log of # of people in State- Occupation-Time cluster	(2) Log of # of people in State- Occupation-Time cluster	(3) Log of # of people in State- Occupation-Time cluster
Licensed occ-state-time (k-means corrected with 2 clusters)	-0.0548*** (0.0076)		
Licensed occ-state-time (k-means corrected with 3 clusters)		-0.1466*** (0.0105)	
Licensed individuals in occ-state-time cluster (self-reported)			-0.0252*** (0.0097)
Average age in an occ-state-time cluster	-0.0003 (0.0003)	-0.0003 (0.0003)	-0.0003 (0.0003)
Proprtion of Male in occ-state-time cluster	0.0029 (0.0080)	-0.0022 (0.0084)	0.0025 (0.0080)
Proportion of White in occ-state-time cluster	-0.0425*** (0.0127)	-0.0401*** (0.0133)	-0.0435*** (0.0127)
Proportion of Black in occ-state-time cluster	-0.0536*** (0.0166)	-0.0539*** (0.0174)	-0.0538*** (0.0166)
Proprtion with high school equi. educ in occ-state-time cluster	-0.0206 (0.0142)	-0.0234 (0.0146)	-0.0213 (0.0142)
Proprtion with some high school educ in occ-state-time cluster	-0.0431*** (0.0141)	-0.0457*** (0.0146)	-0.0456*** (0.0141)
Proprtion with advanced degrees an occ-state-time cluster	0.0026 (0.0179)	0.0018 (0.0187)	-0.0013 (0.0179)
Proprtion of single inds. in occ-state-time cluster	0.0100 (0.0095)	0.0057 (0.0099)	0.0107 (0.0095)
Proprtion of separated inds. in left in occ-state-time cluster	0.0026 (0.0094)	-0.0005 (0.0099)	0.0025 (0.0094)
Average number of children in an occ-state-time cluster	0.0010 (0.0031)	-0.0002 (0.0033)	0.0008 (0.0031)
Proportion of inds owning their home	-0.0097 (0.0077)	-0.0125 (0.0080)	-0.0100 (0.0077)
Average personal income an occ-state-time cluster	0.0002*** (0.0001)	0.0002*** (0.0001)	0.0002*** (0.0001)
Regression	OLS	OLS	OLS
Controls	Yes 8	Yes 8	Yes 8
Time FE	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes
Interacted Time-State FE	Yes	Yes	Yes
Interacted Time-OCC FE	Yes	Yes	Yes
r2	.702799	.713909	.7025603
N	58,093	48,875	58,093

Dependent variable (migration) is a dummy. Constant term is not reported.

* $p < .10$, ** $p < .05$, *** $p < .01$

In-Migration Specification likely to be endogenous due to self-selection

Table 1.15: Affect of Occupational Licensing on Migration, 2016-2020

	(1) Migrated: Moved from a different state	(2) Migrated: Moved from abroad / different nation	(3) Migrated: from a different county within state	(4) Migrated: within county same state	(5) Migrated: No / Stayed in Same House (Non- Mover)
Govt. issued prof cert required for job / Licensed	-.001294* (.0007574)	-.001193*** (.002713)	.0006325 (.000883)	.0008053 (.001423)	.001049 (.001759)
Home: Owned (or being bought)	-.02982* (.0007532)	-.005779*** (.0003106)	-.02976*** (.0008264)	-.0904*** (.00134)	.1558*** (.001651)
Age	-.001579*** (.0001271)	-.002331*** (.00004985)	-.001769*** (.0001425)	-.005628*** (.0002339)	.00921*** (.0002852)
Age Squared	.00001194*** (1.285e-06)	1.566e-06*** (4.935e-07)	.00001382*** (1.447e-06)	.00004193*** (2.371e-06)	-.00006925*** (2.902e-06)
Gender: Male	7.279e-06 (.0006706)	.001067*** (.0002795)	-.0002538 (.0007359)	-.0007212 (.001223)	-.00009887 (.001499)
Race: White	-.001397 (.001123)	-.006304*** (.0006768)	.00185 (.001135)	.0004544 (.001994)	.005396** (.002453)
Race: Black	-.005225*** (.001471)	-.006293*** (.0008013)	-.003045** (.001535)	-.00433 (.002651)	.01889*** (.003256)
Educ: High School completed or equivalent	.005676 (.0009691)	.000414 (.0005206)	.006497*** (.001228)	.01552 (.002162)	-.02811*** (.002575)
Educ: Some college, Associate or Bachelor Degree	.01001*** (.0009976)	.001134*** (.0005175)	.009983*** (.001244)	.0145*** (.002142)	-.03563*** (.002573)
Educ: Has a Master Degree, Professional or Doctorate Degree	.01608*** (.001412)	.003688*** (.0007042)	.009802*** (.001551)	.01795*** (.002639)	-.04752*** (.003242)
Marst: Never married/single	-.001894* (.0008647)	-.001154*** (.0003591)	.00126 (.0009362)	.0003166 (.00157)	.001471 (.001921)
Marst: Separated	.00228*** (.0007475)	.0005254* (.000304)	.005648*** (.0008645)	.01649*** (.001415)	-.02495*** (.001744)
# of own children under 5 in household	-.003436*** (.000651)	-.0007213*** (.0002796)	-.0009418 (.0007549)	.004514*** (.001326)	.0005857 (.001581)
Total Personal Income (in 1000s)	-3.656e-06 (4.128e-06)	-8.112e-06*** (1.486e-06)	7.704e-06* (4.119e-06)	.0000103* (6.143e-06)	-6.241e-06 (8.075e-06)
Regression	OLS	OLS	OLS	OLS	OLS
Controls	Yes (8)	Yes (8)	Yes (8)	Yes (8)	Yes (8)
Time FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes	Yes	Yes
Interacted Occupation-Time FE	Yes	Yes	Yes	Yes	Yes
Interacted Occupation-State FE	Yes	Yes	Yes	Yes	Yes
Interacted State-Time FE	Yes	Yes	Yes	Yes	Yes
r2	.08924	.06965	.07971	.1129	.1515
N	278500	278500	278500	278500	278500

Dependent variable (migration) is a dummy.
* $p < .10$, ** $p < .05$, *** $p < .01$

1.8.2 Migration to Another State (Transformed Data)

Table 1.16: Affect of Occupational Licensing on Migration, 2016-2019

	(1)	(2)	(3)	(4)	(5)	(6)
Licensed (k-means corrected with 2 clusters)	0.0005 (0.0012)	0.0007 (0.0012)	0.0007 (0.0012)	0.0007 (0.0012)	0.0006 (0.0012)	0.0007 (0.0012)
Proportion Licensed (prob. of re-licensing costs)	-0.0017 (0.0015)	0.0000 (0.0015)	0.0001 (0.0015)	0.0001 (0.0015)	-0.0043*** (0.0016)	-0.0039** (0.0016)
Age		-0.0016*** (0.0001)	-0.0016*** (0.0001)	-0.0016*** (0.0001)	-0.0018*** (0.0001)	-0.0012*** (0.0001)
Age Squared		0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)
Gender: Male			0.0001 (0.0006)	0.0002 (0.0006)	0.0005 (0.0006)	0.0005 (0.0006)
Race: White				-0.0027** (0.0011)	-0.0019* (0.0011)	-0.0020* (0.0011)
Race: Black				-0.0005 (0.0015)	0.0008 (0.0015)	0.0006 (0.0015)
Educ: High School completed or equivalent					0.0069*** (0.0010)	0.0056*** (0.0010)
Educ: Some college, Associate or Bachelor Degree					0.0114*** (0.0010)	0.0101*** (0.0010)
Educ: Masters, Professional or Doctorate Degree					0.0198*** (0.0013)	0.0186*** (0.0013)
# of own children in household						-0.0037*** (0.0003)
Regression	OLS	OLS	OLS	OLS	OLS	OLS
Controls	No	Yes (1)	Yes (2)	Yes (3)	Yes (4)	Yes (5)
Time FE	No	No	No	No	No	No
State FE	No	No	No	No	No	No
Occupation FE	No	No	No	No	No	No
r ²	9.38e-06 200,437	.0054968 200,437	.005497 200,437	.005497 200,437	.0068227 200,437	.0077132 200,437
N						

Dependant variable is binary migration variable measuring if a person moved to another state over the next one year.

* $p < .10$, ** $p < .05$, *** $p < .01$

Table 1.17: Affect of Occupational Licensing on Migration, 2016-2019

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Licensed (k-means corrected with 2 clusters)	0.0007 (0.0012)	0.0011 (0.0012)	0.0007 (0.0012)	0.0008 (0.0012)	0.0007 (0.0012)	0.0011 (0.0012)	0.0011 (0.0012)	0.0008 (0.0012)	0.0008 (0.0012)
Proportion Licensed (prob. of re-licensing costs)	-0.0039** (0.0016)	-0.0040** (0.0016)	-0.0036 (0.0068)	-0.0037** (0.0016)	0.0000 (.)	-0.0037 (0.0068)	-0.0041 (0.0068)	-0.0034 (0.0068)	0.0000 (.)
Age	-0.0012*** (0.0001)	-0.0011*** (0.0001)	-0.0013*** (0.0001)	-0.0011*** (0.0001)	-0.0013*** (0.0001)	-0.0013*** (0.0001)	-0.0013*** (0.0001)	-0.0013*** (0.0001)	-0.0013*** (0.0001)
Age Squared	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)
Gender: Male	0.0005 (0.0006)	0.0004 (0.0006)	-0.0009 (0.0007)	0.0004 (0.0006)	-0.0009 (0.0007)	-0.0010 (0.0007)	-0.0010 (0.0007)	-0.0010 (0.0007)	-0.0011 (0.0007)
Race: White	-0.0020* (0.0011)	-0.0020 (0.0012)	-0.0017 (0.0011)	-0.0019 (0.0012)	-0.0017 (0.0011)	-0.0017 (0.0013)	-0.0017 (0.0013)	-0.0016 (0.0012)	-0.0017 (0.0013)
Race: Black	0.0006 (0.0015)	0.0001 (0.0016)	0.0018 (0.0015)	0.0001 (0.0016)	0.0017 (0.0015)	0.0014 (0.0016)	0.0014 (0.0016)	0.0015 (0.0016)	0.0014 (0.0016)
Educ: High School completed or equivalent	0.0056*** (0.0010)	0.0060*** (0.0010)	0.0052*** (0.0010)	0.0060*** (0.0010)	0.0053*** (0.0011)	0.0056*** (0.0011)	0.0056*** (0.0011)	0.0056*** (0.0011)	0.0057*** (0.0011)
Educ: Some college, Associate or Bachelor Degree	0.0101*** (0.0010)	0.0101*** (0.0010)	0.0083*** (0.0011)	0.0101*** (0.0010)	0.0084*** (0.0011)	0.0085*** (0.0011)	0.0085*** (0.0011)	0.0085*** (0.0011)	0.0085*** (0.0011)
Educ: Masters, Professional or Doctorate Degree	0.0186*** (0.0013)	0.0181*** (0.0013)	0.0134*** (0.0015)	0.0181*** (0.0013)	0.0136*** (0.0015)	0.0132*** (0.0015)	0.0132*** (0.0015)	0.0131*** (0.0015)	0.0132*** (0.0015)
# of own children in household	-0.0037*** (0.0003)	-0.0038*** (0.0003)	-0.0036*** (0.0003)	-0.0038*** (0.0003)	-0.0036*** (0.0003)	-0.0037*** (0.0003)	-0.0037*** (0.0003)	-0.0037*** (0.0003)	-0.0037*** (0.0003)
Regression Controls	OLS Yes (5)	OLS Yes (5)	OLS Yes (5)	OLS Yes (5)	OLS Yes (5)	OLS Yes (5)	OLS Yes (5)	OLS Yes (5)	OLS Yes (5)
Time FE	Yes	No	No	Yes	Yes	No	Yes	Yes	Yes
State FE	No	Yes	No	Yes	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	Yes	No	Yes	Yes	Yes	Yes	Yes
Interacted State-Time FE	No	No	No	Yes	No	No	No	Yes	Yes
Interacted Occupation-Time FE	No	No	No	No	Yes	No	No	No	Yes
r2	.0077257	.0114298	.0110022	.0135405	.0166205	.0147039	.0147205	.0168103	.0224478
N	200,437	200,437	200,429	200,437	200,357	200,429	200,429	200,429	200,357

Dependant variable is binary migration variable measuring if a person moved to another state over the next one year.

* $p < .10$, ** $p < .05$, *** $p < .01$

Out-Migration Specification with Full Set of Controls does not change results

Table 1.18: Affect of Occupational Licensing on Migration, 2016-2020

	(1) Moved to another state	(2) Moved to another county within state	(3) Moved to another house within county	(4) Did not move
Licensed (k-means corrected with 2 clusters)	0.0012 (0.0012)	0.0009 (0.0014)	-0.0010 (0.0022)	-0.0011 (0.0027)
Proportion Licensed (prob. of re-licensing costs)	-0.0038 (0.0068)	-0.0050 (0.0079)	0.0062 (0.0126)	0.0027 (0.0155)
Home: Owned (or being bought)	-0.0283*** (0.0009)	-0.0286*** (0.0009)	-0.0879*** (0.0015)	0.1448*** (0.0019)
Marst: Never married/single	-0.0035*** (0.0010)	-0.0011 (0.0011)	-0.0022 (0.0018)	0.0068*** (0.0022)
Marst: Separated	0.0021** (0.0008)	0.0040*** (0.0010)	0.0150*** (0.0016)	-0.0210*** (0.0019)
Total Personal Income (in 1000s)	-0.0000 (0.0000)	0.0000** (0.0000)	0.0000** (0.0000)	-0.0000*** (0.0000)
Age	-0.0013*** (0.0001)	-0.0016*** (0.0002)	-0.0055*** (0.0003)	0.0084*** (0.0003)
Age Squared	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)	-0.0001*** (0.0000)
Gender: Male	-0.0005 (0.0007)	-0.0007 (0.0008)	-0.0003 (0.0014)	0.0015 (0.0017)
Race: White	-0.0004 (0.0012)	0.0001 (0.0013)	0.0008 (0.0022)	-0.0005 (0.0026)
Race: Black	-0.0025 (0.0016)	-0.0035** (0.0017)	-0.0026 (0.0029)	0.0086** (0.0036)
Educ: High School completed or equivalent	0.0065*** (0.0011)	0.0055*** (0.0014)	0.0192*** (0.0024)	-0.0311*** (0.0028)
Educ: Some college, Associate or Bachelor Degree	0.0101*** (0.0011)	0.0090*** (0.0014)	0.0182*** (0.0024)	-0.0373*** (0.0028)
Educ: Masters, Professional or Doctorate Degree	0.0153*** (0.0015)	0.0089*** (0.0017)	0.0232*** (0.0029)	-0.0474*** (0.0035)
# of own children in household	-0.0032*** (0.0003)	-0.0019*** (0.0003)	-0.0026*** (0.0005)	0.0078*** (0.0006)
Regression	OLS	OLS	OLS	OLS
Controls	Yes (8)	Yes (8)	Yes (8)	Yes (8)
Time FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes	Yes
Interacted State-Time FE	No	No	No	No
Interacted Occupation-Time FE	No	No	No	No
r ²	.0234229	.0187814	.056046	.0889917
N	200,429	200,429	200,429	200,429

Dependent variable (migration) is a dummy.

* $p < .10$, ** $p < .05$, *** $p < .01$

Out-Migration Specification limited to continuing workers does not change results

Table 1.19: Affect of Occupational Licensing on Migration, 2016-2020

	(1) Moved to another state	(2) Moved to another county within state	(3) Moved to another house within county	(4) Did not move
Licensed (k-means corrected with 2 clusters)	0.0013 (0.0012)	0.0007 (0.0014)	-0.0014 (0.0024)	-0.0007 (0.0029)
Proportion Licensed (prob. of re-licensing costs)	-0.0022 (0.0067)	-0.0118 (0.0081)	0.0106 (0.0135)	0.0033 (0.0165)
Age	-0.0010*** (0.0001)	-0.0014*** (0.0002)	-0.0057*** (0.0003)	0.0082*** (0.0003)
Age Squared	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)	-0.0000*** (0.0000)
Gender: Male	-0.0009 (0.0007)	-0.0005 (0.0008)	-0.0011 (0.0015)	0.0025 (0.0018)
Race: White	-0.0013 (0.0012)	-0.0016 (0.0013)	-0.0034 (0.0024)	0.0063** (0.0028)
Race: Black	0.0009 (0.0016)	-0.0026 (0.0017)	0.0131*** (0.0032)	-0.0114*** (0.0038)
Educ: High School completed or equivalent	0.0050*** (0.0010)	0.0046*** (0.0014)	0.0161*** (0.0026)	-0.0258*** (0.0031)
Educ: Some college, Associate or Bachelor Degree	0.0068*** (0.0010)	0.0072*** (0.0014)	0.0116*** (0.0026)	-0.0256*** (0.0031)
Educ: Masters, Professional or Doctorate Degree	0.0102*** (0.0014)	0.0069*** (0.0017)	0.0156*** (0.0031)	-0.0327*** (0.0038)
# of own children in household	-0.0033*** (0.0002)	-0.0026*** (0.0003)	-0.0058*** (0.0005)	0.0117*** (0.0006)
Regression	OLS	OLS	OLS	OLS
Controls	Yes (5)	Yes (5)	Yes (5)	Yes (5)
Time FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes	Yes
r2	.0134225	.0107423	.0322574	.0446523
N	175,659	175,659	175,659	175,659

Dependent variable (migration) is a dummy.

* $p < .10$, ** $p < .05$, *** $p < .01$

1.8.3 k-means with 3 clusters

Table 1.20: Summary Statistics: kmean 3 cluster drops “Not Sure category”.

	Mean	Std. Dev.	Min	Max	Universe
Licensed (Self-Reported)	0.158	0.30	0	1	157862
Licensed at beginning of year (k-means corrected with 3 clusters)	0.150	0.36	0	1	157862
Proportion of people licensed in an occupation (3 kmean)	0.150	0.32	0	1	157862
Migrated: Moved to different state	0.018	0.13	0	1	157862
Migrated: Moved to abroad / different nation	0.000	0.00	0	0	157862
Migrated: Moved to different county within state	0.022	0.15	0	1	157862
Migrated: Moved to different house within county	0.066	0.25	0	1	157862
Migrated: No / Stayed in Same House (Non-Mover)	0.895	0.31	0	1	157862
Home: Owned (or being bought)	0.688	0.46	0	1	157862
Home: Rented	0.312	0.46	0	1	157862
Age	43.085	14.41	16	79	157862
Gender: Male	0.515	0.50	0	1	157862
Gender: Female	0.485	0.50	0	1	157862
Race: White	0.817	0.39	0	1	157862
Race: Black	0.092	0.29	0	1	157862
Race: Other	0.091	0.29	0	1	157862
Educ: Has some schooling	0.079	0.27	0	1	157862
Educ: High School completed or equivalent	0.265	0.44	0	1	157862
Educ: Some college, Associate or Bachelor Degree	0.523	0.50	0	1	157862
Educ: Has a Master Degree, Professional or Doctorate Degree	0.134	0.34	0	1	157862
Marst: Married	0.543	0.50	0	1	157862
Marst: Never married/single	0.298	0.46	0	1	157862
Marst: Separated	0.159	0.37	0	1	157862
# of own children in household	0.802	1.12	0	9	157862
Total Personal Income (in 1000s)	61.996	79.92	-13	2085.7	157862
State of Residence at beginning of year			2	52	157862
Occupation at beginning of year			1	443	157862
Beginning of Year (March, 20XX)			2016	2019	157862
Observations	157862				

Source: Current Population Survey (CPS) and Annual Social & Economic Supplemental (ASEC), 2016-2020

Licensed (self-reported) binary variable takes on a value of 1 if the individual requires a govt. issued professional certificate for their primary job, and 0 otherwise. It was created from the three certification questions.

Table 1.21: Affect of Occupational Licensing on Migration, 2016-2020

	(1) Moved to another state	(2) Moved to another county within state	(3) Moved to another house within county	(4) Did not move
Licensed (k-means corrected with 3 clusters)	0.0033 (0.0023)	-0.0018 (0.0024)	-0.0055 (0.0039)	0.0040 (0.0049)
Proportion Licensed (prob. of re-licensing costs)	0.0073 (0.0092)	0.0008 (0.0104)	0.0085 (0.0167)	-0.0166 (0.0208)
Regression	OLS	OLS	OLS	OLS
Controls	Yes (5)	Yes (5)	Yes (5)	Yes (5)
Time FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Occupation FE	Yes	Yes	Yes	Yes
Interacted State-Time FE	No	No	No	No
Interacted Occupation-Time FE	No	No	No	No
r2	.0151968	.011955	.0315694	.0458892
N	157,854	157,854	157,854	157,854

Dependent variable (migration) is a dummy.

* $p < .10$, ** $p < .05$, *** $p < .01$

Observations are fewer than 200,437 due to “Not Sure” cluster values being dropped.

Individual level controls such as age, age squared, gender, race, education and number of children are not reported. Constant term is not reported.

Chapter 2

Pricing and Competition in the Electric Vehicle Charging Infrastructure¹

2.1 Introduction and Motivation

Electric vehicles (EVs) are poised to play a major role in the decarbonization of the transportation sector, and the growing market for EVs will have to be matched by an adequate EV charging infrastructure. Even though EVs can be charged at home, the lack of public charging infrastructure would limit EV adoption to only home dwellers with garages, and would perpetuate range anxieties among EV owners. The US's 17% share in the global stock of 10.2 million EVs (in 2020) is way behind that of China's and Europe's, which stand at 44% and 31% respectively. It is no surprise that the total number of public EV charging stations (EVCS) in the US, is not even a third of the total number of gasoline stations in the US (estimated to be around 150,000). While EV adoption has grown steadily in the US over the last couple of years, with the total number of registered EVs tripling between 2016 and 2020 to 1.8 million cars, new EV sales still only accounts for around 2% of new car sales in the US during this time (Desilver 2021). California has the highest share of EVs

1. Joint work with Dr. Nehan Naim, Assistant Professor of Energy Economics, Regulations, and Finance, Department of Sustainable Resources Management, SUNY ESF. e-mail: nenaim@esf.edu

in the US, thanks to its mandates for EV production and other policies that incentivize buying EVs; over 30% of public EVCS in the US are located in California. Since pricing is a major determinant of the economic feasibility and optimal quantity of the EV charging services, this paper will study the effect of competition among different charging technologies and networks interoperability on the pricing decisions of individual EV charging stations in the US.

As the demand for EVs and the demand for EV charging stations go hand in hand, there are interesting indirect network effects on both sides of the market. Li. et al. (2017) estimate that a 10% growth in EV stock would lead to a 6% increase in charging station deployment, while a 10% increase in the number of public charging stations would increase EV sales by 8%. The paper further elaborates on the significance of public policies incentivizing the deployment of EVCS, as it can be twice as effective in promoting EV adoption as the federal income tax credit of up to \$7500 for EV buyers. Public EV charging infrastructure in the US has so far been developed through a ‘patchwork of grant funding, settlement funds, private investment, and electric company pilots and programs’ (EPRI 2019). Most of these stations are owned and/or operated by ‘charging networks’ like ChargePoint, Blink Charging, ElectrifyAmerica etc. and are characterized by different business models, which result in highly varied pricing structures (often free as well) across different charging technologies (AC, DC) and across different consumer bases (subscribers, non-subscribers).².

Public EVCS optimal number in a location will be a function of station level profit or ‘the increase in profits that free/cheap chargers bring to the host businesses’ or if it is bundled with other service such as parking. While EVCS’ costs depend on technology used (level 2, level 3/fast charging), location of the station, and electricity charges, their profits will be very sensitive to price and quantity/customer base, with quantity being more important because the high number of customers can be monetized through advertisements etc., even in the absence of pricing/free

2. While many level 2/AC chargers are free across the country, there is a lot of variation in pricing at different ‘networks’, particularly for DC fast charging. Unlike ElectrifyAmerica and EVgo who own/operate their stations and decide their own prices, Chargepoint does not own the stations in their network, but rather supports property owners and leaves the pricing formula up to them. Chargepoint supports pricing per kWh. EVgo prices at 30c/minute for their 50KW stations and gives discount/benefits to monthly subscription payers. Electrify America also charges per minute but a different price for different EV models. Tesla also has a preference for pricing by kWh. Source: <https://electrek.co/2019/08/12/kwh-pricing-ev-drivers-miss-benefits/>

service.

Recently, some EVCS networks have signed interoperability contracts, or ‘roaming’ agreements, which ‘allow an EV user to use various charging networks under different operators by taking membership with just one operator’ (IHS Markit 2019). This is not equivalent to bypassing the physical charger plug standards³, but primarily for bringing together backend communication, payments, and information sharing. Members of networks engaging in ‘roaming’ agreements will be conveniently able to locate, access, and pay for charging services at stations of other networks, all through a single app. By entering in roaming agreements (which reduce administrative burden on the customer), networks are improving EV user experience and gaining customer loyalty. Having access to larger customer base results in greater utilization of their EVCS infrastructure, which offers opportunity for enjoying quicker and greater returns of investment. We can say two things about the impact of roaming on pricing decisions: greater customer utilization can reduce the need to rely on competitive pricing to stay profitable; pricing will be heterogeneous even within roaming members based on location/rents, electricity prices, and technology offered at the station.

While existing literature has focused on simulating optimal pricing strategies for electricity pricing at EV charging stations or focused on EVCS competition in specific geographic regions of US, there hasn’t been an empirical estimation of pricing, competition, and effects of roaming for EVCS in the entire country. Using the Alternative Fuel Data, this paper aims to estimate competition among EV charging stations, across different technologies, and studies the impact of ‘roaming’ on pricing decisions of individual stations in the US. Since pricing is an important determinant of optimal location and quantity of EV charging stations, these findings will help facilitate the right regulations and incentives to encourage EVCS deployment and EV adoption.

Rest of the paper is organized as follows: we discuss relevant literature on competition and pricing in Section 2.2, followed by a detailed explanation of data/transformations in Section 2.3,

3. All level 2 charger plugs are compatible. But L3 charger plugs come in three categories (which have intra but not inter compatibility): Chademo (adopted by Nissan, Mitsubishi, Toyoto, Kia, Peugeot, and Citroen); Combo/CCS (serving BMW, GM, Volkswagen, Ford, Chrysler, Daimler); Tesla. Source: <https://afdc.energy.gov/>

two different model specifications and results in Section 2.4 and Section 2.5. Finally we present the conclusion at the end in Section 2.6.

2.2 Pricing and Competition Literature

Rather ambitious discussion around EV charging space have focused on: smart charging in the UK; battery swapping stations in China; and peer-to-peer EV charging with dynamic pricing in Germany (Chen, T. et al 2020; Li and Ouyang 2011; Plenter, F. et al. 2018). Yet there are opportunities to learn from both the research work studying different perspectives on competition in EVCS market at present and competition in gasoline markets in general.

Product differentiation and DC fast charging

In industrial organization, strategic interaction theories can be classified as either spatial or aspatial. Location is the primary strategic choice variable in spatial theories, while aspatial theories could be examining the choice of price, quantity, production capacity, and/or quality/variety (Eckert and West 2006).

In the EVCS market, Luo et al. (2017) highlight importance of how production differentiation (charging speed) softens price competition, and that fosters a preference for clustering ‘clustering’ as opposed to spatial differentiation from EV charging service providers. Using road network and EVCS location data from the SanPedro District of Los Angeles, this papers the problem of multi-stage placement strategy of electric vehicle (EV) charging stations with incremental EV penetration rates. Findings suggest that while L1 and L2 are spaced more evenly and in greater numbers across the entire area (to work around the average delay constraint), L3 chargers tend to be placed in more ‘selective’ locations, making L1 and L3 weak substitutes.⁴ On the other hand, spatial differentiation

4. Note that EVs can be charged with three different connector speeds - L1 (slow, residential use, 120 V), L2 (normal, residential & commercial use, 240 V) and L3 (fast, commercial use, 480 V). L1 are residential overnight chargers (approximately 8 hours give a full battery as they add only 5 miles of charge per hour) and used mostly at home using the standard 120V outlet. L2 is used for both residential and commercial purposes (adds between 20-60 miles per hour of charge) using 240V outlet that can be hardwired (more permanent structure recommended for any outdoor installation for safety purposes and better power connection). L3 are commercial only based on Direct Current (DC) and not Alternative Current (AC) like L1 and L2. Also, L3 is also popularly know as the supercharger

can be very important in the retail gasoline market (Light 2004).

In terms of location specific pricing, it is suggested that DC charging stations located along major highways in the US will be more profitable if they embraced shorter charging sessions and charged higher prices per session (Huang and Kockelman 2020). DC fast charger deployment at a large national scale in the US is estimated to have a much more significant impact on the sales of EVs, increased electrified mileage, and reduced GHG emissions, when compared to level 2 charging stations. This difference holds even if ‘one DC fast charging station is built for every ten level 2 charging stations (Levinson and West 2018). However, the single-family homes with at-home charging capability will enjoy a greater impact of DC charging infrastructure’s national rollout on EV sales and electrified mileage, than non-single family homes.

Charging behavior, parking, and parking fees

The impact of parking behavior and waiting time at the EVCS has been studied well in the literature. Bagerzadeh et al. (2020) suggests that having a ‘parking barrier’ at the station can increase the EV charging service provider’s profits and reduce waiting delays. Also, a study of EVCS charging stations in Switzerland suggests that it takes half the time to break-even when the EVCS operator charges users on the basis of parking fees rather than power sales (Pagani et al. 2019).

Motoaki and Shirk (2017) uses a field experiment to investigate DC station usage efficiency through two period categorization: free; flat-fee. They find that both free and flat-fee DC EVCS may induce inefficient charging behavior as people parked for more than 30 minutes, suggesting that DC stations maybe used as parking spots and kill the fast charging social benefit. People were parking their cars at DC EVCS for longer time in the post-fee to get more bang for the buck. Though overall charging demand decreased post-fee because participants had L2 chargers at home, EVs that did charge were ones with much lower levels of charging left and absolutely needed to use the DC fast charging to complete their trip. This does highlight some substitution between L2 or DCFast (around 30 minutes to full charge since it could be adding 60-100 miles per 20 minute charge, but is much more expensive to install than L2).

and L3 provided that L2 is available at home.

Wolbertus et al. (2018) uses L2 stations data in Netherlands and reaffirm that in urban areas parking stations also double as parking stops. It is important to note that the EVCS pricing policies need to take into account diverse built landscapes: areas with more homogenous dwelling types will experience higher peak demand for EVCS compared to areas with a variation in the types of dwellings (and may require a lower number of stations overall due to different charging time needs). While time based fees may not be appropriate for street parking, it may serve higher pressure parking areas well.

More is not better

Underutilization and overcapacity at EVCS may hurt the profitability. Li and Ouyang (2011) and Zhang et al. (2018) recommend increasing EVCS profitability in China by keeping the cost of electricity low through battery storage use during peak hours, and that centralized operation of EVCS could help with load management and negotiation of electricity price with the grid operators. By keeping the utilization of stations high by not having overcapacity through extra stations, EVCS can achieve strong business feasibility.

Simulations predict that while a large national deployment of DC charging infrastructure in the US will have sharp returns on battery electric vehicles (BEV) sales and green house gas (GHG) emission reduction, these returns will start to diminish after 30,000 stations (if the customers bear an electricity surcharge to pay for the station). Returns may fade off completely at 80,000 stations (Levinson and West 2018). In general, states with zero emission vehicle (ZEV) mandates will see better returns on DC stations investments.

A study EVCS market in Switzerland suggests that new stations profitability will remain very uncertain until EV penetration reaches 10% (Pagani et al. 2019). Another work recommends focusing more on understanding the charging needs of the target customer, like delivery companies etc., instead of focusing on expanding the number of fast charging stations or waiting for more EVs in a region (Markkula et al. 2013).

2.3 Data

The primary data source is the Alternative Fuel Data (AFD), which is maintained by Alternative Fuels Data Center (AFDC)⁵ - a resource of the U.S. Department of Energy's (DOE) Office of Energy Efficiency and Renewable Energy's Vehicle Technologies Office administered by the National Renewable Energy Laboratory. DOE launched the AFDC in 1991 in response to the Alternative Motor Fuels Act of 1988 and the Clean Air Act Amendments of 1990.

We take a cross section of Alternative Fuel Data as of Nov. 26, 2019. In November 2019, electric fuel dominated the alternative fuel (AF)⁶ industry. Almost 80% of the 33,000 AF stations in US were electric (26,306), while only about 10% AF stations were E85, 5% CNG, 3% LPG, 2% BD, 0.5% LNG and .1% HY. Moreover, out of the 26,306 EV charging stations in 2019, almost 90% were accessible to the public (EV charging stations can be accessible to the public or private groups only such as fleet customers only or government only) - see Figure 2.1 in Section 2.7: Appendix. For our purposes, we restrict ourselves to public stations only (90% of all EV stations), or 23,331 observations spanning across US. Note that CA is the clear leader, with almost 25% of the total stations (5,629), followed by NY which has only 6.36% of the total stations (1,485).

Second, we can observe many location characteristics of the station - latitude/longitude, station name/street address/intersection directions, zipcode, city and state. While it is also possible to define a relevant market to be an MSA, we define it to be counties as MSAs are often too big while zip codes are often too small. We face no major issues in mapping zipcodes to counties (only 31 obs lost).

5. AFDC is an online platform that collects and maintains information while disseminating tools of interest to transportation decision makers that are seeking to diversify energy sources domestically in order to make better economic decisions.

6. Energy Policy Act (EPA) of 1992 classifies alcohols like pure methanol or ethanol, blends of gasoline with 85% or more of alcohol (E85), natural gas and its domestic derivatives like liquid fuels, propane, pure biodiesel (B100), hydrogen and electricity as alternative fuels. Furthermore, DOE can classify any other fuels as alternative fuels too as long as "it is substantially non-petroleum, yields substantial energy security benefits, and offers substantial environmental benefits".

Third, we merge basic socioeconomic indicators/demographic from United States Department of Agriculture (USDA) which are released at county level to serve as controls. However, note that there is some time lag with which USDA data is reported, and median income and poverty rate is two years old (unlike other controls that are one year old). So our basic market level controls consist of - (1) Median Income (2017) (2) Population and Labor Force Size (2018) (3) Unemployment Rate (2018) (4) Estimated Percent of People (of all ages) living in Poverty (2017).

Fourth, out of these 23, 298 station level observations, price data on 9.22% of the data is missing. Furthermore, 12,330 stations or 52.92% are free ⁷, while 8,821 stations or 37.86% are paid (Figure 2.2). Out of the 8,821 paid stations, pricing strategy can be regrouped into different formats - based on length of time⁸ (2,960 stations or 12.70% of total dataset), based on quantity⁹ (2,808 stations or 12.05% of total dataset), based on both time and quantity¹⁰ (1,757 station or 7.54% of total dataset), based on parking (840 stations or 3.61% of total dataset), and others¹¹ (456 stations or 1.96% of total dataset) - as displayed in Figure 2.3).

Fifth, in the raw data, we have 12 EVCS networks that are chain affiliated, and the 13th EVCS network is the residual category - it contains all the independent stations (non-chain affiliated/mom and pop stations). Recall, gasoline literature naturally divides stations into branded (chain affiliated) vs non-branded (single independent stores), where branded stations are of higher perceived quality¹² (Sampaio et al. 2010). However, we further break the branded/chain affiliated stations into two more categories - ‘roaming’ vs ‘non-roaming network’, to reflect business partnerships formed amongst some chain affiliated networks.

7. Note that EV Charging stations can gain revenues by membership fees, advertising (displays), higher sales due to customer good will, support by businesses offering employee perks, or even donors. Source: <https://greentransportation.info/ev-charging/range-confidence/chap4-charging/making-money.html>

8. per hour; per minute; per hour subject to minimum and maximum charges; et cetra

9. per kilowat; per kilowat with maximum/minimum limits; kWh Variable Energy Fee; et cetra

10. per kWh and per hour; et cetra

11. price per session; session interacted with time, quantity or both; et cetra

12. Unbranded stations typically compete by charging the lowest price. So, unbranded stations tend to heighten price competition, forcing branded stations to charge lower prices in local markets where independent stations are present.

In particular, we know that Chargepoint has formed a “peer to peer roaming agreement” between EVBox, Flo, Greenlots, Electrify America and EVgo charging networks¹³. Chargepoint has been leading the effort to create peer-to-peer roaming agreements that make it easy for drivers to charge wherever they go. Peer-to-peer roaming doesn’t add surcharges for the driver, eliminates any middleman and provides access to more drivers without any additional effort for station owners (since it is an open network). The driver doesn’t have to sign up for any additional accounts or networks - everything is taken care of by their existing network. This makes it convenient for the driver to charge their EV, gives them confidence in being able to drive electric and increases their choice of where they charge. A useful analogy here is being able to get cash at an ATM machine even when it’s not your bank, and members gain access to thousands more EV charging stations nationwide due to interoperability. Note while do not have EVBox or Flo networks in data, the roaming network category we create will consist of 4 EV networks - ChargePoint Network, Greenlots, Electrify America and EVgo.

Thus, we can re-arrange the 13 EV networks into two distinct business partnership network categories: non-networked (NN) or independents (20% of the EV stations fall in this category), and roaming network (48% of EV stations). Thus, the non-roaming network (32% of EV stations) consisting of chain affiliated stores will consist of 8 EV networks - Blink Network, EV Connect, GE Watt Station, Op Connect, SemaCharge Network, Webasto, Tesla Destination and Tesla, but we are not interested in this grouping even though we can create it. Again, the roaming and non-roaming differ from non-networked because they are chain affiliated, and roaming and non-roaming differ from each other in terms of charging reciprocal agreement between firms/business partnerships formed by EV networks.

Why do we emphasize this roaming dimension? From UK’s experience, it seems to be important. The Automated and Electric Vehicles Bill [9] in October 2017 allowed the British government to regulate if needed for three objective - (1) to improve the consumer experience of EV charging infrastructure, (2) to ensure the provision at key strategic locations, and (3) to require that charg-

13. <https://www.chargepoint.com/products/network/>

ing points have “smart” capability. It is anticipated to enable government to establish regulations, which allow to require operators to provide standardized measure of accessing public charging points in the near future, ensuring that customers can access any charging points without a specific membership.

Lastly, we create the competition variables. Again, note that EVs can be charged with three different connector speeds - L1 (slow, residential use, 120 V), L2 (normal, residential & commercial use, 240 V) and L3 (fast commercial use, 480 V). L1 are residential overnight chargers, L2 is used for both residential and commercial purposes while L3 are commercial only based on Direct Current (DC) and not Alternative Current (AC) like L1 and L2. L3 is also popularly know as the supercharger or DCFast (around 30 minutes to full charge since it could be adding 60-100 miles per 20 minute charge, but is much more expensive to install than L2).

In our alternative fuel data cross section, there are 480 stations with L1 technology (2% of total dataset), 21,111 with L2 technology (90% of total dataset) and 3,288 stations with L3 technology (14% of total dataset). Furthermore, about 5% stations have both L3 and L2 technology, while another 5% stations have both L2 and L1 technology, though there are hardly any stations with L1 and L3 technology. This is depicted in Figure 2.4.

We believe that stations with L2 chargers will compete primarily with neighboring L2 stations (stations with L2 charger only) and ignore stations with L3 chargers. Similarly, L3 stations compete with other L3 stations and ignore L2 stations. Thus, a station will only primarily be concerned about number of same type charger station within one mile radius of itself (Model 1, though we relax this assumption in Model 2 in Section 2.4 and 2.5 respectively). Hence, we create two competition variables measuring intensity of competition, by technology type. Note we will ignore L1 stations as they are primarily residential in nature, and less than 5% stations even have L1 technology.

This also realigns with some research on chargers and competition mentioned in the literature review. One, L2 service providers may be relatively spread out across all area since they

take some time to charge (are relatively slow) but L3 service providers may place their stations at “hot” locations, suggesting different marketing strategies. Two, service providers prefer clustering instead of spatial separation. The service providers have segmented the EV charging market by providing three different products (different charging level services of L1, L2, L3) in terms of voltage, current, charging speed and charging price. Due to product differentiation, they significantly soften the price competition so that they do not need to spatially separate from each other to further relax competition. This observation supports the opinions that firms do not have to maximize differentiation in every characteristic of its production.

2.3.1 Summary Statistics: Key Variables

The summary statistics are attached below.

Table 2.1: Summary Statistics

Statistic	Mean	St. Dev.	Min	Max	N
Positive Price EV Charging Station (mi=.)	0.42	0.49	0.00	1.00	21151
L3 chargers at the station	0.50	1.93	0.00	40.00	23298
L2 chargers at the station	2.44	3.63	0.00	260.00	23298
L1 chargers at the station	0.04	0.55	0.00	51.00	23298
Station belongs to Roaming network	0.48	0.50	0.00	1.00	23298
Station belongs to Non-Roaming network	0.33	0.47	0.00	1.00	23298
Station belongs to Independant network	0.20	0.40	0.00	1.00	23298
Station belongs to any Chain Affiliated Network	0.80	0.40	0.00	1.00	23298
# of L2 Stations within 1 miles of L2 station i	7.93	15.34	0.00	116.00	23298
# of L3 Stations within 1 miles of L3 station i	0.62	3.73	0.00	109.00	23298
Median Household Income, 2017	69078.51	18446.28	25569.00	136191.00	23297
Unemployment Rate, 2018	3.73	0.99	1.60	18.10	23298
Population, 2018	1525584.84	2352657.28	762.00	10105518.00	23298
Poverty Rate, 2017	12.42	4.14	3.00	42.20	23297
Observations	23298				

Source: Alternative Fuel Data, Nov 2019.

Price observations dropped if price information is missing in data.

First, about 42% of the stations charge a positive price in our data. In addition, figure 2.5 and figure 2.6 in appendix show distribution of free or paid EVCS by EV network.

Second, L2 stations (21,111) are much more common than L3 stations (3,288), defined as a station that has one or more L2 charger or L3 charger respectively, as shown in Figure 2.4.

Figure 2.7 shows the distribution of L2 chargers at L2 stations, and figure 2.8 shows the distribution of L3 chargers at L3 stations. Furthermore, figure 2.9 and figure 2.10 show the conditional distribution of L2 chargers at L2 stations and L3 chargers at L3 stations if the station contains 10 or less chargers.

Third, about 20% of the stations belong are independent, and thus about 80% stations are chain affiliated. Additionally, 48% stations belong to roaming network.

Fourth, in a one mile radius of a station, assuming the same type station compete with each other (non-substitution of charging technology), L2 station is 12 times more likely to face competition. than a L2 station.

2.4 Model 1: Service and Product/Technology Differentiation

Again, we have a cross-sectional data of EV charging stations as of November 28, 2019.

We hypothesize that EVCS pricing decision (to charge a price or not) is a function of station-level characteristics (types and number of chargers), market structure (competition), network characteristics (chain affiliated or independent), business partnerships (part of roaming network), and market demographics. We believe the right hand side variables are exogenous since EVCSs charging a price or not is a much more reversible action than changing station level, market structure or demographic decisions in the short run.

The model specification we have is Logit with EV network fixed effects, since our dependent variable is binary.

Following is the first model to be estimated:

2.4.1 Specification 1

$$\begin{aligned}
 positive_price_i = & \alpha_1 net_roaming_i + \beta_1 L3_charger_i + \beta_2 L2_charger_i \\
 & + \delta_1 comp_station_L3_i + \delta_2 comp_station_L2_i \quad (2.1) \\
 & + \psi chain_affiliated_i + \vec{\omega} county_controls_c + \epsilon_i
 \end{aligned}$$

where $positive_price_i$ takes a value of 0 if EV charging station is free and a value of 1 if station is paid access only,

$net_roaming_i$ is a dummy indicator for EV charging station being part of roaming EV network,

$L3_charger_i$ is the number of L3 (DC) chargers at the station, and equals 0 if none are present,

$L2_charger_i$ is the number of L2 (AC) chargers at the station, and equals 0 if none are present,

$comp_station_L3_i$ is the total number of L3 stations within 1 mile radius of L3 station i ,

$comp_station_L2_i$ is the total number of L2 stations within 1 mile radius of L2 station i ,

$chain_affiliated_i$ is a dummy for EV charging station belonging to any chain-affiliated network,

$county_controls_c$ is a set of county level controls including population, median income, poverty & unemployment rate for county of station i , and

ϵ_i is the independent and identically distributed error term.

We have a few hypothesis that we would test based on gasoline and industrial organization literature.

First, we expect chain affiliated stations to charge a higher price than independent stations due to higher consumer perception/brand value/better recall or marketing value ($net_chain_affiliated > 0$). We call this **chain affiliation effect**.

Second, we similarly would expect the chain affiliated stations belonging to roaming network to be able to charge an even higher price as this is an “enhanced service” (more places an individual can charge their vehicle). We call this **roaming service effect** ($net_roaming > 0$).

The second hypothesis is about business partnerships that some chain affiliated EV charging station networks have formed, and is measuring the impact of a new service in the EVCS space. In contrast, the next two null hypothesis are about product or technological differentiation/station level characteristics, with the later one a stronger condition.

Third, more station idle capacity likely implies higher barriers to entry. Also, people do not like waiting, and should be willing to pay higher prices to avoid waiting costs at the station too. Thus, higher capacity at station should enable the station to charge a higher or a positive price. We call this **product/technology capacity effect** ($L3_charger > 0, L2_charger > 0$).

Fourth, offering premium service requires (higher) fixed costs to be recovered. Second, L3 chargers have a higher installation (fixed) cost and the station wants to recoup its investment, and/or because time sensitive and price insensitive customers are more likely to use L3 chargers. We call this **product/technology quality effect** ($L3_charger > L2_charger$).

The fifth, and probably of most interest to industrial organization economists is the hypothesis of competition limiting the ability of EVCS to charge a price. We call this **competition price effect** ($comp_station_L3 < 0, comp_station_L2 < 0$)

We would expect $comp_station_L3 < comp_station_L2 < 0$ due to the fact that L3 charger is more for commercial purposes, while L2 charger is slightly less important for commercial customers. In the same vein, competition should be more intense for serving non-retail clients (U-Haul fleets) or more inelastic customers, though we do not currently test for this.

Main Table I

Table 2.2 reports the results of the estimation.

First, we do find support for our roaming service network hypothesis, as expected. Stations belonging to roaming category are more likely to charge a price than a station that do not belong to the roaming category, and are thus charging for this enhanced service/convenience to charge your vehicle at a wider area. The coefficient is significant at 1% level of significance. The logit coeffi-

Table 2.2: Model 1 Logistic Regression Results

	Positive Price EV Charging Station (mi=.)		
	Model 1	Model 2	Model 3
Station belongs to Roaming network	1.333*** (.03182)	1.227*** (.03195)	.7712*** (.0341)
L3 chargers at the station	.6122*** (.02595)	.6474*** (.027)	.6493*** (.03357)
L2 chargers at the station	.02531** (.01053)	.007432 (.00579)	.006398 (.00518)
# of L3 Stations within 1 miles of L3 station i	-.02444*** (.00703)	-.03711*** (.007853)	-.02925*** (.006905)
# of L2 Stations within 1 miles of L2 station i	.01454*** (.0009732)	.003982*** (.001062)	.0009884 (.001066)
Median Household Income, 2017		.00002597*** (1.415e-06)	.00002459*** (1.450e-06)
Unemployment Rate, 2018		-.1285*** (.02206)	-.122*** (.02311)
Population, 2018		1.487e-07*** (7.399e-09)	1.538e-07*** (7.492e-09)
Poverty Rate, 2017		.06162*** (.00623)	.0567*** (.006513)
Station belongs to Chain Affiliated Network			1.749*** (.06788)
Constant	-1.358*** (.03427)	-3.496*** (.1778)	-4.638*** (.1947)
Logit	Yes	Yes	Yes
Controls	No	Yes	Yes
Observations	21,151	21,150	21,150

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

cient interpretation is the following - if a station belongs to roaming category (a unit increase), it is associated with increased odds of station charging a price by a factor of $e^{.7712} \approx 2.718$. Roaming network characteristics seems to be moderate in magnitude too.

Second, we find some support for our product/technology capacity and product/technology quality effect. The more chargers at a station, more likely that the station charges a price, and this effect is stronger for L3 chargers. If we increase the number of L3 chargers at a station by 1 count, the station's odds of charging a price increases by a factor of $e^{0.6493} \approx 1.914$. However, odds that a station charges a price after adding one more L2 charger is not statistically greater than 1. This makes sense as L3 charger is a higher quality product for which customers have a higher willingness to pay and it is also more costly to provide. Thus, as L3 chargers increase at a station, the station is more likely to charge a price.

There is a potential endogeneity concern with co-efficient on number of L3 chargers though as firms may be selective in whether to install or how many to install (being an expensive and fast charger) at the station based on expected profits. Stations that are likely to have high demand for EV charging may also be the stations more likely to install L3 chargers. Thus, without explicitly controlling for demand/quantity sold at the location, our L3 charger coefficient could be positively biased upwards. We do control for county population of the station to reduce this omitted variable bias, though population density of the county is likely to be a slightly better proxy.

Third, we do find partial support support for price competition effect - L3 stations seems to be competing, while L2 stations do not. For an L3 station, if another L3 station enters in its 1 mile radius, the odds of it to charge a price fall by a factor of $e^{-0.02925} \approx 0.97$, which is significant but not very large in magnitude. However, for an L2 station, if another L2 station enters in its 1 mile region of influence, the station's odds of charging a price increase by a factor of $e^{0.00099} \approx 1.001$. This is neither statistically significant nor economically meaningful. L2 station's pricing decision could be driven by factors other than competition like business type at station premises (shopping mall, highway, school et cetera.) or even real estate price, as it is relatively cheap to install. These

stations could be trying to lock in customers for future (especially since EVCS industry is in its infancy), or likely gaining revenues from alternative sources like super-app advertisement.

Fourth, we find that chain affiliated networks charge a higher price than if the station is non-chain affiliated/non-branded/mom and pop store. This makes sense as chain affiliated stations are perceived of a higher quality by consumers, and can thus charge a premium for their brand name - they have higher recall or marketing value. Thus, chain affiliated stations are exploiting their market power relative to independent stations.

Note that the difference between station belongs to roaming network and station belongs to chain affiliated network is that as long as the station belongs to either of the 4 networks - Charge-Point, Greenlots, Electrify America or EVgo, the station belongs to a roaming network. However, as long as the station does not belong to an independent network, it is chain affiliated. This definition of chain affiliated networks is required over simply keeping in a roaming network dummy and all 13 chain fixed effects because of perfect multicollinearity, since roaming network is simply made up of combining 4 chain fixed effects.

Fifth, the coefficients on demographic variables largely make sense. Counties with higher median household income and higher population are more likely to charge a positive price as one would expect, though the odds increase by a very small factor but are nonetheless statistically significant. Similarly, counties with one percentage point higher unemployment rate are less likely to charge a positive price by a factor of $e^{-0.22} \approx 0.978$. The coefficient on poverty rate is statistically significant but positive - counties with one percentage point higher poverty rate are more likely to charge a positive price by a factor of $e^{0.0567} \approx 1.058$. If areas with high poverty also have high income inequality, and high income equality allows stations to charge a higher price to skim the inelastic consumers, then not including income equality can explain the small positive coefficient on unemployment.

2.5 Model 2: Extended Spatial Competition

We also estimate another model slightly different than the former, with two key differences. First, we add an interaction term for roaming network and number of chargers at a station to combine the effects of roaming as a service and high quality product/technology. Second, we also relax our competition variable definition from stations of the same type (L2 or L3) competing with same type stations only to stations competing with all other stations (easy substitution between different charging technology/charging speeds). For robustness, we also expand the extent of geographic competition that a station faces from 1 mile radius to a wider range of values.

2.5.1 Specification 2

Following is the equation to be estimated:

$$\begin{aligned}
 positive_price_i = & \alpha_1 net_roaming_i + \beta_1 L3_charger_i + \beta_2 L2_charger_i \\
 & + \gamma_1 (net_roaming * L3_charger)_i + \gamma_2 (net_roaming * L2_charger)_i \\
 & + \delta comp_station \\
 & + \vec{\psi} chain_affiliated_i + \vec{\omega} county_controls_c + \epsilon_i
 \end{aligned} \tag{2.2}$$

where $positive_price_i$ takes a value of 0 if EV charging station is free and a value of 1 if station is paid access only,

$net_roaming_i$ is a dummy indicator for EV charging station being part of roaming EV network,

$L3_charger_i$ is the number of L3 (DC) chargers at the station, and equals 0 if none are present,

$L2_charger_i$ is the number of L2 (AC) chargers at the station, and equals 0 if none are present,

$(net_roaming * L3_charger)_i$ is the interaction of roaming indicator and L3 charger discrete variable,

$(net_roaming * L2_charger)_i$ is the interaction of roaming indicator and L2 charger discrete variable,

$comp_station_i$ is the total number of L3 or L2 stations within an x mile radius of station i , $chain_affiliated_i$ is a dummy for EV charging station belonging to any chain-affiliated network, $county_controls_c$ is a set of county level controls including population, median income, poverty & unemployment rate for county of station i , and ϵ_i is the independent and identically distributed error term.

Signs on co-efficients as mentioned in previous section should be the same (chain affiliation effect, roaming-service effect, product/technology capacity and quality effect). The price competition variable ($comp_station$) is a bit differently specified but we would expect it to have a negative coefficient. Also, we would expect the interaction terms between roaming station and number of different technology chargers at the station ($net_roaming * L3_charger$ and $net_roaming * L2_charger$) to have a positive sign.

Main Table II

Table 2.3 gives regression results for second model.

The results are somewhat similar. First, we still find that stations part of roaming networks are more likely to charge a positive price than stations that are not part of roaming network by a factor of about $e^{0.8} \approx 2.22$. In fact, its statistical significance and magnitude are largely similar to what we had previously found in the last model (Table 2.2), and thus we find support for stations charging a positive price for roaming as a service.

Second, L3 chargers at a station remain significant in explaining the station's decision to charge a positive price like in Table 2.2, while presence of L2 chargers do not seem to be affecting the station's decision to charge a positive price. Thus, while we do find partial support for product/technology capacity effect, we do find support for product/technology quality effect as L3 chargers are usually more heavily priced than L2 chargers. Again, this is rationalized by L3 chargers is a higher quality product for which customers have a higher willingness to pay and it is also more costly to provide.

Table 2.3: Model 2 Logistic Regression Results

	Positive Price EV Charging Station (mi=.)				
	Model 1	Model 2	Model 3	Model 4	Model 5
Station belongs to Roaming network	.7967*** (.04677)	.7912*** (.04698)	.8057*** (.04624)	.8059*** (.04585)	.7842*** (.04626)
L3 chargers at station	.8338*** (.04289)	.8318*** (.0428)	.8362*** (.0429)	.8365*** (.04278)	.8328*** (.04279)
L2 chargers at station	-.001586 (.005664)	-.001414 (.005664)	-.002332 (.005664)	-.003278 (.00568)	-.002468 (.005674)
L3 chargers at station X Roaming network	-.5272*** (.07738)	-.5263*** (.07732)	-.5284*** (.07742)	-.5303*** (.07743)	-.5312*** (.07735)
L2 chargers at station X Roaming network	.008642 (.01028)	.008995 (.01038)	.007592 (.01001)	.006631 (.009823)	.007213 (.01002)
# of stations within 1 miles of station i	.0005929 (.00106)				
# of stations within 5 miles of station i		-.0003225 (.0002721)			
# of stations within 10 miles of station i			.0007696*** (.0001647)		
# of stations within 15 miles of station i				.0009765*** (.0001308)	
# of stations within 50 miles of station i					.0003503*** (.00005224)
Station belongs to Chain Affiliated Network	1.81*** (.07041)	1.821*** (.07053)	1.788*** (.07023)	1.785*** (.07032)	1.826*** (.07044)
Logit	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes
Observations	21,150	21,150	21,150	21,150	21,150

Controls include median income (2017), unemployment rate (2018), population (2018), and poverty rate (2017). Constant term is also not reported.

*p<0.1; **p<0.05; ***p<0.01.

Third, if a station is part of roaming network, compared to non-roaming network, as the number of L3 stations increases, the odds that a station charges a positive price is $e^{0.53} \approx 0.589$. Thus, the station is less likely to charge a positive price by about $(0.589 - 1) * 100\% \approx 42\%$.

It is interesting to note that while there are about 48% stations (12,190 stations) in our dataset that are part of roaming network, and about 14% (3288 stations) that have L3 charger, the distribution of L3 chargers is fairly even across roaming (1,912 stations) and non-roaming networks (1,376 stations), reducing concerns about roaming network self-selection into L3 stations. Furthermore, the distribution of L3 chargers across roaming and non-roaming networks is fairly balanced too. Thus, roaming networks could be trying to lure customers, hoping to make profits through non-price means too like advertising revenue through the super app or profits from infrastructure on the station like shopping center or convenience store, et cetera.

Fourth, the co-efficient on competition variables become very small and insignificant when the radius of competition circle is less than 5 miles, but turns positive and significant (though very small in magnitude again) when we increase the radius of competition circle to 50 miles. The co-efficient suggest that stations are not really competing with each other due to their small magnitude. However, as we increase the market size, price competition effects decline in magnitude, suggesting people do not drive too far for charging. Thus, distance does not seem to have a huge bearing on firm's positive pricing decision.

The positive and significant coefficients on competition variables is not what one normally expects in industrial organisation studies - both theoretically and empirically, as competition lowers price in free functioning markets. In fact, in Table 2.2, we did not get significantly positive coefficients either for the one mile competition radius variables. Thus, competition variable co-efficients from Table 2.3 can be explained by some omitted variable that could be correlated with entry of competitors and charging a positive price, like station's current expectation of growing or stable high charging demand, which will bias the competition variable coefficients in the positive direction.

Fifth, if the station belongs to any of the 12 chain affiliated networks, the odds that it charges a positive price increases by a factor of about $e^{1.8} \approx 6.05$. Like in the last model, this suggests that chain affiliated stations are perceived of a higher quality by consumers and are charging a premium for their/exploiting their market power.

Lastly, coefficient signs on all controls are reasonable (like in Table 2.2), except on poverty rate which is positive (presumably due to omitted variable of income inequality). These are not reported in Table 2.3.

2.6 Conclusion and Future Work

This paper contributes to the literature on competition in electric vehicle charging infrastructure market in the US by empirically identifying factors that affect stations' pricing decisions. We also exploit the impact of roaming agreements between charging service networks on their competitive strategies. These findings are valuable in understanding firm's response to competitor's service and product differentiation, and can help inform better regulations and incentives in the EV charging market that promote optimal placement and pricing, eventually leading to higher EV adoption.

Electric vehicles are critical for transport decarbonization, and we need more electric vehicle charging stations to promote EV purchases to solve the indirect network effect - EV and EVCS chicken and egg problem, especially till battery range anxiety remains an issue. Factors that allow a station to charge a positive price would also help policy makers get leverage over tools to promote public EV charging infrastructure through private investment, and 'subsidizing' EVCS is a more cost effective way to promote a clean energy future. than subsidizing EVs.

First, EVCS that are part of peer-to-peer 'roaming' agreements are more likely to charge a positive price. In other words, EV charging stations under 'roaming' are less likely to offer free service. We think this is due to stations charging for this enhanced service/convenience as

consumers can recharge their vehicle at a wider set of stations, and this may not necessarily be welfare reducing for society if it leads to greater utilization of charging infrastructure. Thus, we find strong support for roaming effect, just like we find support for chain affiliation effect (better recall or marketing value).

Second, product/technology quality differentiation softens price competition in EVCS space too, like in gasoline industry. Stations with L3 chargers are more likely to charge a positive price, which makes sense as this is an expensive product. On the other hand, many L2 chargers are free. Furthermore, we only find partial evidence for product/technology capacity effect at stations - while stations with more L3 fast chargers can charge a positive price, stations with more standard L2 chargers are not statistically significantly to charge a positive price.

Third, L3 stations seem to compete with other L3 stations on price but L2 stations do not seem to be competing with other L2 stations on price. Overall, EV charging stations are not competing with each other on price even though as one increase the radius of competition circle faced by each station beyond 1 mile, suggesting that competition price effect diminishes as people are unlikely to drive far off to refuel seeking cheaper charging services. We interpret this as overall weak evidence of spatial competition.

Some improvements that we intend on making soon include modeling spatial competition slightly better since we have latitude/longitude information for every station. We could model L2 charger competition along 17 minute travel distance area instead of a single unit circle, since literature suggests that is the distance people shop around for better prices. ArcGIS helps create 5 mile drive time areas, an improvement for better capturing price competition effect.

Second, we could take a deep dive into L3 (DC fast) charger plug types to add another layer of complexity on top of product/technology differentiation. There are three different/distinct plug types or connector shapes within L3 due to different charging protocols, each supported by few automakers who have committed themselves to a protocol - implying that not all EVs can use any L3 station. Furthermore, there is an interesting welfare component here because mandating

compatibility in charging standards can on one hand decrease duplicative investment in charging stations by car manufacturers and benefit consumers, while on the other hand induce firms to cut back on their investments (negative spill-overs) (Li 2017).

Third, our data allows us to break down pricing by type (pricing by time, pricing by quantity, pricing by session) to estimate a multinomial logit regression, and explore different pricing strategies as an additional layer.

Fourth, we also plan to use more appropriate demographic controls like population density of the county instead of population of county, and income per capita in county instead of median household income of the county. Additionally, we would like to control for income inequality too.

Fifth, our results can be further investigated through separation of different neighborhoods by housing types, rural/urban categorization and other 'points of interest' like business establishment, parking, et cetra..

There are also some natural extensions of this paper. First, we could transform the current static competition model into a dynamic structural model where entry/exit decisions of firms can also be accounted for like Mazzeo (2002) does for entry of high/low quality motel across highways exits to study product and price differentiation. Second, like Seim's (2006) modelling of video rental retail industry, we could focus on spatial differentiation by stations to see if firms use location choices to shield themselves from competition. Third, we could collect some electric vehicle car sales data and combine that with our station level data to conduct some welfare studies about the impact of electric vehicle adoption in US.

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2.8 Appendix

2.8.1 Figures

Figure 2.1: Most EVCS are public.

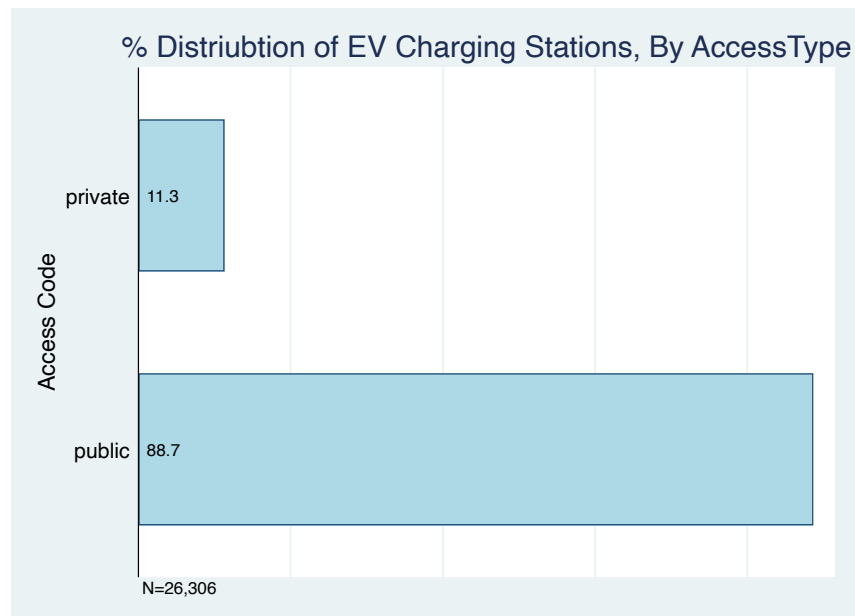


Figure 2.2: Most EVCS are free.

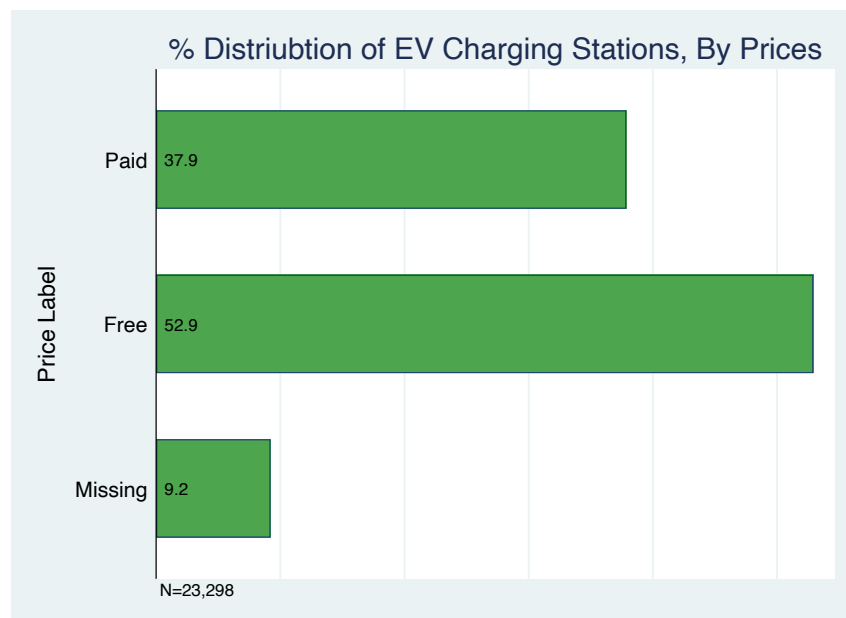


Figure 2.3: Pricing strategy at EVCS.

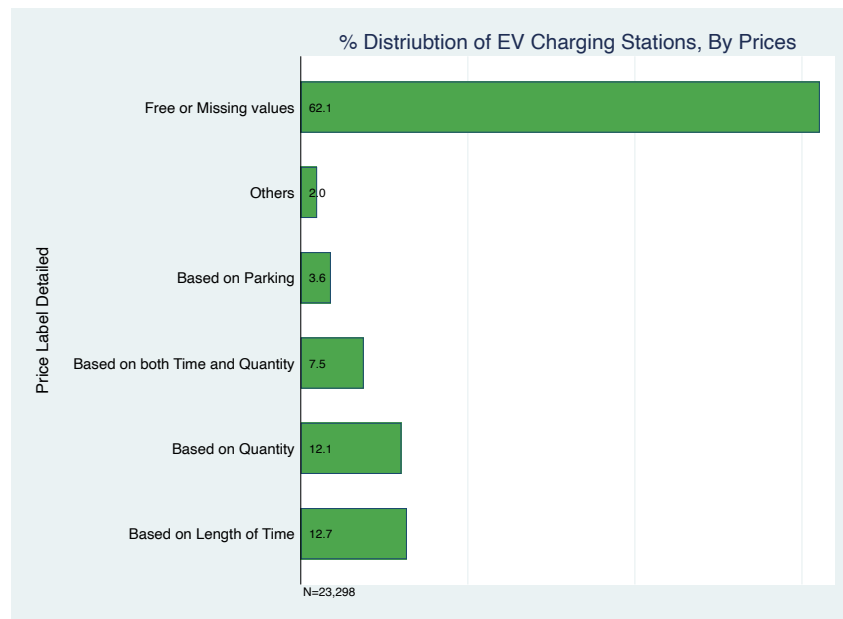
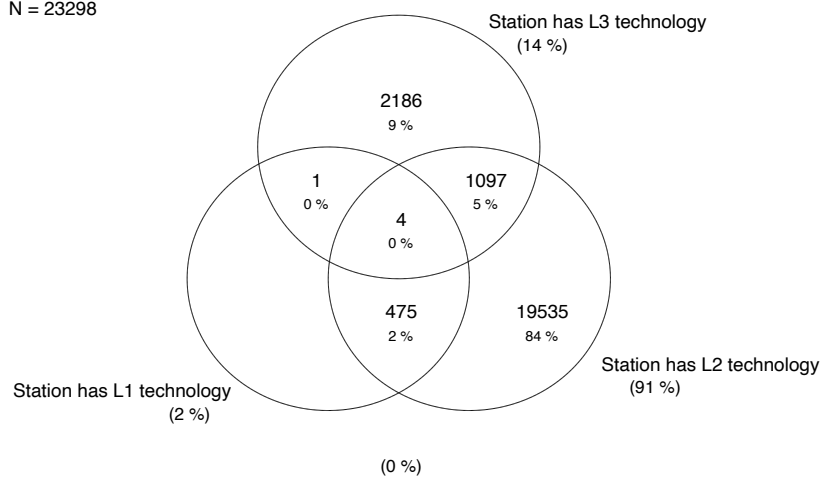


Figure 2.4: Technology at EVCS.

Venn Diagram

N = 23298



16 Mar 2021

% of total

File: clean_cross_Nov 26 2019.dta (16 Mar 2021)

Figure 2.5: Free EVCS by network

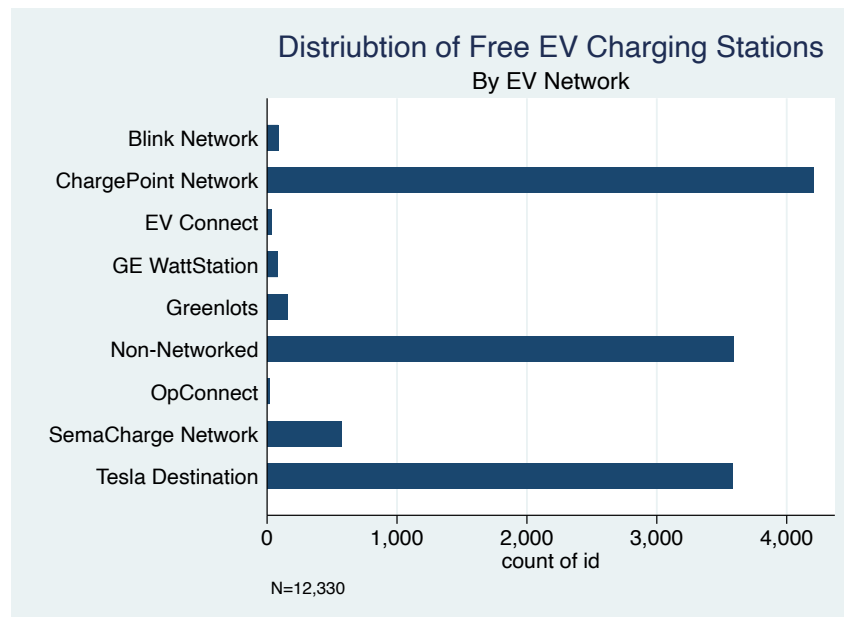


Figure 2.6: Paid EVCS by network

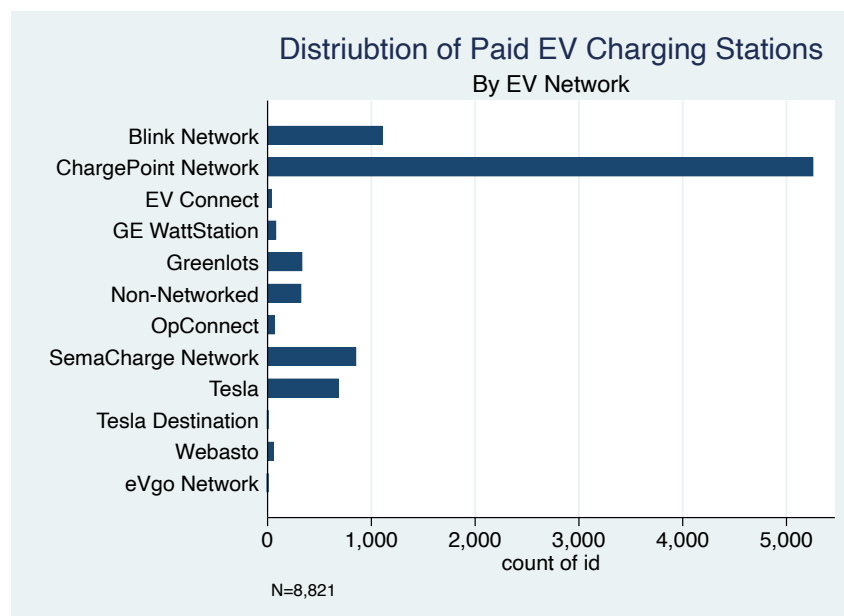


Figure 2.7: Distribution of L2 chargers at L2 stations.

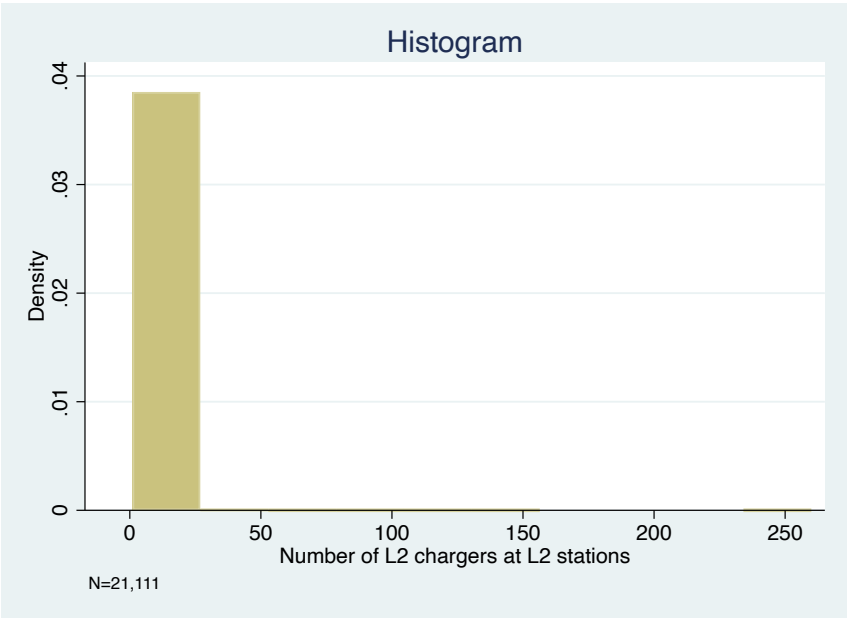


Figure 2.8: Distribution of L3 chargers at L3 stations.

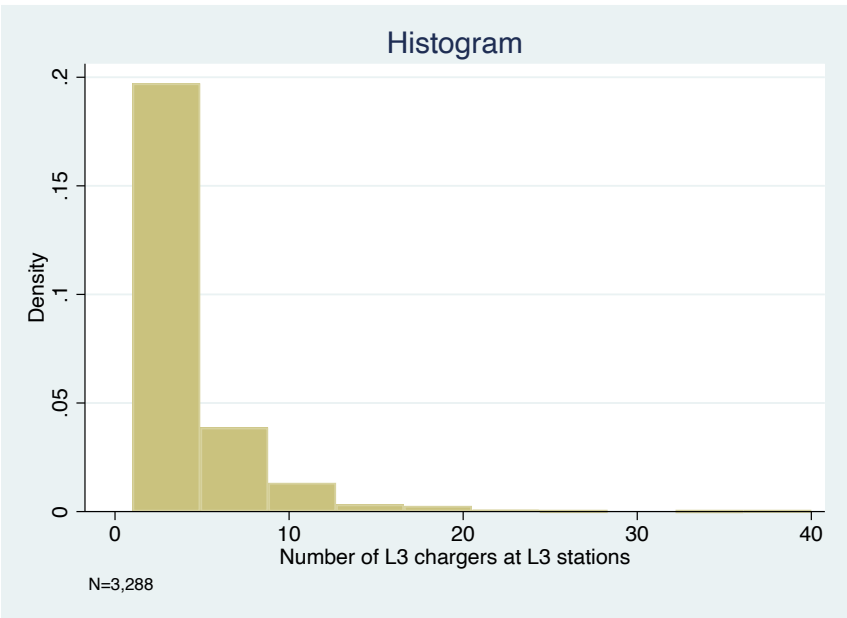


Figure 2.9: Distribution of L2 chargers at L2 stations conditional on L2 station having less than 11 L2 chargers.

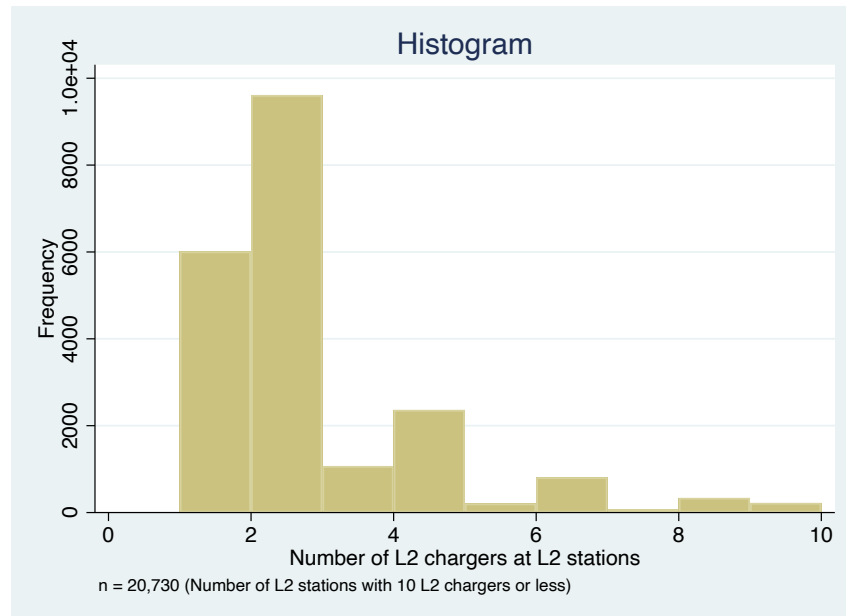
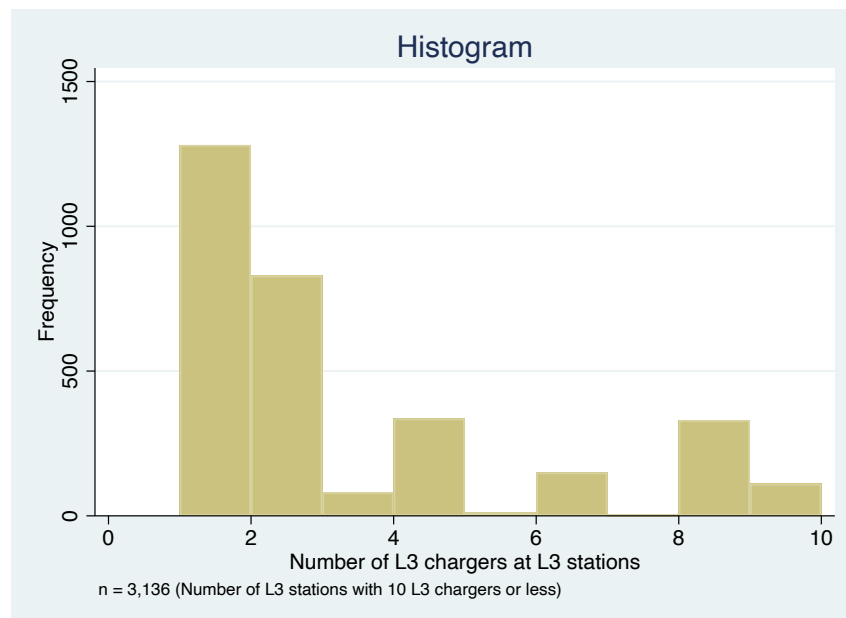


Figure 2.10: Distribution of L3 chargers at L3 stations conditional on L3 station having less than 11 L3 chargers.



Chapter 3

Impact of FinTech Firms on Housing Mortgage Market

3.1 Introduction

Residential Mortgage market is the largest consumer loan market in US (more than 10 trillion in outstanding debt and five times outstanding student loan market) and has evolved dramatically in the years following the 2009 financial crisis (Fuster et al. 2018; FRED 2021). The two important trends in financial intermediation that have been well documented in the US residential mortgage market post 2009 recession are (Buchak et al. 2017):

- (1) shifting of financial intermediation from traditional banks to shadow banks¹, and
- (2) shifting of financial intermediation from “brick and mortar” originators to online originators.

The first trend, or **rise of shadow banks**/nonbanks, is evident from the fact that shadow bank lenders' market share among all residential mortgage lending has grown from roughly 30% in 2007 to 50% in 2015. These nonbanks are often nondepository institutions with a long-standing presence

1. Nonbank (non-depository) institutions falling outside the scope of traditional banking regulation. Nonbanks have no deposit base, which implies that they have no access to the Federal Reserve's discount window. Thus, if nonbanks need funding, they often go to banks too to raise short-term debt collateralized by new loans (Krishnamurthy and Wachter 2018).

in the mortgage market (say firms that existed before the crisis), or new entrants that have increased market share via online lending platforms (like Quicken Loans) (Kim et al. 2018). There are two broad hypotheses explaining the decline of traditional banks, namely increased regulatory burden and disruptive technology. After the 2009 financial crisis, multiple new regulations were passed to enhance risk management practices at financial institutions, including The Dodd-Frank Wall Street Reform and Consumer Protection Act which alone propagated hundreds of new regulations (Linke 2018 ; Tarullo 2016), and can explain up to 60% of why shadow banks are more active in financial intermediation today (Buchak et al. 2017). Another 30% of why shadow banks are more active in financial intermediation can be explained by emergence of disruptive firms called FinTech lenders, and gives rise to our second trend (Buchak et al. 2017).

The second trend, or **rise of online originators**/"FinTech lenders", is evident from their market share having grown from roughly 3% in 2007 to 12% in 2015 (Buchak et al. 2017). FinTech² firms are stand-alone (online) mortgage originators that primarily securitize mortgages and operate without deposit financing or a branch network. FinTech lenders have comprehensive end-to-end online mortgage application and approval process backed by centralized underwriting operations instead of the traditional local brokers network or "brick-and-mortar" branches (Fuster et al. 2018). They automate a labor-intensive process (gathering/verifying supporting material), speed up information transfer (bank-end database checks), improve accuracy (transcription errors through pattern recognition software) and efficiency (Fuster et al. 2018; Nash and Beardsley 2015). FinTech firms have an originate to distribute model i.e. they originate mortgage loans, hold them for a short period of time, then pool them together and issue mortgage-backed securities to investors (Krishnamurthy and Wachter 2018). Most FinTech lenders sell their loans through established channels, just like other nonbanks, to government guarantee programs like Veterans Association (VA), Fannie

2. Since the term "FinTech" is very general, we will restrict ourselves to firms at technology frontier in mortgage loans markets. Thus, we are excluding innovative firms in non-loan/mortgage category such as wallets and mobile payments (example Venmo), cards and other traditional payments, savings and checking accounts, value added services, or wealth and asset management services (example Stash) (EY 2019). This restriction allows us to get to a precise working definition discussed more formally in Section 3.1.1 where we will see that FinTech firms are just a subset of shadow banks using a lot of technology for decision making so that they can pre-approve applications online, unlike convention/traditional nonbank mortgage originators that may have a basic online presence too.

Mae and Freddie Mac, or pack them in Ginnie Mae or Government Sponsored Enterprise (GSE) securities so that they can transfer the interest rate risk to the investor instead of holding it themselves (Fuster et al. 2018; Krishnamurthy and Wachter 2018). Certain market segments have seen tremendous growth, like refinances and home purchase mortgages insured by the Federal Housing Administration (FHA) (Fuster et al. 2018).

There are two forces that are associated with disruption caused by online technology, namely the ease of online origination³, and the use of different information acquired through lending process to set interest rates. Thus, financial intermediation is increasingly being provided by online originators, which are replacing the “brick and mortar” originators.

This paper measures the impact of online originators in the housing mortgage market by looking at three different types of loans, and thus it is an extension of the second trend. The rest of the paper is organized as following - Section 3.1.1 talks about the three types of financial intermediation firms in US residential mortgage markets, Section 3.2 touches upon some important economic literature in the domain of mortgage lending and FinTech firms, Section 3.3 describes the three data sources used for the study, Section 3.4 discusses the instrument variables specification (Section 3.4.1) and summary statistics (Section 3.4.2) while Section 3.5 discusses the regression results, and finally Section 3.6 concludes.

3.1.1 Classifying FinTech Firms

Recall that that FinTech firms are just a subset of shadow banks (mortgage originators using high technology), do not have any physical network of branches and pre-approve applications online. For our research purpose, we will take the FinTech classification by Fuster et al. (2018) as given. They look at the top 20 mortgage originators in 2016 and classify all the lenders into 3 types of firm, namely banks, nonbank mortgage lender and FinTech firms. Classifying firms into banks/depository institution and non-depository institutions is relatively standard, but classifying

3. FinTech firms originate housing loans online completely, unlike most banks or mortgage originators which require a physical visit to branch or telephonic conversation with a loan officer at some stage of the loan approval process.

non-depository institutions/shadow banking firms as FinTech or nonbank mortgage originator may not always have an unanimous consensus.

Figure (3.1) in Appendix 3.8 shows the classification table generated by Fuster et al. (2018), and has three key takeaways.

First, there are only 3 FinTech firms in the top 20 mortgage originators, and the remaining firms include 9 banks and 8 nonbank mortgage lenders.

Second, the mortgage origination market is competitive at the national level, with most firms having a very small market share nationally. In fact, there are over 6,000 residential mortgage financial intermediation firms in the US, and the largest 20 firms account for the less than 35% of the total market share, with the largest two firms accounting for more than 11% of the market share and largest 4 firms accounting for about 17% of the market share.

Third, there are not a lot of FinTech firms in US residential mortgage market, and the authors were able to identify only 18 FinTech firms in total by examining the websites of the top 100 nonbank lenders for both home purchase loans and home refinances over the 2011 to 2017 time period (135 lenders in total). For each of these 135 lenders, a loan application was submitted manually and the research analysts recorded if the pre-approval could be obtained online or not. Majority of the lenders stopped the online application process and requested the applicant to contact a broker or loan officer. Lenders were classified as FinTech only if an applicant was able to continue with the online process until they got to the pre-approval decision that is based on a hard credit check of the applicant's credit score (Fuster et al. 2018).

Given these three observations, we will restrict ourselves to only the largest 20 firms by market power (as of 2016) instead of the approximately 6000 financial intermediaries to see whether FinTech firms are having any impact on the housing mortgage market.

Lastly, this classification comes very close to what many industry experts believe, and very similar to the other major work that has tried to classify financial intermediation firms into the three

categories. In particular, the classification is very similar to one proposed by Buchak et al. (2017), with only minor differences “with respect to the classification of a few smaller lenders” (Fuster et al. 2018).

3.2 Literature Review

There are a few important results that look at FinTech firms and mortgage lending.

First, FinTech lending model represents a technological innovation that reduces frictions in mortgage lending, such as processing times and scalability (Fuster et al. 2018). However, Fuster et al. (2018) find no evidence that FinTech lenders target marginal borrowers, even when controlling for detailed loan, borrower, and geographic variables. Furthermore, Fuster et al. (2018) find that in areas with more FinTech Lending, borrowers refinance more, especially when the current mortgage interest rate falls below the original rate by a certain margin (to cover refinance costs). In fact, this paper is quite heavily motivated from this last finding, but wants to improve upon the modeling by instrumenting for FinTech presence in counties where more refinancing takes place. More details in specification Section 3.4.

Second, Buchak et al. (2017) exploit geographical heterogeneity caused by 4 particular changes in regulatory burden in a diff-in-diff study, namely increase in capital requirements, increase in mortgage servicing rights, more mortgage related lawsuits, and the shift of supervision due to shutting down of Office of Thrift Supervision to Office of Comptroller and Currency to find that shadow bank fill in gaps where traditional banks contracted. Furthermore, FinTech firms appear to provide convenience to borrowers rather than cost savings to them - they serve more credit worthy borrowers although charging them a premium, and are more active in refinancing market (Buchak et al. 2017).

Third, FinTech lenders increase mortgage credit lending after demand shocks caused by natural disasters, unlike traditional banks and non-FinTech (Allen, Yu and Yao 2022). This increased

lending is not associated with changes in mortgage rates to impacted borrowers, lending standards, or loan performance.

Fourth, increase in internet access due to the entry of Google Fiber in Kansas City (Missouri) after late 2012 did not help increase FinTech borrowing. In fact, not only were the results not statistically insignificant, but had the opposite sign to expectation in the few cases where they were significant, arguably due to a migration in mortgage lending from nonbanks to banks (Fuster et al. 2018).

3.3 Data

3.3.1 Sources

The primary data source is the Loan Application Register (LAR) which is maintained by the government due to The Home Mortgage Disclosure Act (HMDA) of 1975. HMDA mandates most lenders (regardless of their depository or non-depository status) to both gather and publicly disseminate information about housing-related loans and applications (Federal Financial Institutions Examination Council 2009).

Thus, LAR contains lender reported information for each individual transaction (loan application, origination, and purchase) in a calendar year⁴. In particular, one can observe the following information from LAR (Federal Financial Institutions Examination Council 2009) - (1) the loan purpose (home purchase, home improvement, or refinancing) and loan amount, (2) the property location and type⁵, (3) the outcome of the application, such as whether it was denied⁶ or resulted in an origination, (4) basic applicant demographic information (ethnicity, race, sex, and income), (5) lien

4. Fuster et al. (2018) being part of the Fed have access to monthly data, while general public like myself have access to yearly data only.

5. There are 3 property types in HMDA LAR - multifamily, manufactured housing, and one-to-four family (other than manufactured housing)

6. 9 reasons for denials are also reported, such as Debt-to-Income ratio, Employment history, Credit history, Collateral, Insufficient cash (down-payment, closing costs), Unverifiable information, Credit application incomplete, Mortgage insurance denied, Other.

status only for purchased loans applications and originations (secured by a first lien, a subordinate lien, not secured, and not applicable).

Unfortunately, HMDA LAR does not report some important and relevant factors used by firms in loan origination process, such as Debt-To-Income (DTI), Loan-To-Value (LTV), or FICO score of the applicant (Federal Financial Institutions Examination Council 2009). This restricts the type of questions that can be answered. For example, taste-based discrimination⁷ in lending process cannot be ascertained, since one cannot know if a person from the specific demographic group was denied due to prejudice or because of their individualistic characteristic. Fuster et al. (2018) largely mitigate this issue by merging HMDA data with Equifax's Credit Risk Insight Servicing McDash (CRISM) consumer level (paid) dataset, which I do not yet use.

Additionally, LAR does not report the interest rate charged on the loan. This restricts our ability to do welfare analysis as only quantity (number of approved loans) is observable while price (interest rate, cost or other terms of the loan) are not reported.

Nonetheless, HMDA LAR is still useful data for a couple of reasons (Federal Financial Institutions Examination Council 2009). For instance, it helps in finding out whether financial institutions are serving the residential needs of their communities, enables public officials in policy making like incentivizing private investment to localities where it is needed most leading to optimal public-sector investment, and even in recognizing potential discriminatory lending patterns.

HMDA LAR is complemented with two data sources. First, county population estimates and unemployment rates from United States Department of Agriculture (USDA). Second, percentage of households with high speed internet access within a county was constructed from Federal Communications Commission (FCC) Form 477. This measure of residential internet penetration was constructed by dividing "Residential Fixed High-Speed Connections over 200 kbps in at least one direction, in thousands" by "Households, in thousands⁸" released in Form 477 for years 2016 and 2017. It is continuous in more recent years. However, in 2015 and the years before that, FCC re-

7. Based on gender or race, as opposed to statistical discrimination

8. Estimates collected from *Census: 5-year ACS*

leased “Residential Fixed High-Speed Connections over 200 kbps in at least one direction per 1000 Households” directly but in categorical terms ranging from 0 to 5⁹. Thus, we have converted the more recent residential internet access (not usage) data in numerical format to categorical format to make it usable for the time period ranging from 2011 to 2017.

Furthermore, I could have used internet data from American Community Survey (ACS) internet supplemental but that data exists only for 400 counties about every two years, though it has many more accurate internet speed measures. I could have used a logistical adoption curve to impute values for missing years by combining the plethora of internet usage questions asked (latent variable model) in the survey, but prefer FCC data as it covers most counties albeit in a more coarser measurements.

Lastly, we are looking at 2011 to 2017, even though HMDA LAR exists in an easy to use format from 2007 because we are looking at post financial crisis time frame. There is a theoretical justification to focus on this time frame - the originate to distribute model is different in post crisis period. We can add 2010 but the results have converged, and also there was only 1 FinTech firm back in 2010 (2010 was its entry year too). Also, we are not looking at post 2017 because HMDA reporting style becomes different (LAR has to be downloaded individually for each firm, unlike in the past where complete LAR could be downloaded as a single csv file).

3.4 Specification

Recall the raw data is pooled cross-sectional data of approved applications by top 20 firms by national market share in 2016. We will collapse this individual level approved loan information to a county level for every year from 2011 to 2017, and thus get a **county-year panel data**.

9. A value of 0 meant that there are 0 connections per 1,000 households, a value of 1 meant that there are more than 0 but upto 200 connections per 1,000 households, a value of 2 meant that there are more than 200 but upto 400 connections per 1,000 households, a value of 3 meant that there are more than 400 but upto 600 connections per 1,000 households, a value of 4 meant that there are more than 600 but upto 800 connections per 1,000 households, & a value of 5 meant that there are more than 800 connections per 1,000 households.

Our null hypothesis that FinTech firms will cause more refinancing loans because of their originate to distribute business model, for which there has been some evidence presented by Fuster et al. (2018). However, the goal of this study is to extend and improve upon their work.

We cannot run a simple Ordinary Least Squares (OLS) regression to test for the simple market expansion theory due to a fundamental endogeneity issue. While the ‘presence’ of FinTech firms varies across counties, surely in these markets where there are more FinTech present more refinancing could take place as consumers may trust “digital-first platforms” or may have a desire for higher tailored product experience/personalization (Wtakins 2020). However, when consumers need more refinancing say in booming real estate markets in certain areas, FinTechs can be more present there too if they are better at identifying credit constrained areas/consumers. In other words, FinTech firms could be simply serving the financing needs of individuals, not increasing the financing of home loans. So, a reverse causality argument exists. In fact, Fuster et al. worry about endogeneity between refinancing and FinTech market share, and are concerned that FinTech firms could have predicted which counties have the highest potential for refinancing and simply advertised or expanded more heavily there. Thus, instead of running OLS like like Fuster et al., I try to find some better causal evidence, because if FinTechs essentially are filling in the refinancing need rather than driving/causing refinancing, OLS would be biased inspite of increasing sample size.

To solve this endogeneity problem, I use an instrument variables approach (by using a state-level Bartik instrument to pick out the “good” variation in county level FinTech presence (our key independent variable)). Initially, I had tried to use internet speed in county as an instrument for FinTech loans made in the county and was expecting a positive coefficient in the first stage regression - counties with faster average internet speeds should have more approved FinTech loans as faster internet is likely correlated with more widely available internet too, both of which would make the cost of accessing FinTechs lower. However, I got statistically significant negative coefficient in first stage, and thus decided to not use county internet speed as an instrument but only as a control.

Therefore, I use the average number of approved FinTech loans in all counties within the state of county c except county c to instrument for the number of approved FinTech loans in county c . This Bartik style instrument is common outside finance and industrial organization too, and especially popular in development economics (trade and migration literature) since 1991, and the basic idea is that county shares will measure the differential exogenous exposure to the common state level shock (Pinkham, Sorkin and Swift 2019).

A valid instrument needs to meet the two basic conditions - instrument relevance and instrument exogenous.

(1) *Instrument Relevance/Correlation Argument*: If certain macro trends are driving the popularity of FinTech firms, say artificial intelligence (AI) advances that are increasing customer engagement on FinTech websites/apps and thus lowering customer acquisition costs, or labor costs are getting lower as machine learning skillsets and other technology gets more mainstream so that setting up/maintaining the automated databases is lower, then FinTech activity/presence in all counties outside county c within the state is likely to be correlated with FinTech presence in county c . There is likely some regional variation across states due to different economics situations across states including different minimum wages, regulatory/banking laws, zoning and residential construction laws, et cetera.

However, there is a theoretical possibility that the instrument may not be relevant or a weak instrument at best. For example, if counties within the state too have strong legislative powers to affect the functioning/operations/regulatory costs of FinTech firms and also draft their own laws, then the idiosyncratic preferences of counties might be more important than general trend in the state - resulting in no/low correlation between what is happening in a particular county and what is happening in rest of the state. Such insulated counties are likely to be rare though, and we would expect the instrument coefficient on the first stage (equation 3.1 in Section 3.4.1) to have a positive coefficient as many banking laws are made a state level (though there are federal banking laws too).

(2) *Instrument Exogeneity Argument*: FinTech activity outside county c should affect refinancing activity in county c only through the channel of FinTech activity/firm behaviour in county c and no other channel. In other words, the correlation between the instrument and error term is zero in second stage (equation 3.2 in Section 3.4.1). Omitted variables (variables that could impact both our key parameter and dependant variable in the second stage) like more detailed risk profile of individuals at county level could invalidate our instrument. For example, if we are not able to completely capture the risk profile of individuals in the county, and this risk profiles significantly affect both FinTech loan origination and refinancing loan origination, then our instrument would be biased.

3.4.1 Empirical Methodology: IV with Time and County FE

The model specification we have is Two Stage Least Squares (2SLS) with county and year fixed effects.

c will index counties, and t will index years. The estimating equations are the following:

First Stage Least Squares (FSLs):

$$\begin{aligned} N_FinTech_{ct} = & \alpha_1 insS_N_FinTech_{-ct} + \alpha_2 N_mtg_{ct} + \alpha_3 internet_{ct} \\ & + \vec{\alpha}_4 CountyControls_{ct} + \vec{\alpha}_5 ApplicationControls_{ct} \\ & + Time_t + County_c + \epsilon_{ct} \end{aligned} \quad (3.1)$$

where $N_FinTech_{ct}$ is number of approved loans by FinTech firms in county c at time t , $insS_N_FinTech_{-ct}$ is the average number of approved FinTech applications in all other counties in home state of county c in year t (*instrument*),

N_mtg_{ct} is number of approved loans by nonbank mortgage lending firms in county c at time t , $internet$ is percentage of households with high fixed speed internet,

$CountyControls_{ct}$ include county unemployment rate and population estimate in year t ,

$ApplicationControls_{ct}$ include **approved applicants** demographic data as race, ethnicity or gen-

der, co-applicant information, loan type, *loan application amount*, and *gross income* aggregated to county level in year t ,

$Time_t$ represents time fixed effects (important to control for nation wide common shocks like Fed Funds rate hike that may impact loans, or other aggregate conditions),

$County_c$ represents county fixed effects (important to control for differences in counties like refinancing speeds that may impact loans).

Second Stage Least Squares (2SLS):

$$\begin{aligned} N_{refinance_{ct}} = & \beta_1 N_{\hat{FinTech}_{ct}} + \beta_2 N_{mtg_{ct}} + \beta_3 internet_{ct} \\ & + \vec{\beta}_4 CountyControls_{ct} + \vec{\beta}_5 ApplicationControls_{ct} \\ & + Time_t + County_c + \eta_{ct} \end{aligned} \quad (3.2)$$

where $N_{\hat{FinTech}_{ct}}$ is the instrumented number of approved loans by FinTech firms in county c at time t , and

$N_{refinance_{ct}}$ is number of approved refinanced loans in county c at time t .

3.4.2 Summary Statistics

Summary statistics of key variables (Table 3.1) and county-level controls (Table 3.2) are presented here.

First, there are almost twice as many approved refinanced loans (about 470) than approved home purchase loans in a “typical” year for an average county. Also, only 1.7% of the total approved/originated loans are for home improvement purposes.

Second, there are almost three and a half times as many approved loans by banks (about 470) than approved loans by FinTech firms in a given year for an average county. Banks are big players (in number of loans, though not so much in the dollar amount of loans - recall rise of shadow banks). Also, the total number of three types of loans (refinance, home improvement and home purchase) is

Table 3.1: Summary Statistics: Key Variables

	mean	sd	min	max	count
Total Refinance Loans in a county	473.360	2047.47	0.00	91952.00	21280
Total Home Improvement Loans in a county	12.172	66.41	0.00	3558.00	21280
Total Home Purchase Loans in a county	230.656	856.18	0.00	22149.00	21280
Total Loans by Depository Institution firms in a county	470.902	2053.39	0.00	94226.00	21280
Total Loans by Non-Bank Mortgage Lender firms in a county	118.774	530.70	0.00	16040.00	21280
Total Loans by FinTech firms in a county	126.513	489.80	0.00	17375.00	21280
State Bartik Instrument for FinTech loans in county c	124.995	173.59	3.89	1710.08	21273
County Mean of Residential Fixed High-Speed Connections	3.656	0.82	1.00	5.00	21280
County Mean: Population Estimate	93499.618	248776.29	676.00	5257481.00	19117
County Mean: Unemployment Rate	6.428	2.60	1.10	20.20	19117
County Mean: Loan Applicant's Gross Annual Income (in 1000's)	79.350	23.38	14.00	631.55	21272
County Mean: (Approved) Loan Amount (in 1000s)	149.717	62.18	15.00	1007.46	21280
Observations	21280				

3040 counties tracked over the 7 year period, 2011 to 2017.

Data shows only Approved Loans. For example, County Mean of (Approved) Loan Amount (in 1000s), or County Mean of (Approved) Loan Applicant's Gross Annual Income (in 1000's).

Source: HMDA, FCC, USDA (2011 to 2017).

equal to the total number of loans made by the three different financial intermediation institutions (banks/depository institutions, nonbank mortgage lenders and FinTechs), as expected.

Third, most counties have residential high speed internet data, though our definition of high speed (over 200 kbps in at least one direction) is not very strong. For our purposes, having 200kbps in one direction is enough to submit an online mortgage application.

Fourth, it is interesting to note that average approved loan applicant's annual income in a county was just under \$80,000, while the average mortgage loan amount in a county was almost twice (\$149,707.)

I also attach the summary statistics of other loan-level and borrower-level characteristics that will serve as control variables.

Most of the loan types are conventional (63%). Another 20% residential mortgages are issued by FSA/RHS (Farm Service Agency or Rural Housing Services), 14% VA-guaranteed (Veterans Administration) and 3% Federal Housing Administration (FHA) insured (mostly low income people who get a loan but at worse conditions). Preapproval are for home purchase loans only, and thus there are many NAs, but we can see that pre-approval is generally not requested. Also, most

Table 3.2: Summary Statistics: Controls

	mean	sd	min	max	count
County Mean of Home Loan: Conventional	0.630	0.16	0.00	1.00	21280
County Mean of Home Loan: FHA_insured	0.200	0.11	0.00	1.00	21280
County Mean of Home Loan: VA guaranteed	0.139	0.11	0.00	1.00	21280
County Mean of Home Loan: FSA or RHS guaranteed	0.031	0.05	0.00	1.00	21280
County Mean of Preapproval: Requested	0.009	0.02	0.00	1.00	21280
County Mean of Preapproval: Not Requested	0.153	0.11	0.00	1.00	21280
County Mean of Preapproval: not applicable	0.838	0.12	0.00	1.00	21280
County Mean of Applicant Ethnicity: Hispanic or Latino	0.040	0.09	0.00	1.00	21280
County Mean of Applicant Ethnicity: Not Hispanic or Latino	0.834	0.12	0.00	1.00	21280
County Mean of Applicant Ethnicity: not known	0.127	0.08	0.00	1.00	21280
County Mean of Applicant Race: American Indian or Alaska Native	0.010	0.03	0.00	1.00	21280
County Mean of Applicant Race: Asian	0.010	0.03	0.00	0.48	21280
County Mean of Applicant Race: Black or African American	0.037	0.07	0.00	1.00	21280
County Mean of Applicant Race: Native Hawaiian or Other Pacific Islander	0.002	0.01	0.00	1.00	21280
County Mean of Applicant Race: White	0.807	0.13	0.00	1.00	21280
County Mean of Applicant Race: not known	0.134	0.09	0.00	1.00	21280
County Mean of Applicant Sex: Male	0.662	0.11	0.00	1.00	21280
County Mean of Applicant Sex: Female	0.237	0.09	0.00	1.00	21280
County Mean of Applicant Sex: not known	0.101	0.08	0.00	1.00	21280
County Mean: Loan has a Co-Applicant	0.517	0.12	0.00	1.00	21280
Observations	21280				

Source: HMDA, 2011 to 2017

not known category includes not applicable responses and responses that were not provided by applicant in mail, internet or telephone application

approved applicants are white, male, and had a co-applicant.

3.5 Results

Table 3.3: Affect of FinTech on Different Types of Loans, 2011-2017

	(1)	(2)	(3)	(4)
	N_FinTech	N_refinance	N_home_imp	N_home_pur
insS_N_FinTech	0.462*** (0.116)			
N_FinTech		1.629* (0.882)	-0.010 (0.023)	0.358*** (0.124)
Regression	IV_FOLS	IV_2SLS	IV_2SLS	IV_2SLS
Relevance	YES (15.94)			
Controls	YES	YES	YES	YES
Time FE	YES	YES	YES	YES
County FE	YES	YES	YES	YES
R2		0.201	0.454	0.732
N	19102	19102	19102	19102

Controls include approved loan applications demographic information aggregated to county level.

* $p < .10$, ** $p < .05$, *** $p < .01$

First Stage Least Squares (FOLS):

From column 1 in Table 3.3, we can see that instrument ($insS_N_FinTech_{-ct}$) has a “correct” positive signed coefficient - when average number of applications to FinTech firms in all other counties in the county’s state is increasing by 1 application, the number of applications to FinTech firms in county c is increasing by 0.462 or roughly $\frac{1}{2}$ application. Instrument is strong/relevant (F-test statistics > 10), and thus seems to have enough variation over time and counties too. As expected, the instrument coefficient on the first stage (equation 3.1) has a positive coefficient as many banking laws are made a state level, though there are federal banking laws too.

Second Stage Least Squares (2SLS):

From column 2 in Table 3.3, we can see that if the number of approved loan applications by FinTech firms in a county increases by 1 unit, ceteris paribus, the number of approved refinance loans in the county are likely to increase by 1.63 units. This is a large in magnitude, and statistically significant

at 10% significance level, but makes sense as word of mouth can easily explain the high elasticity of refinancing loans due to higher presence of FinTech firms in the county. Also, since people who are refinancing their home are likely to have had a home purchase loan or even a prior home refinancing loan, their higher financial literacy can explain their willingness to refinance.

From column 3 in Table 3.3, we can see that if the number of approved loan applications by FinTech firms in a county increases by 1 unit, *ceteris paribus*, the number of approved home improvement loans in the county are not going to be statistically different from 0. This makes sense as hardly any, or only 1.7% of the total approved/originated loans are for home improvement purposes.

From column 4 in Table 3.3, we can see that if the number of approved loan applications by FinTech firms in a county increases by 1 unit, *ceteris paribus*, the number of approved home purchase loans in the county are likely to increase by 0.36 units. While this is not large in magnitude, it is statistically significant at 1% significance level, and makes sense if FinTech firms really simply the home purchase decision for first time home buyers.

Also, the coefficients in column 2 and column 4 in Table 3.3 make sense in magnitude as we know that FinTech firms are more than twice as likely to be active in refinancing market (70%) than home purchase market (30%), suggesting that magnitude in column 2 should be higher than column 4. as we indeed find.

In Appendix 3.8.2, I attach the bigger version of Table 3.3 so that with the co-efficients on control variables can be seen. We can see from Table 3.4 in Appendix 3.8.2 that signs on many control variables are as expected.

First, when unemployment rate increases in county c by one percentage point (one unit), approved home purchase applications in the county c seems to be decreasing by more than 4 applications (column 2), approved home improvement applications seem to decreasing by .32 applications (column 3) and approved refinancing applications do not seem to be changing statistically significantly (column 4), keeping everything else constant. This makes as higher unemployment rate

should on average not induce people to take out loans when incomes are falling.

Second, as internet speed in county increases, it does not statistically significantly decrease the number of approved refinanced loans, home improvement loans or home purchase loans. Nonetheless, internet speed does not have a statistically significant positive impact on financial intermediation.

Third, non-bank mortgage originators seem to have a significant impact on home purchases (and home improvement) but not on refinancing, suggesting they have found their own niche and usually stay out of refinancing business dominated by FinTech firms.

Fourth, as county population increases, approved refinancing loans decrease significantly - maybe due to pressure on income of loan applicants.

Fifth, as county average gross income increases, number of approved home improvement loans increase as people may want to refurbish their homes, but surprisingly approved home purchases decrease.

Lastly, as the county average loan amount increases, approved number of home purchases also significantly increases, suggesting people are buying more houses and entering the residential mortgage market.

3.6 Conclusion

In this paper, the key result that I find is that when there are larger the number of approved applications by FinTech firms, *ceteris paribus*, there is a larger number of approved refinancing and home purchase loans. Both these results are statistically significant, and we find no impact of FinTech firms in home improvement market. This can be attributed to the originate-to-distribute business model of Fintech firms.

While like Fuster et al. (2018) I find that FinTech firms cause more refinancing in home refinancing market, I do use a different methodology (IV over OLS) and specification (number of

approved applications at a county level over refinancing propensity¹⁰. In spite of lower frequency publicly available HMDA data (Fuster et al. 2018 have access to monthly HMDA) and slightly less detailed controls from external sources like CRISM, and small differences in sample size (I have all counties but only largest 20 firms look while they examine only the largest 500 counties with 12,000 applications each (number of applications in the smallest county), we get similar results. The primary contribution of this paper is that we mitigate the endogeneity problem between refinancing activity and FinTech presence.

3.6.1 Further Work

While this is an improvement, nonetheless there are a few things that can be done to further strengthen my results. First, I would like to run some formal tests of instrument exogeneity to strengthen the methodology validity.

Second, I would like to mitigate concerns about the Bartik instrument being driven by demand shifters. For example, it may not be clear whether universal lowering of cost, or increased consumer preference for online loan origination can explain sign on Bartik instrument in first stage. Disentangling the two effects will provide more insights, and I plan to mitigate the concerns about demand side effects driving the results by controlling for demand side factors. One quick thing I can do is to utilize Google search trends for key words like “online loan” or “Quicken Loans” to account for demand changes.

Third, I would like to control for market expansion and business stealing effects, since my panel data currently looks at only top 20 firms by market share within counties. I can use total amounts of loans originated in a county as a control for total market size. for business stealing, I will have to expand by dataset to the entire set of firms,. However, note that there are not a lot of FinTech firms anyways in the data anyways (only 18), and most of them lie outside top 500 originators by market share (though market seems to be competitive at national level after the four

10. Created by identifying refinances, and then calculating refinance propensities at the county level, defined as the percentage of loans from month $t - 1$ that are refinanced in month t .

largest firms).

Fourth, the current analysis can be completed by obtaining data on declined applications. Furthermore, obtaining more detailed (paid) dataset on borrower and loan characteristics like FICO score, interest rate/payments, or even HMDA at a higher frequency could further strengthen our analysis.

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3.8 Appendix

3.8.1 Figures

Figure 3.1: Top 20 mortgage originators by market share classified into 3 types of firm. Note that residential mortgage origination market is competitive at national level, with each firm having a very small market share.

Table 1: FinTech and Top 20 Mortgage Originators in 2016, based on Home Mortgage Disclosure Act (HMDA)

Rank	Type of Lender	Lender Name	Volume (Bn)	Market Share (%)	FinTech Since
1	Bank	Wells Fargo	132.58	6.62	2010
2	FinTech	Quicken Loans	90.55	4.52	
3	Bank	JPMorgan Chase	75.52	3.77	
4	Bank	Bank Of America	60.24	3.01	
5	FinTech	Loandepot.com	35.94	1.80	2016
6	Mtg	Freedom Mortgage	32.17	1.61	
7	Bank	US Bank	30.69	1.53	
8	Mtg	Caliber Home Loans	27.94	1.40	
9	Bank	Flagstar	27.31	1.36	2010
10	Mtg	United Shore	22.93	1.15	
11	Bank	Citibank	21.73	1.09	
12	FinTech	Guaranteed Rate	18.44	0.92	
13	Mtg	Finance of America	17.63	0.88	2014
14	Mtg	Fairway Independent	16.10	0.80	
15	Bank	USAA Federal Savings	15.52	0.78	
16	Mtg	Guild Mortgage	15.07	0.75	
17	Mtg	Stearns Lending	14.93	0.75	2016
18	Bank	Suntrust Mortgage	14.77	0.74	
19	Bank	Primelending	13.87	0.69	
20	Mtg	Nationstar Mortgage	13.50	0.67	
...					
23	FinTech	Movement Mortgage	11.61	0.58	2014
39	FinTech	Everett Financial (Supreme)	7.62	0.38	2016
534	FinTech	Avex Funding (Better.com)	0.49	0.02	2016

Bank = Depository Institution, Mtg = Nonbank Mortgage Lender, FinTech = FinTech Lender

Market share is based on dollar volume of originations in HMDA. Table reports all lenders classified as Fintech at the end of 2016. Our classification also identified 12 additional nonbank lenders who began offering an online application platform in the first half of 2017, after the end of our HMDA sample period: FBC Mortgage Inc.; RPM Mortgage Inc.; ARK-LA-TEX Financial Services (Benchmark Mortgage); Envoy Mortgage; Skyline Financial Corp.; Evergreen Moneysource Mortgage; American Neighborhood Mortgage; Mortgage Investors Group; 21st Mortgage Corp; Amerisave Mortgage Corp.; Homeward Residential Inc. and American Internet Mortgage.

3.8.2 Tables

Table 3.4: Affect of FinTech on Different Types of Loans, 2011-2017

	(1) N_FinTech	(2) N_refinance	(3) N_home_imp	(4) N_home_pur
insS_N_FinTech	0.462*** (0.116)			
N_FinTech		1.629* (0.882)	-0.010 (0.023)	0.358*** (0.124)
N_Mtg (Nonbank Mortgage Lender)	0.458** (0.189)	-0.861 (0.653)	0.053*** (0.016)	0.586*** (0.096)
Internet Speed	-6.444*** (2.173)	6.164 (13.666)	0.205 (0.157)	2.261 (2.220)
County Unemployment Rate	0.264 (1.489)	16.891 (14.504)	-0.319* (0.184)	-4.096*** (1.178)
County Population Estimate	0.004 (0.003)	-0.031** (0.014)	0.000 (0.000)	0.001 (0.001)
County Avg. Gross Income	0.013 (0.061)	0.794 (0.493)	0.015* (0.008)	-0.092* (0.052)
County Avg. Loan Amt	-0.130 (0.104)	-3.703*** (0.990)	-0.030** (0.015)	0.336*** (0.113)
Regression	IV_FOLS	IV_2SLS	IV_2SLS	IV_2SLS
Relevance	YES (15.94)			
Controls	YES	YES	YES	YES
Time FE	YES	YES	YES	YES
County FE	YES	YES	YES	YES
R2		0.201	0.454	0.732
N	19102	19102	19102	19102

Controls include approved loan applications demographic information aggregated to county level.

* $p < .10$, ** $p < .05$, *** $p < .01$